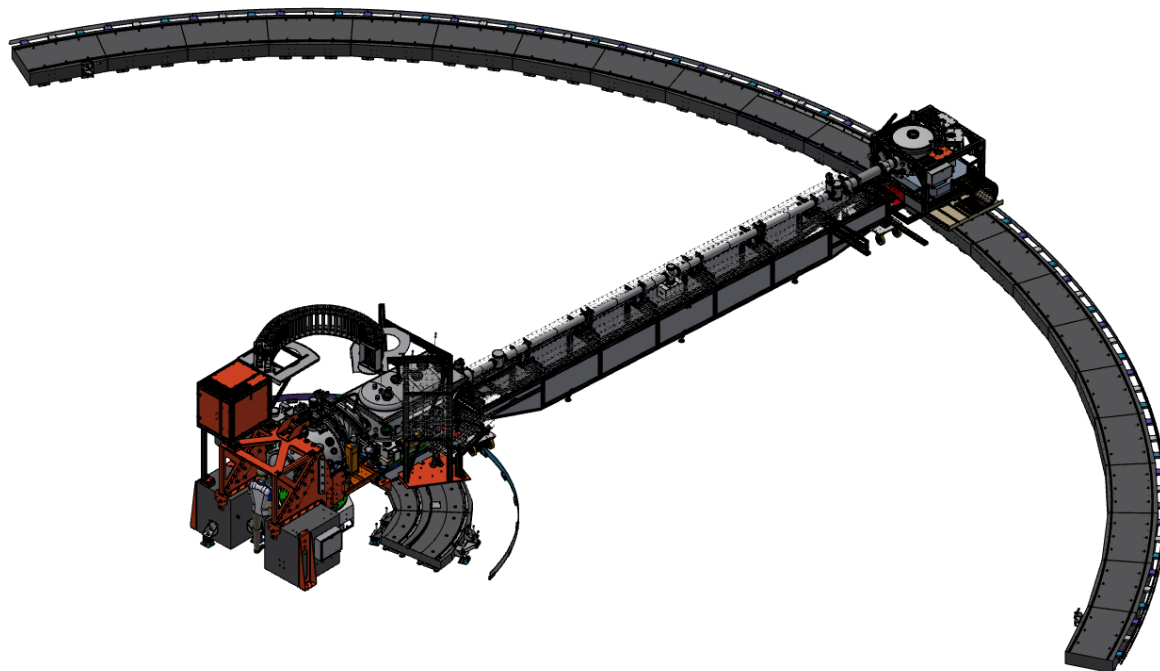


Revised Instruction Manual

Spectrometer Arm System for SIX Beamline
at the National Synchrotron Light Source II
at Brookhaven National Laboratory



BESTEC Revised Instruction Manual - Project P442

Version 2.0

October 8, 2020

CEO: Dr. Rainer Hammerschmidt
Municipal court Charlottenburg HRB 34406
VAT ID DE 137202757



BESTEC GmbH
Am Studio 2b
12489 Berlin
GERMANY

Phone: +49 30 677 43 76
Fax: +49 30 677 57 18
Web: <http://www.bestec.de>
Email: info@bestec.de (administrative tasks)
service@bestec.de (all service reasons)

All rights reserved. No part of this manual may be reproduced without the prior permission of BESTEC GmbH. Other product and company names mentioned in this document may be trademarks or registered trademarks of their respective owners and are used for identification purposes only.

1 Inhalt

2	Introduction	14
3	Intended use	15
4	Coordinate system and conventions	16
5	Design overview of the SIX end station	18
5.1	Provisions for EPS	19
6	SC - sample chamber	21
6.1	Safety information	21
6.1.1	Used symbols	21
6.1.2	Safety instructions	22
6.1.3	Modifications and service	24
6.2	Design overview of the sample chamber	25
6.3	Maintenance	29
6.3.1	Bakeout procedure	29
6.4	Appendix	31
6.4.1	Overview of technical specification	31
6.4.2	List of supplied wiring schemes	33
6.4.3	List of supplied assembly drawings	34
6.4.4	List of recommended spare parts	35
6.4.5	List of supplied subcontractors documents	35
7	Triple rotation flange (TRF)	36
7.1	Safety information	36
7.1.1	Used symbols	36
7.1.2	Safety instructions	37
7.1.3	Modifications and service	39
7.2	Design overview of the triple rotation flange	40
7.3	Installation of the triple rotation flange	42
7.4	Maintenance	56
7.4.1	Bakeout procedure	56
7.5	Appendix	57
7.5.1	Overview of technical specification	57
7.5.2	Rotation angles of the TRF flange	58
7.5.3	Possible collision regions	60
7.5.4	List of supplied wiring schemes	63

7.5.5	List of supplied assembly drawings	64
7.5.6	List of recommended spare parts	64
7.5.7	List of supplied subcontractors documents.....	65
8	M5 mirror system.....	66
8.1	Safety information	66
8.1.1	Used symbols	66
8.1.2	Safety instructions.....	67
8.1.3	Modifications and service	69
8.2	Design overview of the M5 mirror system.....	70
8.3	M5 mirrors assembly: overview and installation of optical elements	74
8.4	Appendix.....	82
8.4.1	Overview of technical specifications	82
8.4.2	Location of in-vacuum components and sensors.....	86
8.4.3	List of supplied wiring schemes.....	86
8.4.4	List of supplied assembly drawings	88
8.4.5	List of recommended spare parts	89
8.4.6	List of supplied subcontractor documents.....	89
9	3AA - 3 m arm and optics tank assembly	90
9.1	Safety information	90
9.1.1	Used symbols	90
9.1.2	Safety instructions.....	91
9.1.3	Modifications and service	93
9.2	Design overview of the 3AA system	94
9.3	Design overview of the optics tank	95
9.3.1	M6 mirror assembly: overview and installation	101
9.3.2	M7 mirror and gratings mechanics: overview and installation	107
9.4	Maintenance	113
9.4.1	Bakeout procedure.....	113
9.4.2	Sealing of the vacuum chamber main flange.....	114
9.4.3	Lubrication schedule	116
9.5	Appendix.....	117
9.5.1	Overview of technical specifications	117
9.5.2	Sine-bar mechanism	125
9.5.3	Location of in-vacuum components and sensors.....	128
9.5.4	List of supplied wiring schemes.....	128

9.5.5	List of supplied assembly drawings	131
9.5.6	List of recommended spare parts	133
9.5.7	List of supplied subcontractor documents.....	134
10	DAA - detector arm and stand assembly.....	135
10.1	Safety information	135
10.1.1	Used symbols	135
10.1.2	Safety instructions.....	136
10.1.3	Modifications and service	138
10.2	Design overview.....	139
10.3	Appendix.....	148
10.3.1	Overview of technical specifications	148
10.3.2	List of supplied wiring schemes.....	151
10.3.3	List of supplied assembly drawings	152
10.3.4	List of recommended spare parts	153
10.3.5	List of supplied subcontractor documents.....	153
11	Instructions and hints for synchronized 2-theta motion of the system.....	155
12	Ambient condition and media supply.....	156
13	Vacuum scheme	157
14	Declaration of conformity	160

List of Figures

Figure 1: Conceptual overview of the SIX beamline	15
Figure 2: Illustration of the coordinate system	16
Figure 3: Sketch of the two M5 mirrors on their common baseplate and their coordinate system. $X_{M5}Y_{M5}Z_{M5}$ are defined with respect to the baseplate of the M5 mirrors	17
Figure 4: Design overview of the spectrometer arm system	18
Figure 5: Design overview of the sample chamber	25
Figure 6: Design overview of the sample chamber	26
Figure 7: Maintenance door at the sample chamber	27
Figure 8: Bottom view of the sample chamber	27
Figure 9: Cryostat manipulator unit	28
Figure 10: Differentially pumped flange at the sample chamber	29
Figure 11: Differentially pumped rotary feedthrough at the cryostat manipulator	30
Figure 12: Triple rotation flange	40
Figure 13: Triple rotation flange	41
Figure 14: Triple rotation flange in transport crate.	42
Figure 15: Triple rotation flange fixed on support frame	43
Figure 16: Center of gravity of the TRF + support frame assembly for the transportation position (left) and for installation position (right), respectively.	43
Figure 17: Top view of the sample chamber. The blue arrow indicates the recommended moving direction of the spectrometer arm	44
Figure 18: Exit flange of the sample chamber	45
Figure 19: Internal mechanics of the sample chamber	45
Figure 20: Initial washer and nut (left) must be replaced be special nut (right)	46
Figure 21: Anchor point at the TRF before moving the TRF1 axis by 90°	46
Figure 22: Angle bracket at the drag chain	47
Figure 23: Screws to be removed	47
Figure 24: Screws to be removed at the star-shaped flange	48
Figure 25: Motor at the TRF1 motion axis	48
Figure 26: Position of the differential pumping ports at the TRF before (top) and after (bottom) its rotation to match the holes at the sample chamber.	49
Figure 27: Angle brackets at the sample chamber	50
Figure 28: Figure 29: Angle brackets at the sample chamber	50
Figure 30: Figure 31: Angle brackets at the sample chamber	51

Figure 32: Angle bracket holding cable drag chain	51
Figure 33: Encoder readheads at the TRF1 motion axis.....	52
Figure 34: Gate valve at the TRF	52
Figure 35: Metal sheet that activates the limit switches at the TRF3 motion axis	53
Figure 36: Top view of the sample chamber and 3AA optics tank.....	53
Figure 37: Gravity compensation spring at the gate valve	54
Figure 38: Collision detection switches at the TRF3 motion axis	54
Figure 39: Connection between the TRF and 3AA optics tank.....	55
Figure 40: Collision detection switches.....	55
Figure 41: Encoder cables at the triple flange.....	56
Figure 42: 2Theta and TRF swivel angles ranges.....	59
Figure 43: Possible collision region	60
Figure 44: Possible collision position	61
Figure 45: Possible collision position	62
Figure 46: Sample chamber internal mechanics and support structure overview.....	70
Figure 47: Sample chamber internal mechanics and support structure overview.....	71
Figure 48: Sample chamber internal mechanics and support structure cross section	72
Figure 49: Top view on sample chamber internal mechanics (left)	73
Figure 50: Pivot translation assembly	73
Figure 51: M5 mirrors assembly	74
Figure 52: M5 mirrors assembly	75
Figure 53: M5 mask assembly (upstream view).....	76
Figure 54: Thermocouple plug (left), motor/limit switches plug (right)	76
Figure 55: Upper part of the tripod.....	77
Figure 56: Collision switch at the M5 mirror assembly	77
Figure 57: M5 mirror holder assembly top view (left) and side view (right).....	78
Figure 58: Mirror holder assembly	78
Figure 59: Removal of the fixing clamps at the mirror holder assembly	79
Figure 60: Upper mirror holder (left), cross section of a fixing clamp (right)	79
Figure 61: Fixing screws which have to be loosened.....	80
Figure 62: Sliding procedure of the optical mirror through the fixing claps	80
Figure 63: Lower mirror holder (left), cross section of a fixing clamp (right)	81
Figure 64: Fixing screws which have to be loosened.....	81

Figure 65: Design overview of the 3AA system	94
Figure 66: Design overview of the optics tank (inboard)	95
Figure 67: Design overview of the optics tank (outboard)	96
Figure 68: Gratings slits assembly (M7 mask), upstream view (left) and downstream view (right).....	97
Figure 69: M7 mirror mask assembly (downstream view)	97
Figure 70: 2-theta rotation stage of the optics tank	98
Figure 71: Internal mechanics of the optics tank.....	99
Figure 72: Internal mechanics of the optics tank (side view)	100
Figure 73: M6 mirror assembly	101
Figure 74: Vertical aperture assembly	102
Figure 75: M6 mirror mask (upstream view)	102
Figure 76: M6 mirror base plate.....	103
Figure 77: Top view on the M6 mirror assembly	104
Figure 78: M6 mirror holder assembly	105
Figure 79: View on the M6 dummy surface.....	105
Figure 80: Internal mechanics (M7 and gratings) of the PGM unit	107
Figure 81: M7 mirror cradle	108
Figure 82: Top view of the mirror cradle	109
Figure 83: Sine-bar drive for mirror rotation	110
Figure 84: Grating cradle, top view	110
Figure 85: Grating holder assembly (side view).....	111
Figure 86: M7 mirror holder assembly	112
Figure 87: Sealing of the main flange	114
Figure 88: Turnbuckle at the M1 vacuum chamber main flange.....	115
Figure 89: Sketch of the sine-bar mechanism driving M6 mirror pitch.....	125
Figure 90: Sketch of the sine-bar mechanism driving M7 mirror pitch.....	126
Figure 91: Sketch of the sine-bar mechanism driving gratings pitch	127
Figure 92: Design overview of the DAA system	139
Figure 93: Design overview of the detector chamber	140
Figure 94: Aperture unit at the entrance of the detector chamber, downstream view	141
Figure 95: Arrangement of the aperture blades at the entrance of the detector chamber....	141
Figure 96: Linear actuator assembly.....	142
Figure 97: Detector arm assembly, side view	142

Figure 98: Upstream carriage, upstream view (left) and downstream view (right)	143
Figure 99: Downstream carriage, upstream view	143
Figure 100: Collision switches (2x) at the downstream carriage (left), collision switches (4x) at the detector arm (upstream)	144
Figure 101: Free switches at the spectrometer arm	144
Figure 102: Stacking order of the detector chamber support structure	145
Figure 103: Encoder and switches at the DT motion axis	146
Figure 104: Free switches at the translation axis of the detector	146
Figure 105: 2-theta rotation stage of detector chamber	147
Figure 106: Vacuum scheme of the spectrometer arm system	157
Figure 107: Vacuum scheme of the spectrometer arm system	158
Figure 108: Vacuum scheme of the spectrometer arm system	159

List of Tables

Table 1: Provision for EPS at the sample chamber.....	19
Table 2: Provision for EPS at the M5 mirror system	19
Table 3: Provision for EPS at the 3AA - 3 m arm and optics tank assembly	20
Table 4: Provision for EPS at the DAA - detector arm and stand assembly	20
Table 5: Overview of technical specifications	31
Table 6: List of wiring schemes for the cryostat manipulator.....	33
Table 7: List of assembly drawings for the sample chamber.....	34
Table 8: List of recommended spare parts.....	35
Table 9: List of provided subcontractors documents.....	35
Table 10: List of wiring schemes for the triple rotation flange.....	63
Table 11: List of assembly drawings for the triple rotation flange.....	64
Table 12: List of recommended spare parts.....	64
Table 13: List of provided subcontractors documents.....	65
Table 14: Overview of technical specifications (M5)	82
Table 15: Location of in-vacuum components and sensors	86
Table 16: List of wiring schemes for the complete spectrometer arm system	86
Table 17: List of assembly drawings for the M5 mirror system.....	88
Table 18: List of recommended spare parts.....	89
Table 19: List of provided subcontractors documents.....	89
Table 20: Overview of technical specifications (M6)	117
Table 21: Overview of technical specifications (vacuum vessel, M7, gratings)	119
Table 22: Location of in-vacuum components and sensors	128
Table 23: List of wiring schemes for the 3AA system.....	128
Table 24: List of assembly drawings for the 3AA system.....	131
Table 25: List of recommended spare parts.....	133
Table 26: List of provided subcontractor documents.....	134
Table 27: Overview of technical specifications (DAA).....	148
Table 28: List of wiring schemes for the complete spectrometer arm system	151
Table 29: List of assembly drawings for the complete spectrometer arm system.....	152
Table 30: List of recommended spare parts.....	153
Table 31: List of provided subcontractor documents.....	153

List of Abbreviations

UHV	Ultra High Vacuum
SAS	Spectrometer Arm System
EPS	Equipment Protection System

SC - sample chamber

MHT1	Manipulator Horizontal Translation 1
MHT2	Manipulator Horizontal Translation 2
MVT	Manipulator Vertical Translation
MR	Manipulator Rotation

M5 mirror system

M5R2T	M5 Mirror 2-Theta Rotation
M5T1->M5T6	M5 Mirror Tripod Axis 1 ->6
PTX	Pivot Translation X
PTY	Pivot Translation Y
M5M	M5 Mirror Mask
TC1	Thermocouple at Mirror Holder
TC2	Thermocouple at Mirror Holder
TC3	Thermocouple at Rotating Support
TC4	Thermocouple at Rotating Support
TC5	Thermocouple at Support Pillar
TC6	Thermocouple at Support Pillar
CSW-M5T1	Collision Switch at M5 Holder
PD1	Photodiode
PD2	Photodiode
PD3	Retractable Photodiode

3AA - 3 m arm and optics tank assembly

3AA	3m Arm and Optics Tank Assembly
M6P	M6 Mirror Rotation (pitch)
M6TZ	M6 Mirror Translation
M6M	M6 Mirror Mask
M7P	M7 Mirror Rotation (pitch)
GP	Grating Rotation (pitch)

GT	Grating Translation
BTZ	Optics Base Plate Translation
M7M	M7 Mirror Mask
3AAR2T	3AA 2-Theta Rotation
OVY	Wedge Leveler Outboard
IVY	Wedge Leveler Inboard
TC7	Thermocouple at Bearing Block Left
TC8	Thermocouple at Bearing Block Right
TC9	Thermocouple at Gratings
TC10	Thermocouple at M7 Grating Support
TC11	Thermocouple at M6 Grating Support
TC12	Thermocouple at 2Theta Traveler
CSW-GP	Collision Switch Grating Rotation (Pitch)
FSW-GT	Free Switch Grating Translation
CSW12-3AAR2T	Collision Switch
CSW34-3AAR2T	Collision Switch
FSW-3AAR2T	Free Switch

DAA - detector arm and stand assembly

DAA	Detector Arm and Stand Assembly
DAAHUL	Detector Arm Height Upstream Left
DAAHUR	Detector Arm Height Upstream Right
DAAHD	Detector Arm Height Downstream
DAAR2T	DAA 2-Theta Rotation
DT	Detector Translation
CSW-DAAR2T	Collision Switch
AT	Aperture Top
AB	Aperture Bottom
AL	Aperture Left
AR	Aperture Right
FWD KSW	Forward Kill Switch
REV KSW	Reverse Kill Switch
FWD LSW	Forward Limit Switch
REV LSW	Reverse Limit Switch

BESTEC GmbH holds the rights for the design of the system. All given technical information shall be handled careful and secure.



2 Introduction

This instruction manual contains information necessary for installing and operating the spectrometer arm system built by Bestec GmbH for the Soft Inelastic X-ray Scattering (SIX) beamline at the National Synchrotron Light Source II (NSLS-II) at Brookhaven National Laboratory. The manual describes the main components of the spectrometer system, its specification and operation parameters. This document is part of the system and must be available for any person who is working with the system.

Manuals of the individual components are supplied as well. It is the responsibility of the user to read all instructions carefully before using the system or individual components. Any damage of components caused by incorrect user operation will void the warranty.

This document contains also caution and warning statements which must be complied with in order to prevent inaccurate measurement, equipment damage, or personal injury. The spectrometer arm system should be installed and operated only by authorized personnel.

**PLEASE READ THE INSTRUCTIONS
BEFORE ATTEMPTING TO OPERATE THE SYSTEM!!!**

3 Intended use

The Soft Inelastic X-ray scattering (SIX) beamline is an ultrahigh resolution inelastic scattering beamline. It is an insertion device beamline, whose source is a variable polarization elliptically polarized undulator, EPU57, and will operate between the nominal energies of 165 to 2300 eV. The beamline intends to reach an unprecedented energy resolution (10 meV at 1000 eV), which is an order of magnitude higher than a typical soft x-ray beamline operating in this energy range.

The conceptual configuration of the SIX beamline is outlined in Figure 1. The first plane mirror (M1) deflects the beam horizontally and reduces the power transmitted to the monochromator. The PGM contains a plane premirror (M2) which bounces the beam vertically onto one of three different VLS gratings. The beam is vertically focused by the VLS grating, and then horizontally focused by an elliptical cylinder mirror (M3), yielding a stigmatic focus at the exit slit. The exit slit defines the beamline energy and acts as a secondary source, which is imaged at the sample position using a refocusing ellipsoidal mirror (M4). In the spectrometer (not shown in Figure 1), the beam scattered off from the sample is horizontally collimated by a pair of parabolic mirrors (M5-A and M5-B) and vertically focused by a circular cylindrical mirror (M6). The beam is then wavelength dispersed vertically focused, and directed at a constant output direction by a premirror (M7) - VLS grating combination, using Riemer-Torge mechanics similar to those used in modern plane grating monochromators. The detector collects the beam at a grazing angle of 25° to 90°.

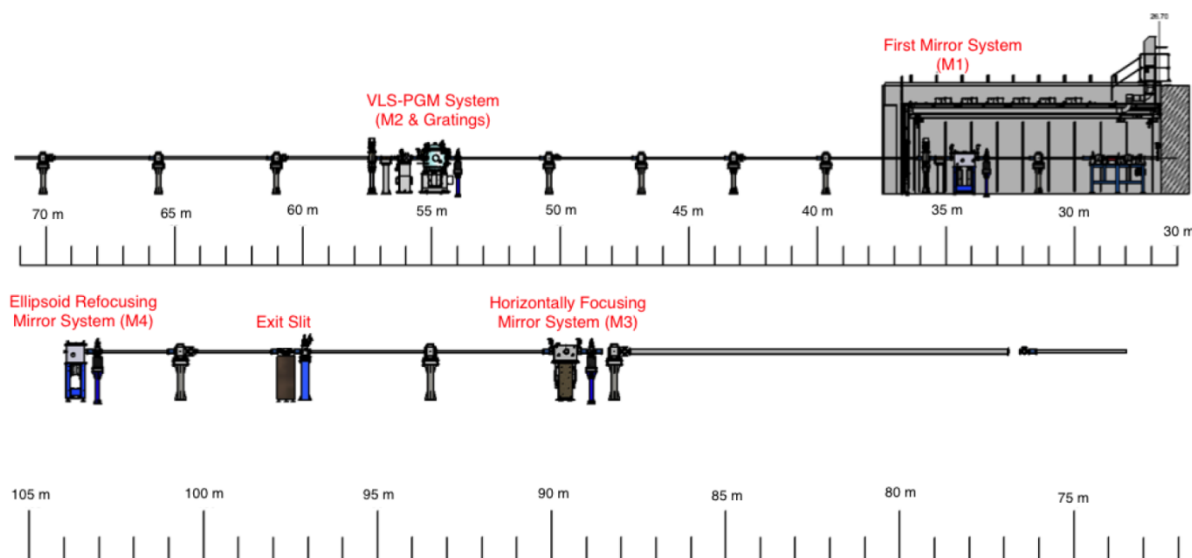


Figure 1: Conceptual overview of the SIX beamline

The beamline occupies a long high- β straight section of the NSLS-II storage ring. The Spectrometer Arm System (SAS) and Sample Chamber (SC) are housed in the experimental area of a dedicated satellite building located on the outboard side of the ring building.

The SC is immediately downstream of the M4 ellipsoidal mirror system. The SAS rotates around the sample location (the 2θ rotation) towards the outboard side of the experimental area about the axis Y defined in Figure 2.

4 Coordinate system and conventions

This document uses a global Cartesian coordinate system to specify various orientations and directions. The coordinate system illustrated in Figure 2 is described as:

- The height (Y) is set as the height of the electron beam in the storage ring.
- The origin is located in the horizontal plane (X, Z) at the ideal location of the sample θ axis of rotation, the 3-m arm 2θ axis of rotation, and the detector arm 2θ axis of rotation, the three of which ideally coincide.

The positive Z axis runs parallel to the inboard wall of the experimental area in the nominal direction of beam propagation. The positive X axis points outboard and is in the horizontal plane. The positive Y axis points vertically upward. 2θ angles are measured positive as counterclockwise from the positive Z axis when looking from the top.

One set of rotation axes is defined as rotation about each of the Cartesian axes. The designation is (ax, ay, az), meaning “angle about x,” etc. The sign of these rotations is right-handed, except for the angle ϕ , which is taken as positive in the upwards direction, with the horizontal taken as the zero reference.

A second set of rotation axes are the named rotations (e.g., pitch, theta, etc.), depending on the requirements of the component. The relationships between these rotations and ax, ay, and az are listed in each of the tables of motions, where applicable. Note that the detector local coordinate system ($x_{det}y_{det}z_{det}$) rotates with 2θ .

The optical components along the spectrometer arm system have individual local coordinate systems, as illustrated in Figure 2 for M6, M7 and the grating and Figure 3 for M5. The origin of the M5, M6, M7 and grating coordinate systems is set at the pole of the respective optic.

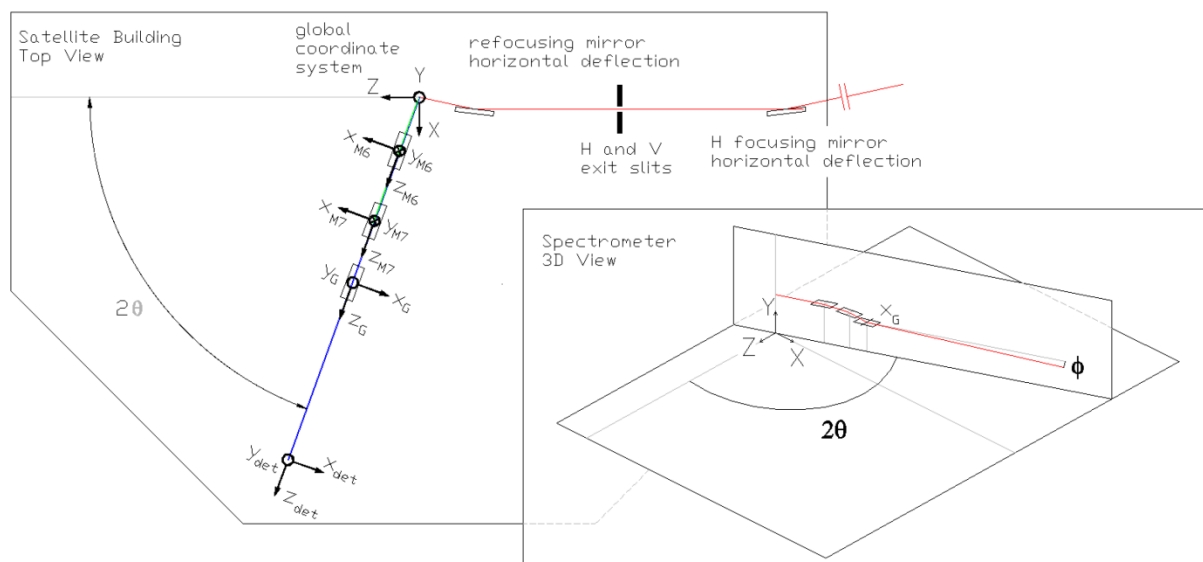


Figure 2: Illustration of the coordinate system

The global coordinate system is referred to as XYZ, the M6 coordinate system as $x_{M6}y_{M6}z_{M6}$, the grating coordinate system as $x_Gy_Gz_G$ and the detector coordinate system (which rotates

with 2θ and stays at $\phi = -1$) as $x_{\text{det}}y_{\text{det}}z_{\text{det}}$. The M6 and the grating coordinate systems are understood to be oriented with the optical face of the element. The angle the detector makes in a vertical plane about the center of pitch rotation of the gratings (x_G) is denoted by ϕ and is taken as positive upwards with the zero at horizontal.

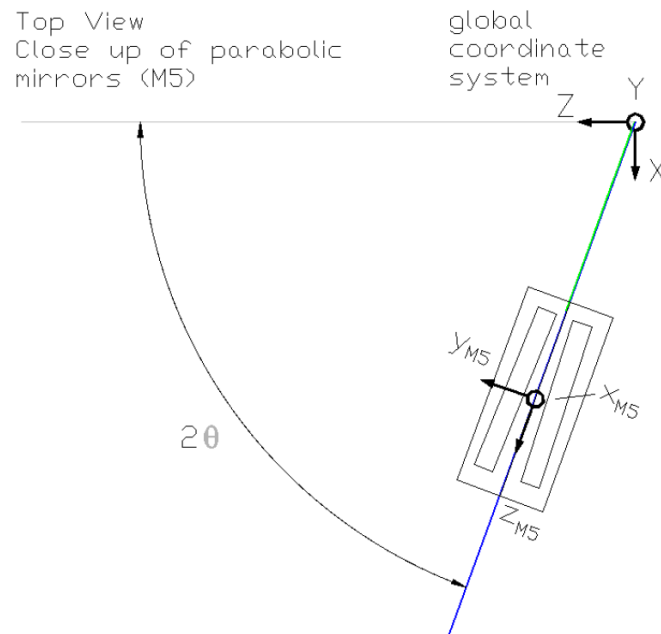


Figure 3: Sketch of the two M5 mirrors on their common baseplate and their coordinate system.
 $x_{M5}y_{M5}z_{M5}$ are defined with respect to the baseplate of the M5 mirrors

5 Design overview of the SIX end station

Figure 4 shows the SIX end-station design overview. The SIX end-station is subdivided into the sample chamber unit (SC) and the spectrometer arm system (SAS). The SAS consists of the 3m arm and optics tank assembly (3AA) and the detector arm and stand assembly (DAA). The motion control system is not included in the scope of delivery, i.e. the SIX end station delivered from Bestec GmbH is an incomplete machine according to the European machine directive 2006/42/EG.

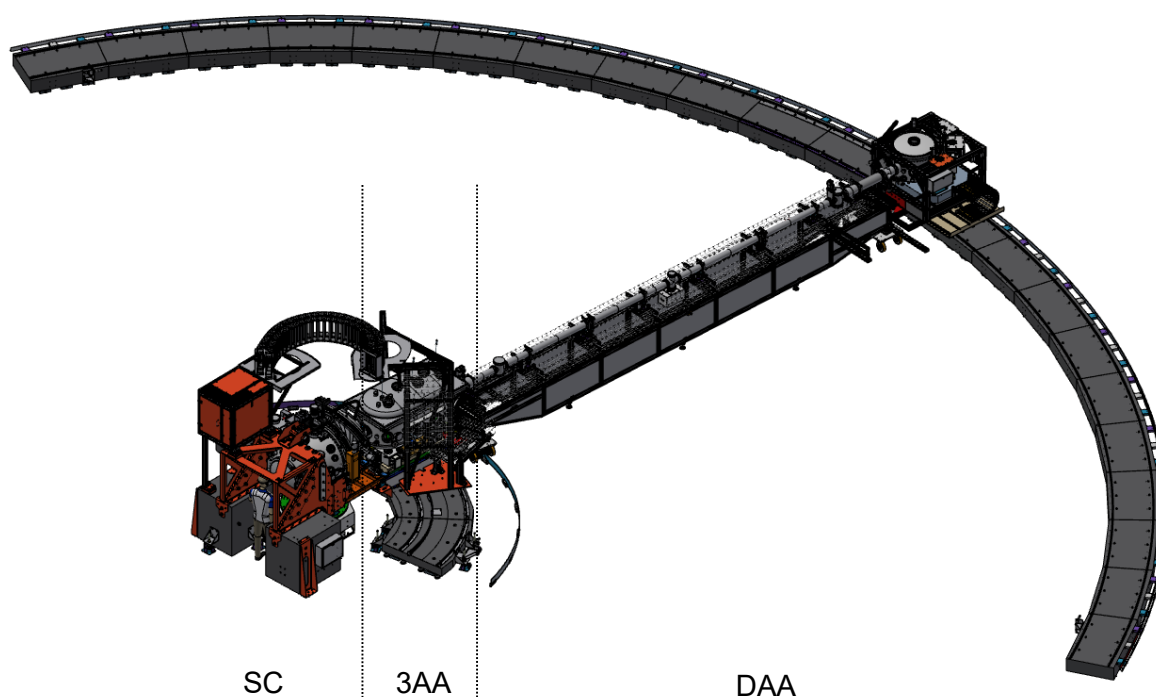


Figure 4: Design overview of the spectrometer arm system

5.1 Provisions for EPS

For an overview of the EPS relevant functional elements please refer to the wiring schemes listed in Table 1 to Table 4. Their position is indicated in the corresponding assembly drawings; see Table 1 to Table 4.

Table 1: Provision for EPS at the sample chamber

SC - sample chamber			
Element	Interface	Wiring scheme	Assembly drawing
Kill switches	Interface sockets at corresponding axes	449 - 7M - 500a	---
Collision switches	Interface sockets at corresponding axes	449 - 7M - 500a	---
Limit switches	Patch panel	449 - 7M - 500a 449 - 7M - 500b	N4432C0045 N4431D0241

Table 2: Provision for EPS at the M5 mirror system

M5 mirror system			
Element	Interface	Wiring scheme	Assembly drawing
Kill switches	Interface sockets at corresponding axes	443 - 7M - 500a	N4432C0002
Collision switches	Interface sockets at corresponding axes	443 - 7M - 500b	N4433C0000
Limit switches	Patch panel	443 - 7M - 500a 443 - 7M - 500b 455 - 7M - 510	N4432C0002 N4433C0000 N4433C0001 N4433C0007
Thermocouples	PLC Box 1	443 - 7M - 300	N4431C0007 N4433C0000 see Table 15

Table 3: Provision for EPS at the 3AA - 3 m arm and optics tank assembly

3AA - 3 m arm and optics tank assembly			
Element	Interface	Wiring scheme	Assembly drawing
Kill switches	Patch panel	444 - 7M - 500a 443 - 7M - 500c	N4440C0000 N4442C0000 N4442C0009
Collision switches	Patch panel	444 - 7M - 500a 443 - 7M - 500c	N4440C0000 N4443C0039 N4443C0003
Limit switches	Patch panel	444 - 7M - 500a 444 - 7M - 500b 443 - 7M - 500c	N4440C0000 N4441C0006 N4442C0009 N4442C0000 N4443C0012 N4443C0023
Thermocouples	PLC Box 2	444 - 7M - 300	see Table 22

Table 4: Provision for EPS at the DAA - detector arm and stand assembly

DAA - detector arm and stand assembly			
Element	Interface	Wiring scheme	Assembly drawing
Kill switches	Patch panel	445 - 7M - 500a	N4453C0000
Collision switches	Patch panel	445 - 7M - 500a	01-0744-00
Limit switches	Patch panel	445 - 7M - 500a	N4453C0000 01-0744-00

6 SC - sample chamber

6.1 Safety information

6.1.1 Used symbols

The following symbols found in the present document are used to identify hazards. An additional purpose of them is to draw users' attention to safety instructions or other useful information.

	Information on preventing any kind of physical injury.
	Information on preventing extensive equipment and environmental damage.
	Information on correct handling or use. Disregard can lead to malfunctions or equipment damage.
	General warning
	Pinch point hazard warning
	High voltage hazard warning
	Hot surface hazard warning
	Watch your head hazard warning
	Pinch point/entanglement hazard warning
	Gloves required
	Eye protection required
	Hard hat required

6.1.2 Safety instructions

General

In normal operation, there is no danger when working with the sample chamber system. The operator of the system should however be aware of potential dangers that can occur by improper handling of the system and that can cause personnel injury or damage to the equipment.

When the system is operated without fulfilling the recommendations described in this document and the legal health and safety regulation, safety of personnel and equipment can be affected. Potential dangers and measures to avoid them are described in this section.

Electrical safety

In normal operation no hazards can occur associated with electrical voltage and current used by the equipment. Following accidents may happen if the electrical installation of the mirror unit is not properly operated: electric shock and burns, fire and electric arcing.

⚠ DANGER

Electrical voltage



1. Voltage and high voltage that may cause injury or death are used in the setup. Service or maintaining work on the electrical installation must only be performed by qualified trained personnel for this type of work.
2. The statutory regulations regarding electrical installations must be respected.
3. The entire electrical equipment supplied as part of the system is provided with a protective ground. Grounding cables must not be removed or manipulated. It is very important that the vacuum system is properly grounded at all times.
4. Fuses may not be bridged, electrical connections must not be separated or otherwise manipulated, protective covers and other electrical insulating material may not be removed or modified.
5. Do not attempt to administer first aid to someone who has suffered electric shock until the source of the shock has been isolated.

Safe handling of gases

In normal operation no dangers through use of gases may appear as long as the recommended precautions are followed. In case of gas handling contrary to the instructions, unconsciousness and associated injuries may occur.

⚠ DANGER

Gases



1. If pure nitrogen is used for venting the vacuum system one has to be aware that its inhalation can induce an oxygen deficiency after only few breaths, which may lead to unconsciousness.
2. Areas in which the oxygen concentration is less than 18% must be abandoned immediately.
3. For venting the vacuum system it is recommended to use dry air with normal oxygen concentration instead of pure nitrogen.

Safe handling of mechanical components

In normal operation only few mechanical works on the sample chamber unit is required. For service and maintenance extensive mechanical work may be necessary. Only qualified personnel should attempt to dismantle, modify or repair equipment. The following hazards may arise when dealing with mechanical components: injury when transporting heavy loads (mirror unit, chamber lid), danger of crushing by moving parts (motors at the motion axes), injury by protruding parts at the mirror unit, severe damage of internal mechanics and optical elements (when changing motors at hexapod motion axes).

⚠ DANGER

Heavy loads



1. Only qualified personnel should lift and carry heavy loads.
2. Use a hard hat when working with heavy loads.
3. Lifting eyes are available at the individual components of the system and they must be used for transportation.
4. Parts of the equipment that have to be lifted may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

⚠ DANGER

Moving parts



1. At the mirror unit there are a number of moving parts, which are powered by electrical motors. Make sure that your way of working does not include any safety risk. Do not leave long hair untied, do not wear loose clothing or jewelry as this may involve the danger of becoming caught in the moving mechanisms.
2. Do not remove or modify covers of the moving parts.

⚠ CAUTION

Components in UHV



1. In-UHV components can only be touched with powder/lint-free gloves.
2. Tools use for work on UHV components must be clean and free of grease.
3. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

⚠ CAUTION

Optical elements



1. Use protective gloves and clothing when working with optical elements.
2. Be careful not to drop anything on/touch the optical surface.
3. A suitable cover for the optical surface is recommended.

Safe handling of hot surfaces

During bakeout procedure the system or parts of it become hot ($>100^{\circ}\text{C}$) and can cause burns when touched.

** DANGER Hot surfaces**

1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system.
2. Wear protective gloves when touching the system while bakeout and cooling down of the system.

Safe handling of vacuum systems

By appropriate use of and compliance with the safety information during normal operation, no dangers occur from the vacuum system of the mirror unit. The following hazards may arise when dealing with vacuum systems: injury from “flying” pieces caused by exploding or imploding parts of the vacuum system; if the vacuum chamber is over pressurized, viewports may shatter and cause person injury. Use viewport covers and wear eye protection glasses.

** DANGER Vacuum system**

1. The walls of the vacuum system can withstand a pressure difference from outside to inside of at least 1 bar, from inside to outside only a pressure difference of about 0.1 bar. An increase in internal pressure of 1.1 bar can lead to the damage of the view ports and must therefore be avoided. Never exceed an overpressure of 0.1 bar inside the vacuum system.
2. The viewports destruction of a chamber under vacuum leads to implosion. There is a risk of injury from flying glass shards. Use viewport and eye protection.
3. Areas in which the oxygen concentration is less than 18% must be immediately abandoned.

6.1.3 Modifications and service

Safe operation, reliability and performance of the equipment may be impaired if assembly operations, extensions, re-adjustments, modifications or repairs are not carried out in accordance with the instructions provided in this document and with any other instructions issued by the manufacturer. If you wish to modify the equipment, please contact Bestec GmbH for future advice.

6.2 Design overview of the sample chamber

Figure 5 illustrates the design overview of the sample chamber and its support structure.

- | | |
|------------------------------------|---|
| 1 Movable support structure | 5 Patch panel |
| 2 Drag chain | 6 Beam entrance flange |
| 3 Sample chamber | 7 Exit flange (6x) with manual gate valve |
| 4 Sample chamber support structure | |

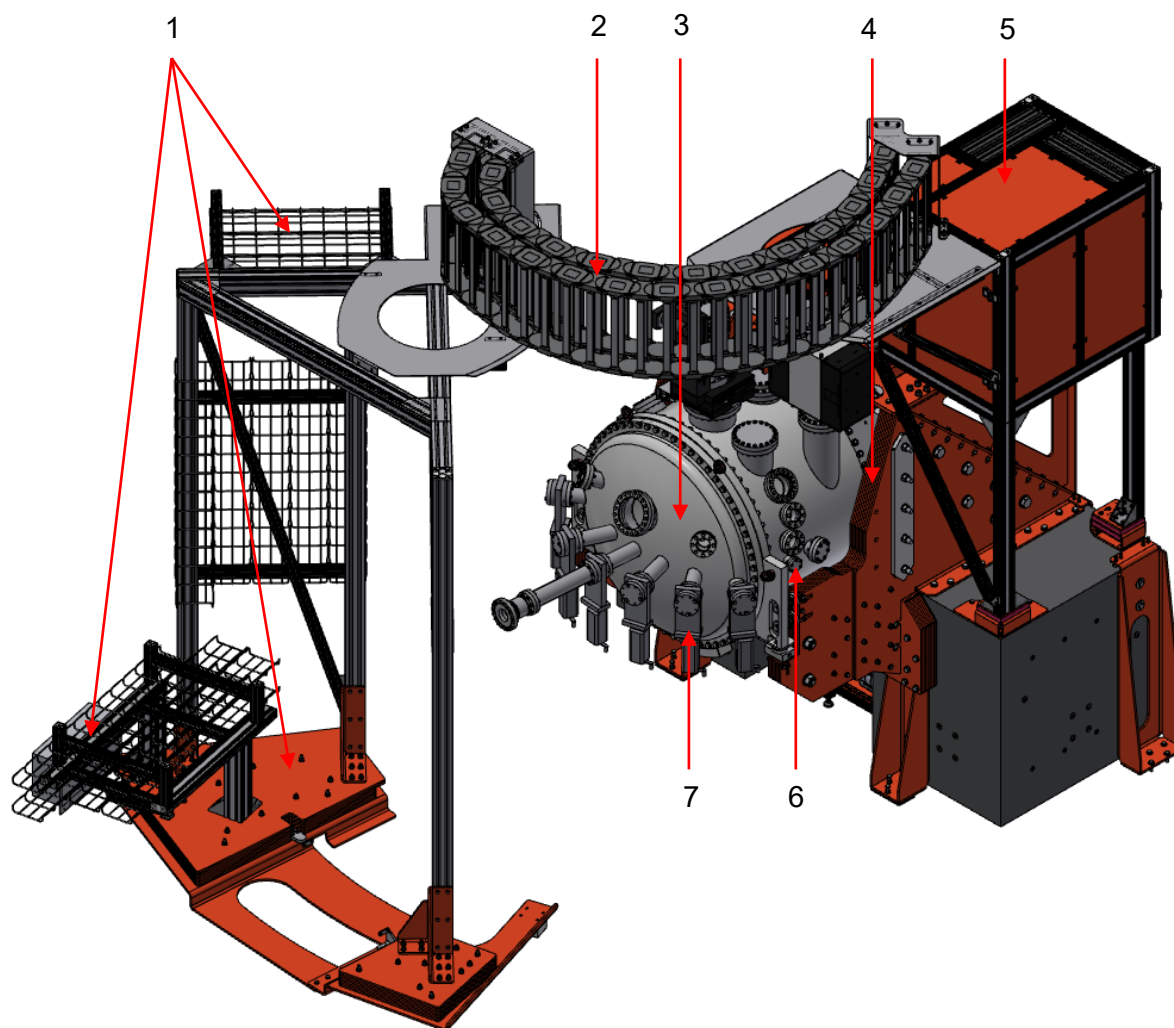


Figure 5: Design overview of the sample chamber

- | | |
|--------------|--------------------|
| 1 TC-PLC box | 3 Granite blocks |
| 2 PLC box | 4 Maintenance door |

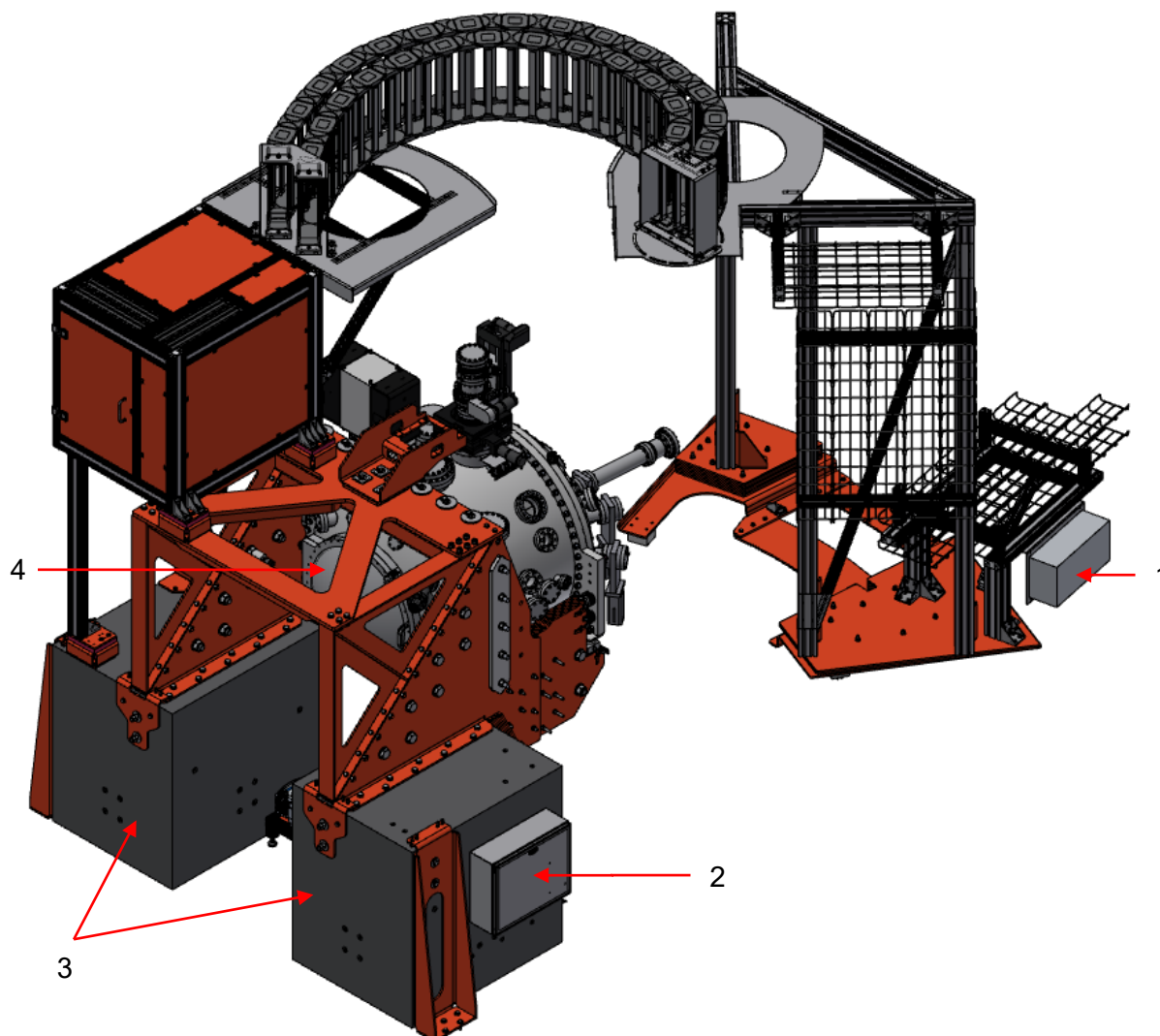


Figure 6: Design overview of the sample chamber

At the back side of the sample chamber a maintenance flange with a differentially pumped door is available (#11 in Figure 9). Two Viton gaskets are used for sealing of the maintenance door.

The rotary feedthrough (#3 in Figure 7) at the cryostat manipulator is also differentially pumped.

- | | |
|-------------------|--------------------|
| 1 Star grip screw | 3 Maintenance door |
| 2 Sample chamber | 4 Star grip screw |

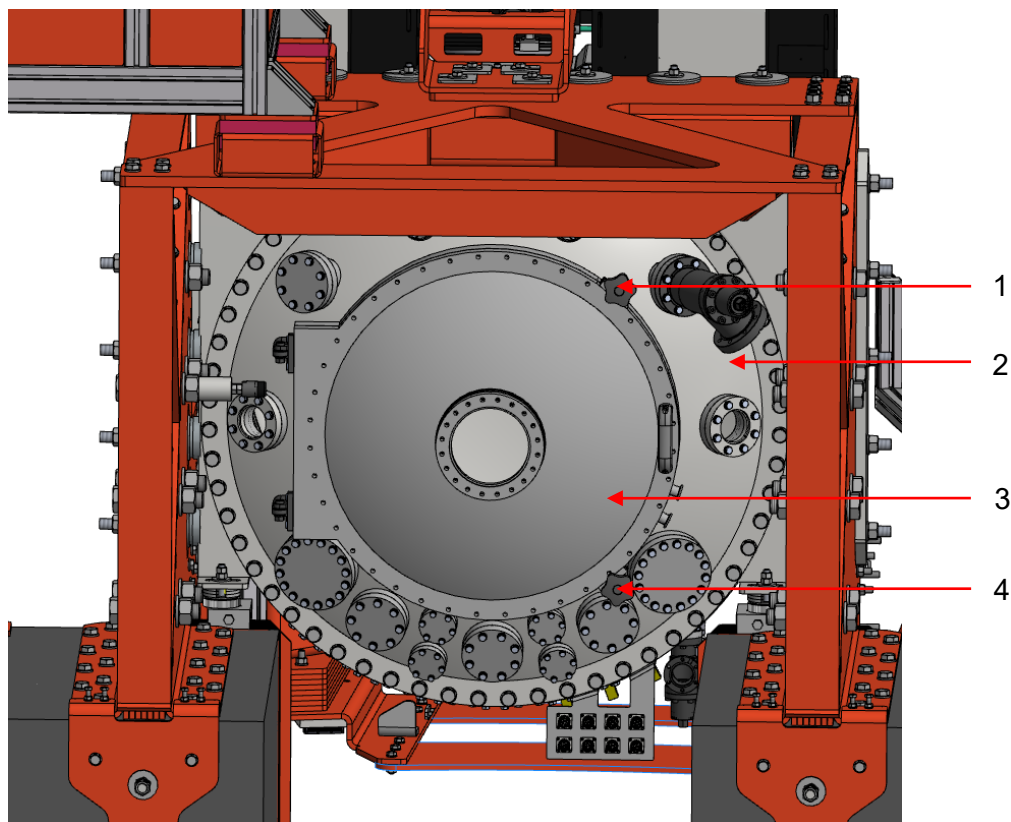


Figure 7: Maintenance door at the sample chamber

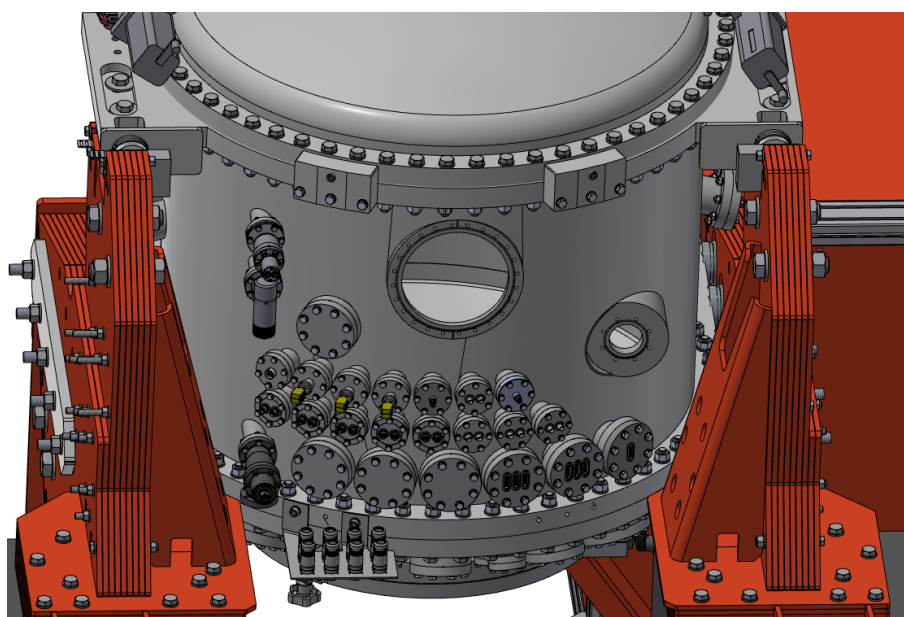


Figure 8: Bottom view of the sample chamber

For a complete assignment of all feedthroughs shown in Figure 8 please refer to the provided assembly drawing N4431C0007.

- | | |
|---|--|
| 1 Manipulator vertical translation actuator (MVT) | 7 Manipulator horizontal translation actuator (MTH2) |
| 2 Encoder scale (MVT) | 8 Patch panel (encoders and motors plugs) |
| 3 Encoder readhead (MVT) | 9 Manipulator horizontal translation actuator (MTH1) |
| 4 Manipulator rotation actuator (MR) | 10 Cross table (MTH1, MTH2) |
| 5 Encoder scale (MTH1) | 11 Rotary feedthrough |
| 6 Encoder readhead (MTH1) | 12 Port aligner |

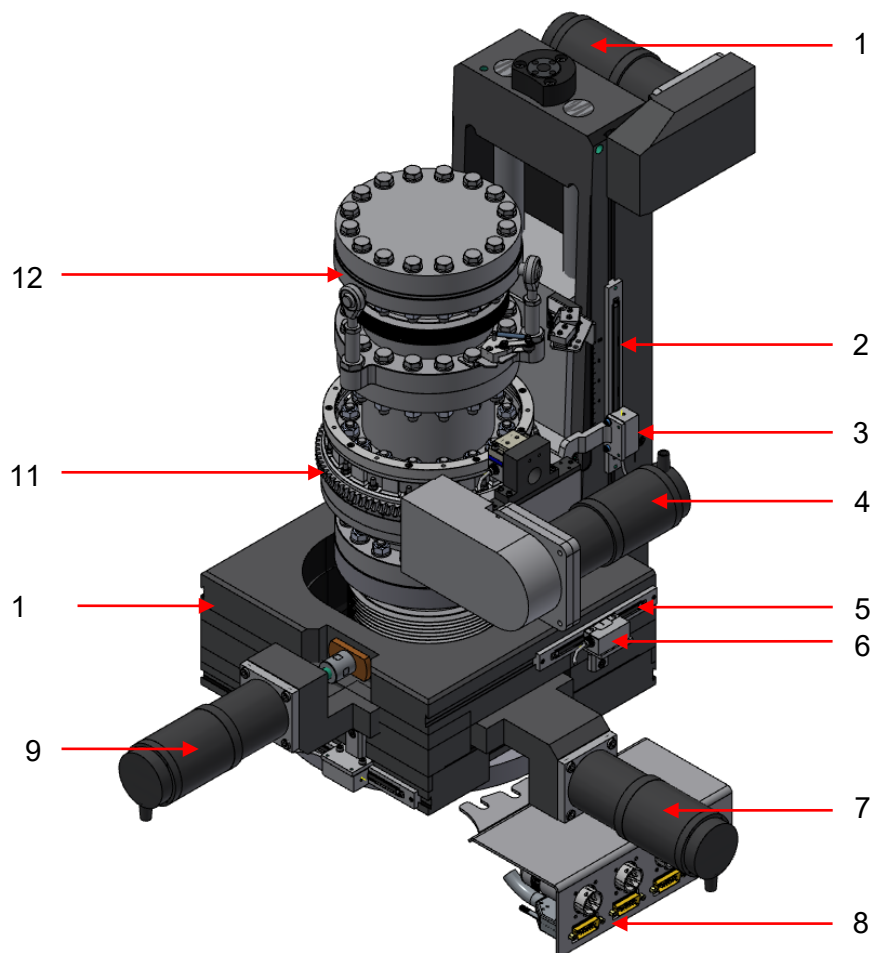


Figure 9: Cryostat manipulator unit

6.3 Maintenance

6.3.1 Bakeout procedure

⚠ DANGER Hot surfaces: system bakeout



1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system.
2. Wear protective gloves when touching the system while bakeout and cooling down of the system

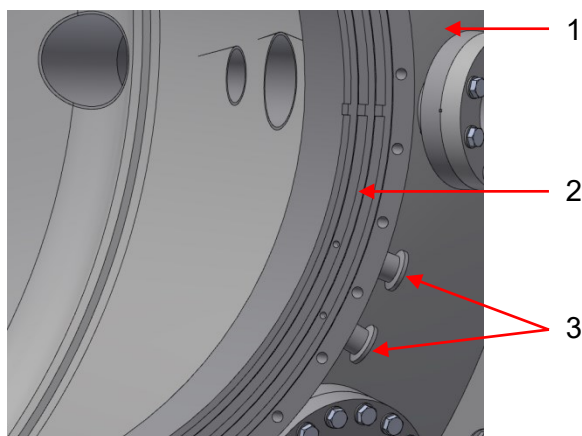
⚠ CAUTION Components in UHV



1. In-UHV components can only be touched with powder/lint-free gloves.
2. Tools use for work on UHV components must be clean and free of grease.
3. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

Two Viton gaskets are used for sealing the maintenance door (#2 in Figure 7) at the sample chamber. A successful bakeout procedure requires following the next steps:

- A. Clean the Viton gaskets with a suitable “lint-free” cloth and alcohol.
- B. The main flange and the bottom side of the chamber lid should be cleaned of possible particle remnants.
- C. The two Viton gaskets should be gently pressed into the corresponding grooves at the flange.



- 1 Sample chamber
- 2 Grooves for Viton gaskets
- 3 Flanges for the differentially pumping stage

Figure 10: Differentially pumped flange at the sample chamber

- D. Carefully close the door and tighten the two star grip screws (#1, 4 in Figure 7).
- E. Connect the scroll pump to the differentially pumping stages at the maintenance door (#3 in Figure 10) and the manipulator rotary feedthrough (#2 in Figure 11).
- F. Remove all cables connected to the chamber.
- G. Connect a RGA to a suitable place.
- H. Wrap edge welded bellows and viewports with thin Aluminum foil.
- I. Wrap the chamber in one or two layers of thicker Aluminum foil.
- J. Start the scroll pump for the two differentially pumped stages.
- K. Start pumping the chamber (scroll pump/turbo pump). The bakeout procedure can be started when the pressure is lower than 1×10^{-6} mbar.
- L. Ramp the system temperature up in stages to ensure the heating process is under control. When a certain temperature level is reached, leave the system at that heating level until the entire system is stable and at constant temperature before continuing.
- M. When the system reaches the target bakeout temperature (max. 120°C), keep the system at that temperature for a period of time of at least 48 hours, or until the total pressure falls below 2×10^{-7} mbar.
- N. Stop the heating of the ion getter pump at least 6 hours before starting to ramp down the system temperature.
- O. Switch on the mass spectrometer.
- P. Ramp the system temperature down in stages to ensure the cooling process is under control. At each stage leave the system at that heating level until the entire system is stable and at a uniform temperature distribution before proceeding to the next stage.

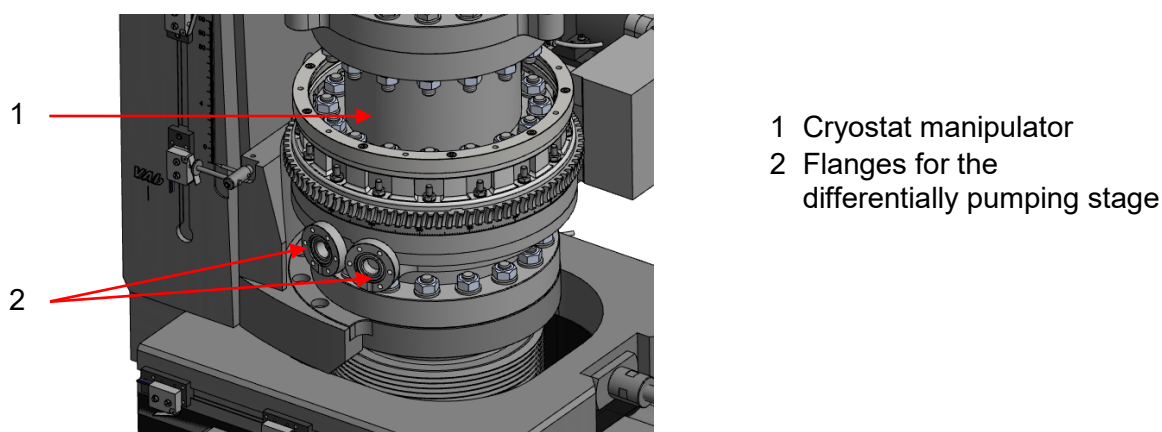


Figure 11: Differentially pumped rotary feedthrough at the cryostat manipulator

6.4 Appendix

6.4.1 Overview of technical specification

Table 5: Overview of technical specifications

General	
Viton sealed service door	CF600
Basis flange cryostat manipulator	CF200
Manipulator Horizontal Translation 1 (MHT1)	
Range	-25.15 mm to 25.04 mm
Offset (middle position)	38809547 counts (0 mm)
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Reverse limit switch	38306628 counts
Forward limit switch	39310292 counts
Manipulator Horizontal Translation 2 (MHT2)	
Range	- 25.17 mm to 25.77 mm
Offset (middle position)	36448322 counts (0 mm)
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880-W3B6k
Reverse limit switch	35944922 counts
Forward limit switch	36963852 counts

Manipulator Vertical Translation (MVT)	
Range	0 mm to 100.2 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Step resolution (full step)	0.2 μ m
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μ m/count
Limit switches	2 x Schaltbau S880-W3B6k
Reverse limit switch	30125156 counts
Forward limit switch	32128803 counts
Manipulator Rotation (MR)	
Range	363.97°
Motor	Phytron ZSS57.200.2,5-GPL52/50
Step resolution (full step)	0.0001° /step
Encoder	Renishaw Resolute Encoder scale: RESA30USA172B Readhead: RA32BAA172B15A
Encoder resolution	0.0838190 $\times 10^{-6}$ °/count
Limit switches	2 x Schaltbau S880-W3B6k
Reverse limit switch	2319726089 counts
Forward limit switch	2367083008 counts

6.4.2 List of supplied wiring schemes

Table 9 contains a list of the provided wiring schemes. For detailed schemes, please refer to Customer Documentation folder.

Table 6: List of wiring schemes for the cryostat manipulator

Scheme number	Description
449 - 7M - 500b	Motion Axes - Overview
449 - 7M - 560d	MHT1 - Motor + Limit switches
449 - 7M - 571a	MHT1 - Encoder
449 - 7M - 560e	MHT2 - Motor + Limit switches
449 - 7M - 571b	MHT2 - Encoder
449 - 7M - 560f	MVT - Motor + Limit switches
449 - 7M - 571c	MVT - Encoder
449 - 7M - 560g	MR - Motor + Limit switches
449 - 7M - 571d	MR - Encoder

6.4.3 List of supplied assembly drawings

Table 10 contains a list of the provided assembly drawings. For detailed drawings, please refer to Customer Documentation folder.

Table 7: List of assembly drawings for the sample chamber

Drawing number	Description
N4430C0001	Sample chamber unit
N4431C0000	Sample chamber
N4432C0045	Manipulator unit
N4431D0241	Plug holder
N4431C0002	Fixed exit flange
N4431C0007	Sample chamber body
N4431C0008	Maintenance flange
N4434C0005	Frame
N4434D0077	Support frame
N4434C0015	Support frame
N4437D0079	Dampener assembly
N4434C0027	Cryostat holder
N4437C0002	Patch panel
N4437C0013	Holder cable chain - mobile end
N4437D0089	Holder cable chain
N4437C0000	Cable management
01-3014-00	Mobile support platform
N4437C0004	Support platform
N4437C0008	Large counterweights
N4437C0010	Small counterweights

6.4.4 List of recommended spare parts

If replacement of system components is necessary, please contact Bestec GmbH. Please note that self-repair of system components without Bestec GmbH consultation can lead to loss of warranty claim.

Table 8: List of recommended spare parts

Spare part	Manufacturer	Part number
Encoder readhead	Renishaw	RL32BAT050B30A
		RA32BAA172B15A
Limit switch	Schaltbau	S880

6.4.5 List of supplied subcontractors documents

Table 13 contains a list of provided subcontractors' documents. For detailed manuals please refer to the provided Customer Documentation folder.

Table 9: List of provided subcontractors documents

Part	Manufacturer	File name
Heating cable	BriskHeat	Heating cable.pdf
Stepper motor	Phytron	ZSS.pdf
Limit switch	Schaltbau	Snap action switches.pdf
Encoder	Renishaw	Resolute L-9517-9448-04-G Resolute L-9517-9399-03-A Resolute M-9553-9735-02-B Resolute M-9553-9433-06-C

7 Triple rotation flange (TRF)

7.1 Safety information

7.1.1 Used symbols

The following symbols found in the present document are used to identify hazards. An additional purpose of them is to draw users' attention to safety instructions or other useful information.

	Information on preventing any kind of physical injury.
	Information on preventing extensive equipment and environmental damage.
	Information on correct handling or use. Disregard can lead to malfunctions or equipment damage.
	General warning
	Pinch point hazard warning
	High voltage hazard warning
	Hot surface hazard warning
	Watch your head hazard warning
	Pinch point/entanglement hazard warning
	Gloves required
	Eye protection required
	Hard hat required

7.1.2 Safety instructions

General

In normal operation, there is no danger when working with the triple rotation flange system. The operator of the system should however be aware of potential dangers that can occur by improper handling of the system and that can cause personnel injury or damage to the equipment.

When the system is operated without fulfilling the recommendations described in this document and the legal health and safety regulation, safety of personnel and equipment can be affected. Potential dangers and measures to avoid them are described in this section.

Electrical safety

In normal operation no hazards can occur associated with electrical voltage and current used by the equipment. Following accidents may happen if the electrical installation of the mirror unit is not properly operated: electric shock and burns, fire and electric arcing.

⚠ DANGER

Electrical voltage



1. Voltage and high voltage that may cause injury or death are used in the setup. Service or maintaining work on the electrical installation must only be performed by qualified trained personnel for this type of work.
2. The statutory regulations regarding electrical installations must be respected.
3. The entire electrical equipment supplied as part of the system is provided with a protective ground. Grounding cables must not be removed or manipulated. It is very important that the vacuum system is properly grounded at all times.
4. Fuses may not be bridged, electrical connections must not be separated or otherwise manipulated, protective covers and other electrical insulating material may not be removed or modified.
5. Do not attempt to administer first aid to someone who has suffered electric shock until the source of the shock has been isolated.

Safe handling of gases

In normal operation no dangers through use of gases may appear as long as the recommended precautions are followed. In case of gas handling contrary to the instructions, unconsciousness and associated injuries may occur.

⚠ DANGER

Gases



1. If pure nitrogen is used for venting the vacuum system one has to be aware that its inhalation can induce an oxygen deficiency after only few breaths, which may lead to unconsciousness.
2. Areas in which the oxygen concentration is less than 18% must be abandoned immediately.
3. For venting the vacuum system it is recommended to use dry air with normal oxygen concentration instead of pure nitrogen.

Safe handling of mechanical components

In normal operation only few mechanical works on the sample chamber unit is required. For service and maintenance extensive mechanical work may be necessary. Only qualified personnel should attempt to dismantle, modify or repair equipment. The following hazards may arise when dealing with mechanical components: injury when transporting heavy loads (mirror unit, chamber lid), danger of crushing by moving parts (motors at the motion axes), injury by protruding parts at the mirror unit, severe damage of internal mechanics and optical elements (when changing motors at hexapod motion axes).

⚠ DANGER

Heavy loads



1. Only qualified personnel should lift and carry heavy loads.
2. Use a hard hat when working with heavy loads.
3. Lifting eyes are available at the individual components of the system and they must be used for transportation.
4. Parts of the equipment that have to be lifted may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

⚠ DANGER

Moving parts



1. At the triple rotation flange there are a number of moving parts, which are powered by electrical motors. Make sure that your way of working does not include any safety risk. Do not leave long hair untied, do not wear loose clothing or jewelry as this may involve the danger of becoming caught in the moving mechanisms.
2. Do not remove or modify covers of the moving parts.

⚠ CAUTION

Components in UHV



1. In-UHV components can only be touched with powder/lint-free gloves.
2. Tools use for work on UHV components must be clean and free of grease.
3. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

⚠ CAUTION

Optical elements



1. Use protective gloves and clothing when working with optical elements.
2. Be careful not to drop anything on/touch the optical surface.
3. A suitable cover for the optical surface is recommended.

Safe handling of hot surfaces

During bakeout procedure the system or parts of it become hot ($>100^{\circ}\text{C}$) and can cause burns when touched.

** DANGER Hot surfaces**

1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system.
2. Wear protective gloves when touching the system while bakeout and cooling down of the system.

Safe handling of vacuum systems

By appropriate use of and compliance with the safety information during normal operation, no dangers occur from the vacuum system of the TRF. The following hazards may arise when dealing with vacuum systems: injury from “flying” pieces caused by exploding or imploding parts of the vacuum system; if the vacuum chamber is over pressurized, viewports may shatter and cause person injury. Use viewport covers and wear eye protection glasses.

** DANGER Vacuum system**

1. The walls of the vacuum system can withstand a pressure difference from outside to inside of at least 1 bar, from inside to outside only a pressure difference of about 0.1 bar. An increase in internal pressure of 1.1 bar can lead to the damage of the view ports and must therefore be avoided. Never exceed an overpressure of 0.1 bar inside the vacuum system.
2. The viewports destruction of a chamber under vacuum leads to implosion. There is a risk of injury from flying glass shards. Use viewport and eye protection.
3. Areas in which the oxygen concentration is less than 18% must be immediately abandoned.

7.1.3 Modifications and service

Safe operation, reliability and performance of the equipment may be impaired if assembly operations, extensions, re-adjustments, modifications or repairs are not carried out in accordance with the instructions provided in this document and with any other instructions issued by the manufacturer. If you wish to modify the equipment, please contact Bestec GmbH for future advice.

7.2 Design overview of the triple rotation flange

Figure 12 and Figure 13 show the triple rotation flange which will be used as exit flange at the SIX spectrometer sample chamber. For detailed description and assignment of limit /kill switches, collision switch, motors and encoders please refer to the assembly drawing 03-1356-00.

- | | |
|-------------------------------|-------------------------------|
| 1 TRF1 rotation actuator | 5 TRF3 rotation actuator |
| 2 Energy chain | 6 Encoder readhead (ENC-TRF3) |
| 3 Energy chain | 7 TRF2 rotation actuator |
| 4 Heater terminal for bakeout | |

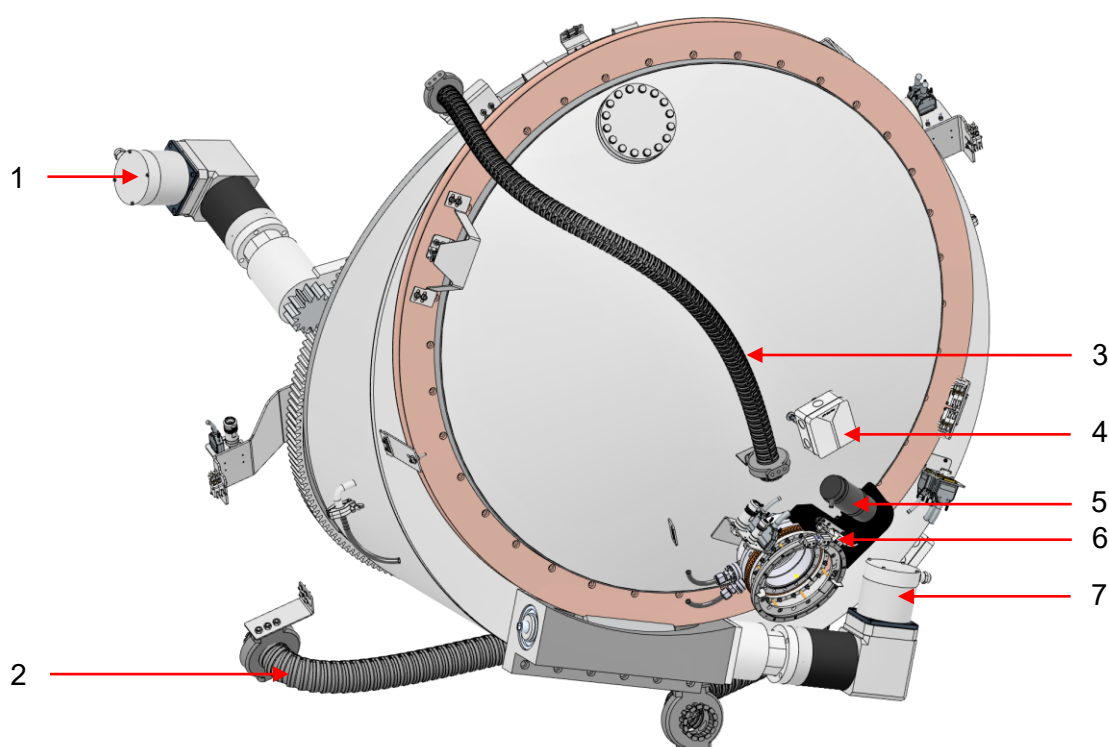


Figure 12: Triple rotation flange

- | | |
|--------------------------------|--------------------------------|
| 1 Encoder readhead (ENC2-TRF2) | 3 Encoder readhead (ENC1-TRF2) |
| 2 Encoder readhead (ENC1-TRF1) | 4 Encoder readhead (ENC1-TRF2) |

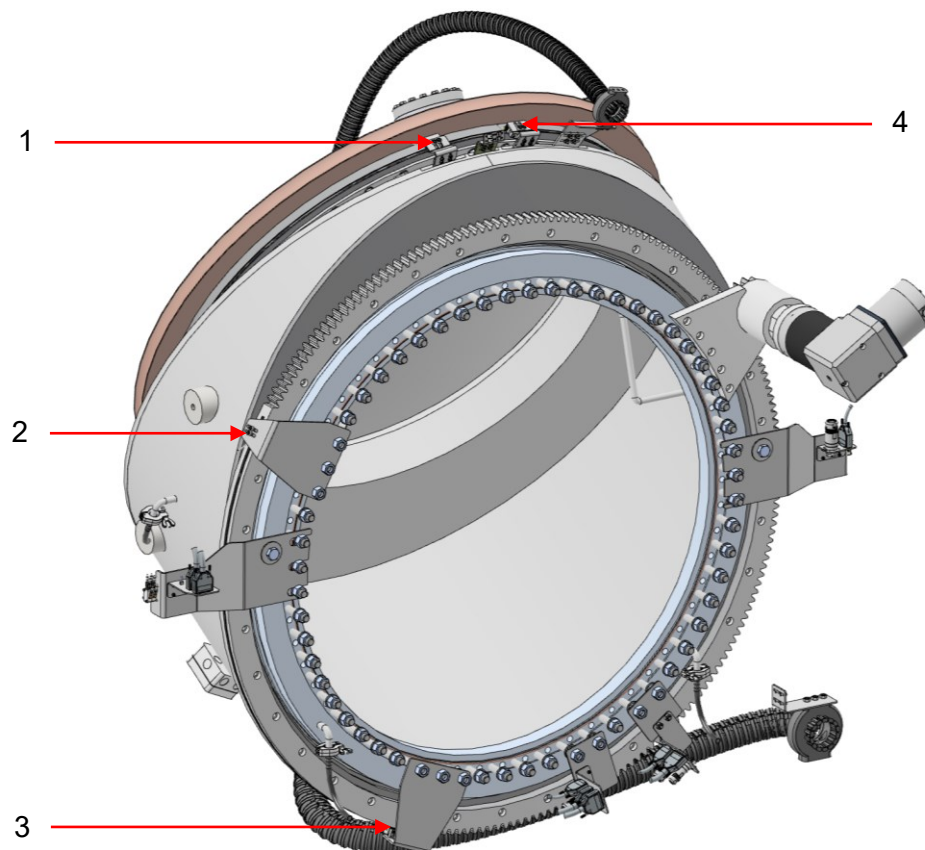


Figure 13: Triple rotation flange

7.3 Installation of the triple rotation flange

In this paragraph the installation of the TRF is described. Please carefully read the safety information (paragraph 7.1) and the instructions given here (paragraph 7.2 and 7.3) before starting the installation procedure.

1. Figure 14 shows the TRF in its transport crate. The flange is attached to a support frame, which is fixed to the crate by wooden beams.

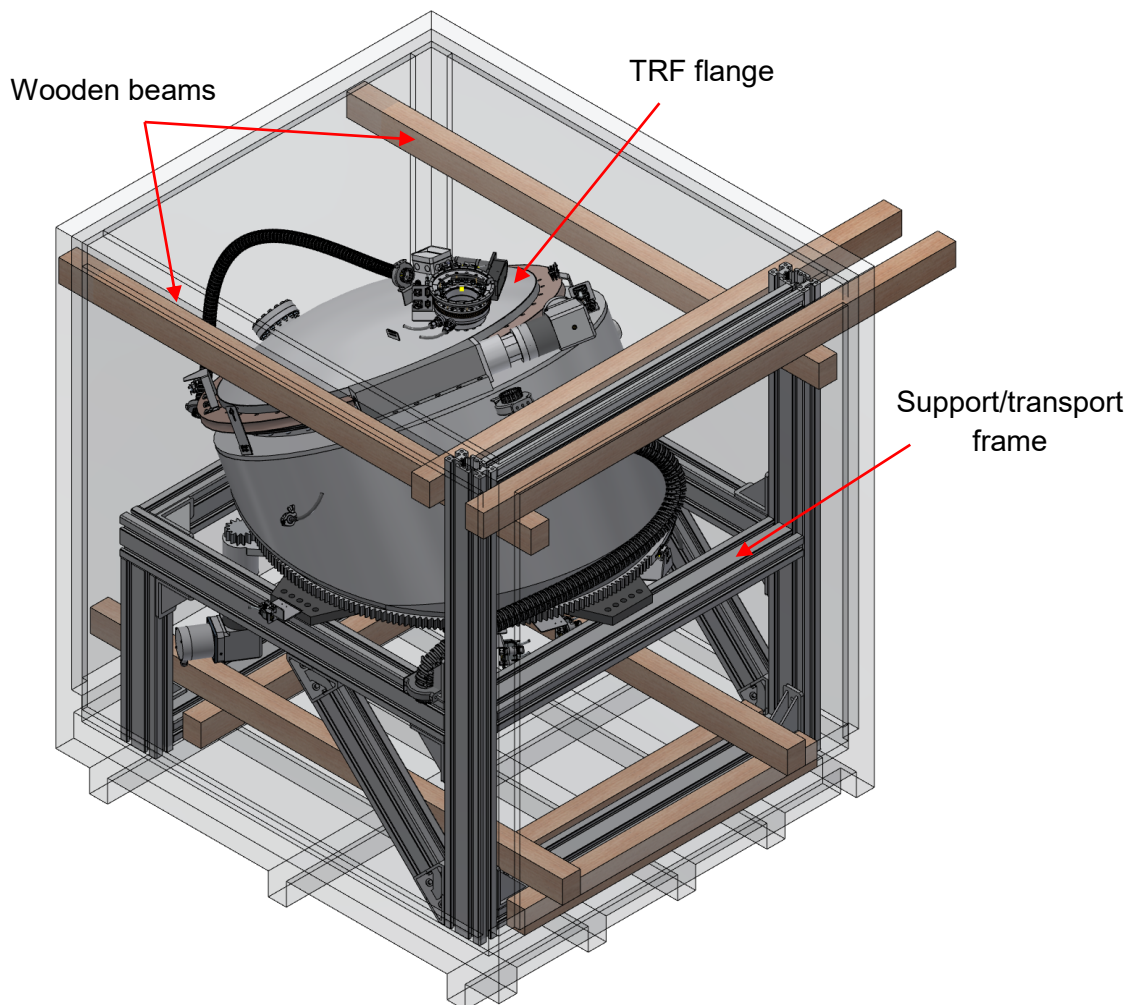


Figure 14: Triple rotation flange in transport crate.

Please note, that for its installation at the sample chamber, the TRF has to be turned by 90° as shown in Figure 15. This should be done while the TRF is attached to its transport frame using two cranes and heavy duty belts. The center of gravity for both positions is indicated in Figure 16. The weight of the TRF + support frame is ca. 1800 kg.

Please make sure that there is enough space around the system for the turning procedure.

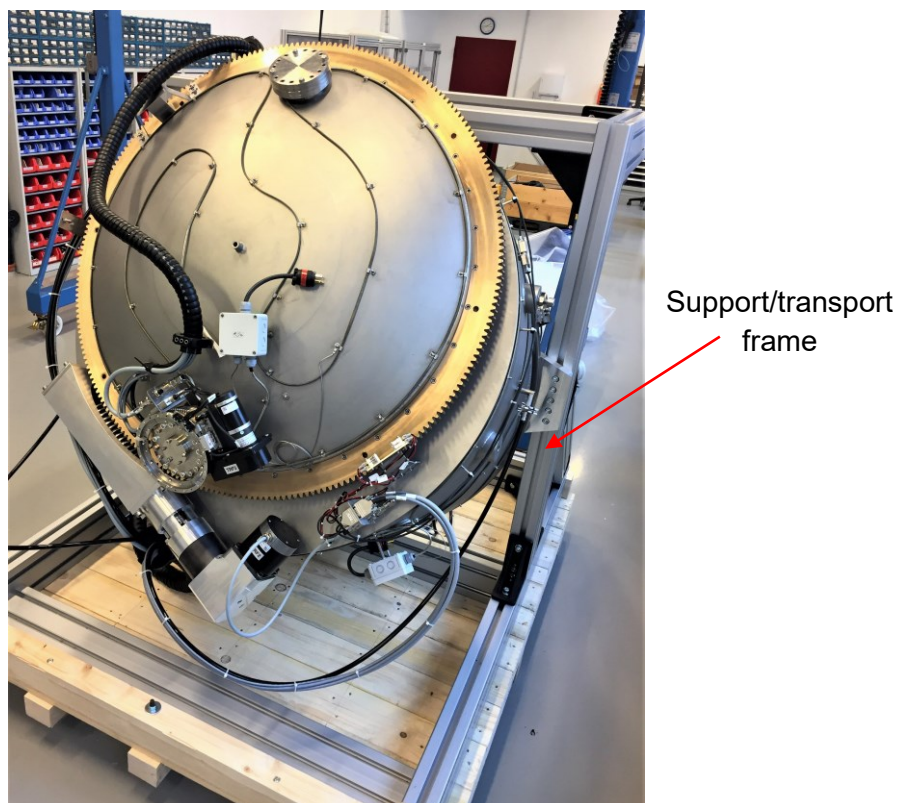


Figure 15: Triple rotation flange fixed on support frame.

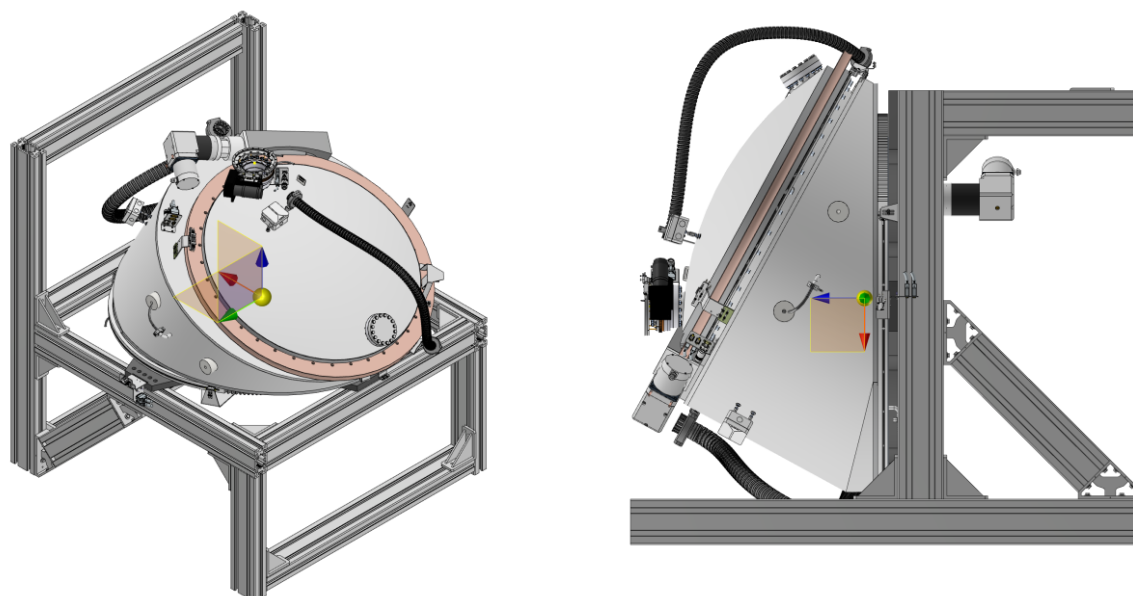


Figure 16: Center of gravity of the TRF + support frame assembly for the transportation position (left) and for installation position (right), respectively.

2. Before proceeding to the next step, the spectrometer arm has to be moved out of the way in the direction indicated by the blue arrow in Figure 17. This moving direction should be preferred because it allows better access to the crane.

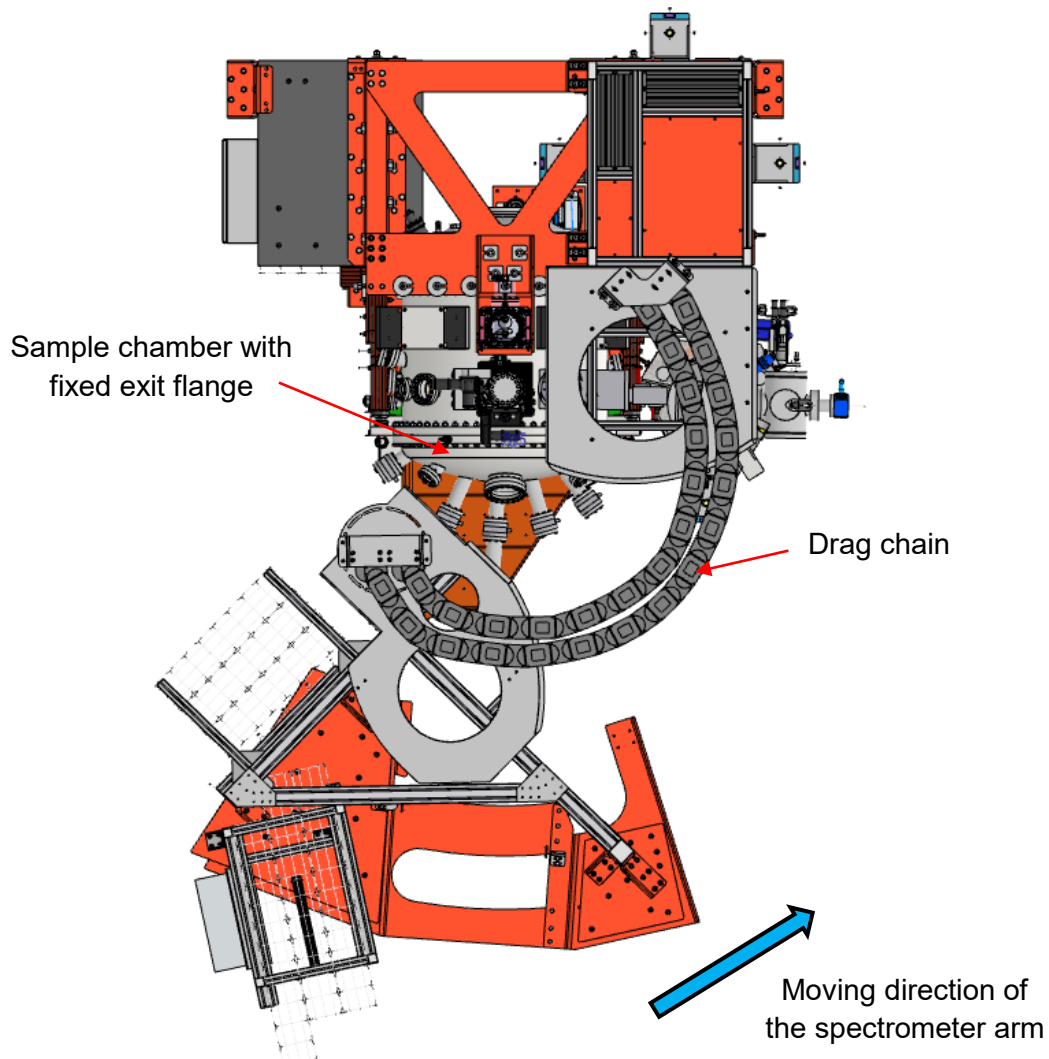


Figure 17: Top view of the sample chamber. The blue arrow indicates the recommended moving direction of the spectrometer arm.

3. The next step is to remove the exit flange of the sample chamber, see Figure 18. Please follow the steps listed below:
 - disconnect the sample chamber from the optics tank assembly 3AA
 - remove bellow and vacuum pipe from the exit flange
 - remove all six manual gate valves from the exit flange
 - use a heavy duty belt and two lifting eyes (M16) as shown in Figure 18 to secure the flange
 - loosen and remove the bolts fixing the exit flange at the sample chamber
 - carefully remove the lid and place it on a clean surface
 - remove the two support blocks of the exit flange, see Figure 19.

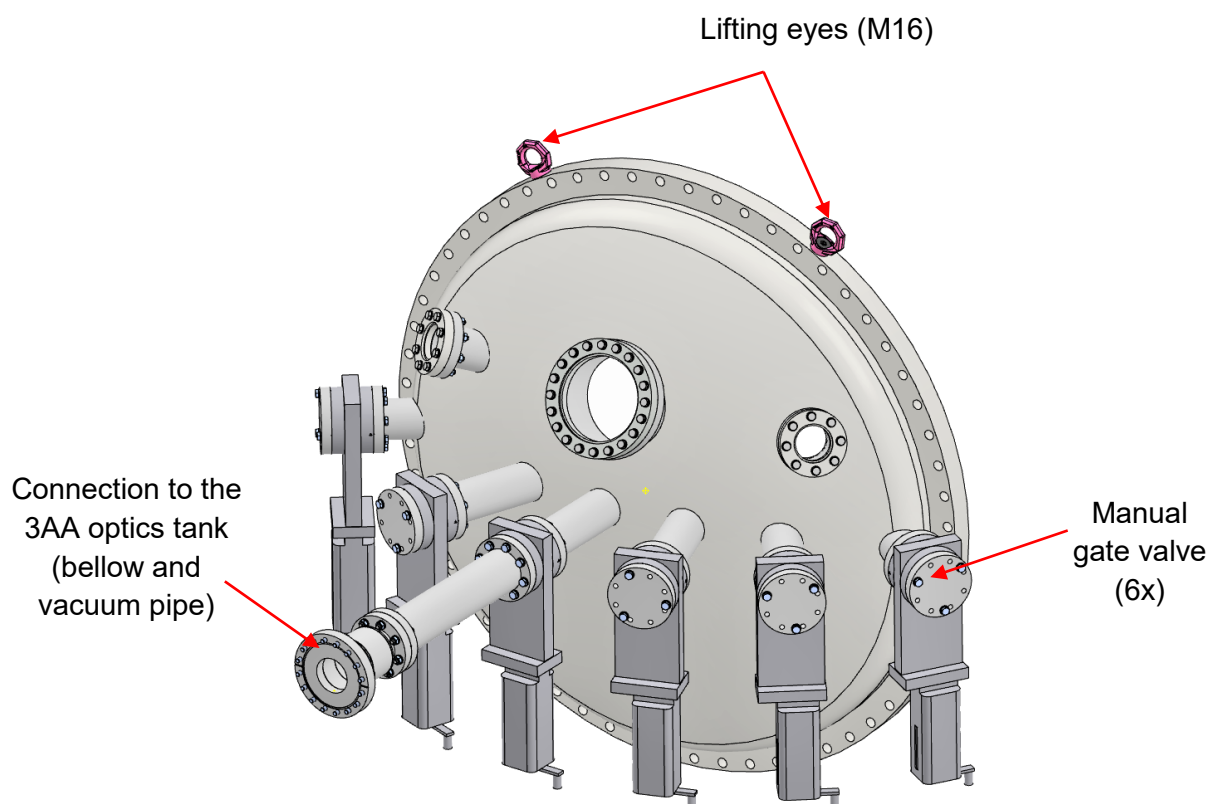


Figure 18: Exit flange of the sample chamber.

CAUTION

Before proceeding to the next step please survey the position of the internal mechanics. The next step will affect this position and realignment to the initial values will be required.

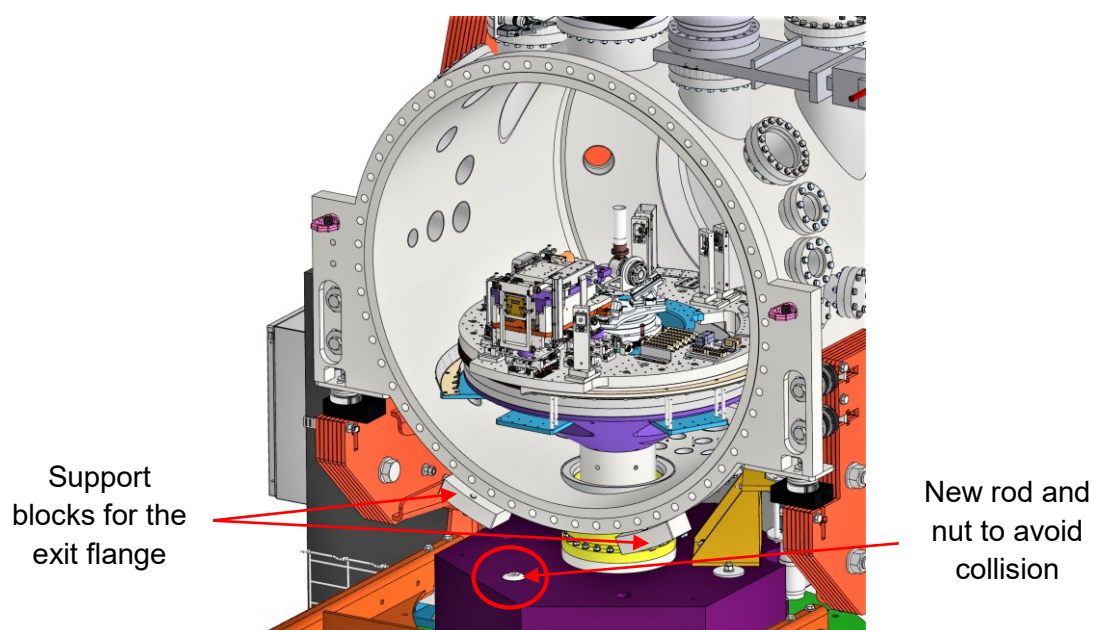


Figure 19: Internal mechanics of the sample chamber

4. In the current state of the system, the upper granite of the sample chamber is fixed to the lower granite in three different points by assemblies consisting of: a threaded rod (M12, 350 mm long), one large washer, one small washer and a nut as can be seen in the left hand side of Figure 20. To avoid collision, one of these assemblies - indicated in Figure 19 - must be replaced by a new assembly consisting of: a shorter threaded rod (M12, 325 mm long) and another nut (part number 0001-10263), see the right hand side of Figure 20. The required components were delivered together with the TRF.

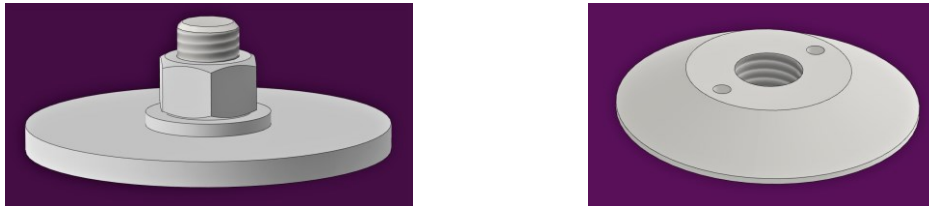


Figure 20: Initial washer and nut (left) must be replaced by special nut (right)

After mounting, the end of threaded rod should be flush with the upper side of the nut. This nut can be tightened with an adjustable caliper pin wrench flat with 3 mm pins. If not available, please purchase it for example here: <https://www.hoffmann-group.com/US/en/hus/Fastening-tools-for-screws/Torx-L-wrenches-hexagon-L-wrenches/Adjustable-caliper-pin-wrench-flat/p/628600-3>

5. Survey the position of the internal mechanics again at the end of the exchange procedure described above. Iteratively adjust the tightness of the three nuts at the upper granite until the initial position of the internal mechanics is achieved.
6. Connect the TRF1 motion axis (#1 in Figure 12) and move it by 90° so that the anchor points at the TRF are located at the top, see blue arrow in Figure 21.

CAUTION

Rotate the TRF only under vacuum.

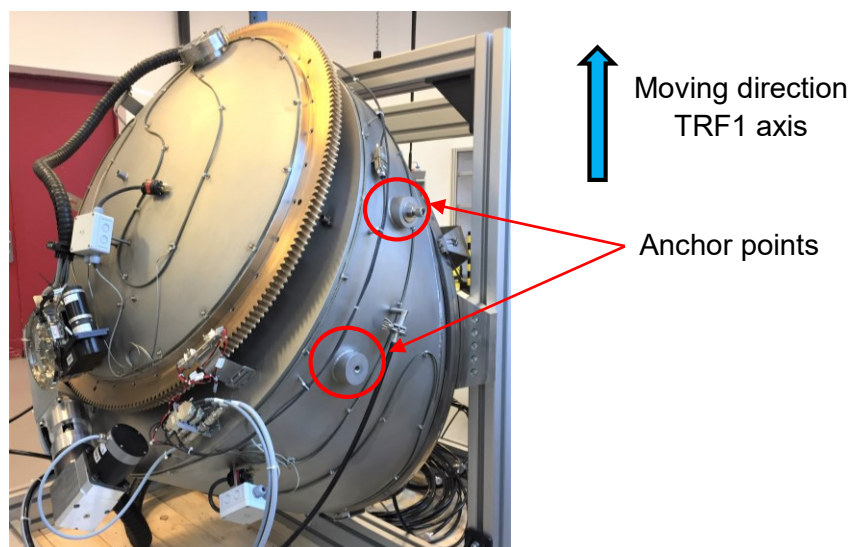


Figure 21: Anchor point at the TRF before moving the TRF1 axis by 90°

7. The angle bracket at the cable drag chain shown in Figure 22 must be replaced. The required part was delivered together with the TRF. Figure 22 shows the exchanged bracket.

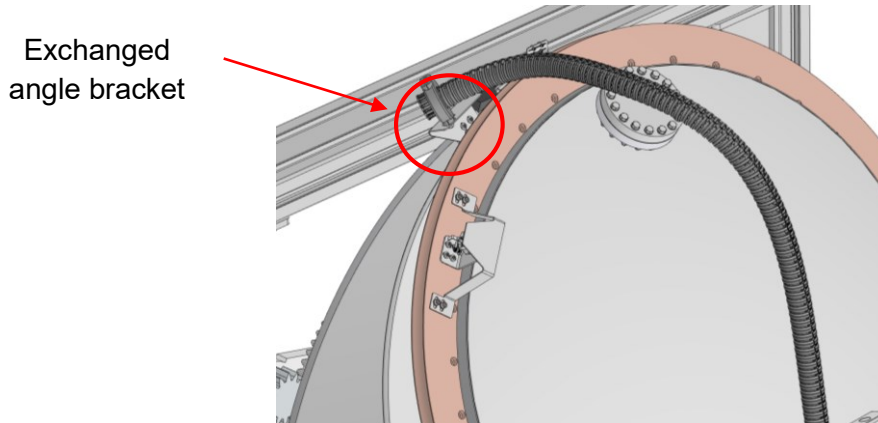


Figure 22: Angle bracket at the drag chain

8. Secure the TRF with the help of a crane using the anchor points shown in Figure 21. Carefully bring the lifting gear under light tension.
9. In the next step, the TRF flange has to be disconnected from its support frame. At first, the angle brackets holding encoder cables, limit switches, etc must be detached from the supporting frame. The screws/nuts holding them are indicated in Figure 23. Please loose (only!) these screws/nuts and temporary fix the brackets at the TRF, for example with duct tape.

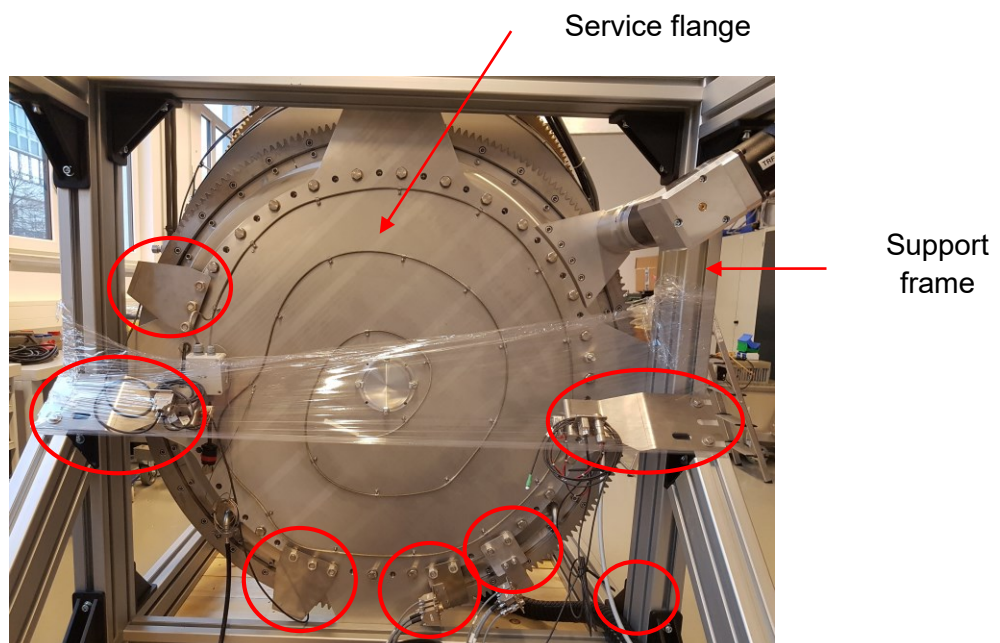


Figure 23: Screws to be removed.

Please refer to the assembly drawing 03-1356-00-X1 for the correct assembly of the parts at the TRF.

10. Please carefully remove all screws holding the TRF to the star-shaped service flange.

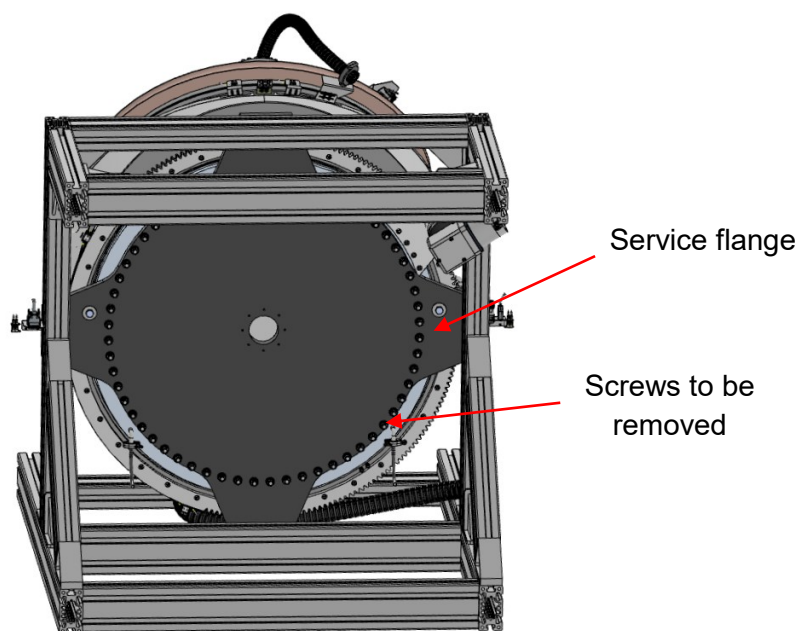


Figure 24: Screws to be removed at the star-shaped flange

At this point, the TRF is free and can be transported to the installation side.

CAUTION

Please make sure that the motor in Figure 25 does not get stuck at the frame

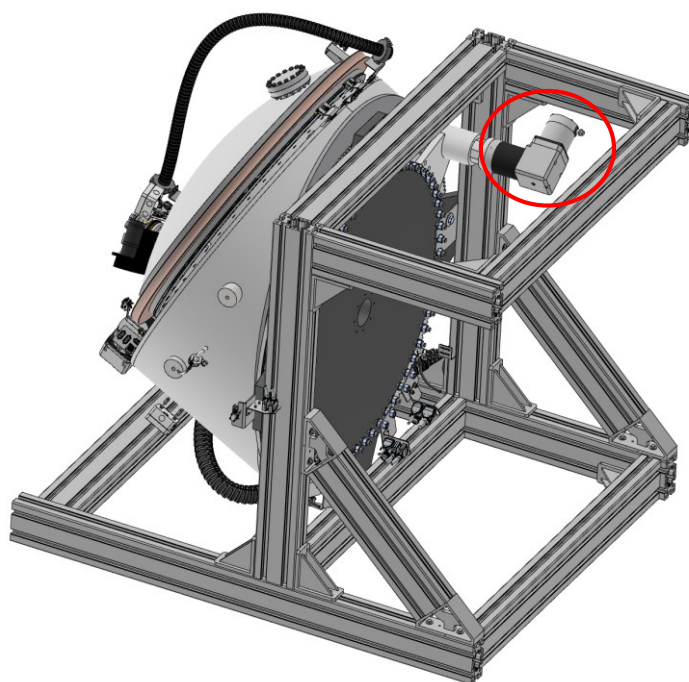


Figure 25: Motor at the TRF1 motion axis.

At this point the TRF is free and can be transported to the installation side.

11. Position the TRF in the front of the sample chamber with the help of the crane.
12. Insert the provided stud screws (DIN938, M16, 75 mm long) into the TRF. Nuts and washers will be used at a later point.
13. Place a new Copper gasket (DN1000) at the sample chamber.
14. Carefully bring the TRF close to the sample chamber and insert the stud screws from the TRF in the corresponding holes at the sample chamber. Please be aware that the holes at the TRF have an angle offset relative to the holes at the sample chamber. Therefore, the TRF has to be slightly rotated in order to match the holes - clockwise relative to the downstream direction.

The position of the differential pumping ports at the TRF will be slightly different after this rotation, see Figure 26.

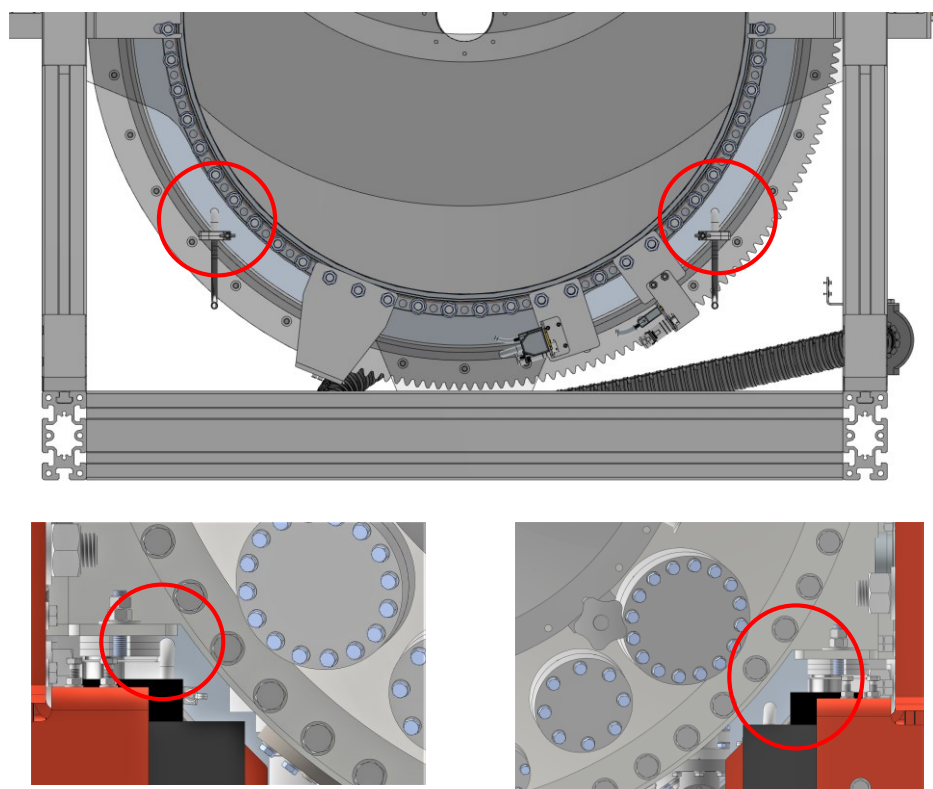


Figure 26: Position of the differential pumping ports at the TRF before (top) and after (bottom) its rotation to match the holes at the sample chamber.

15. Figure 27, Figure 28 and Figure 30 show the position of the angle brackets (see step #9 in the description above) holding encoder cables, limit switches, etc at the sample chamber. Note that for easier maintenance, some of the brackets are fixed below the nuts holding the TRF, some of them above.

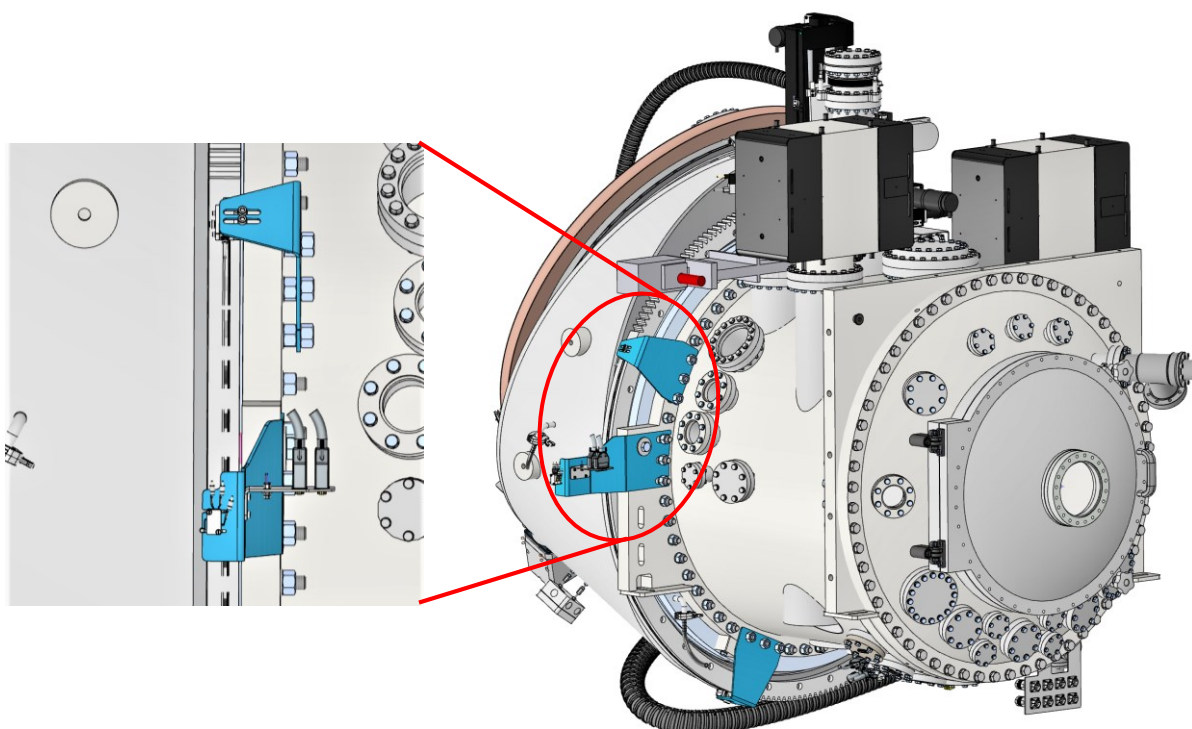


Figure 27: Angle brackets at the sample chamber

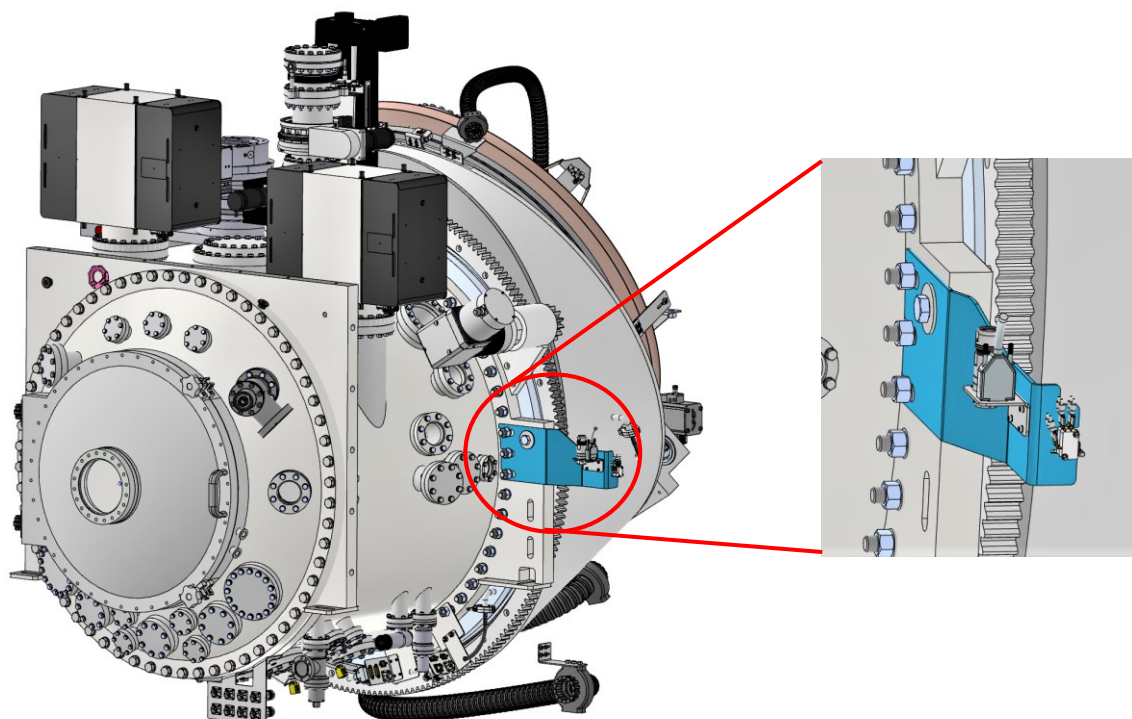


Figure 28: Figure 29: Angle brackets at the sample chamber

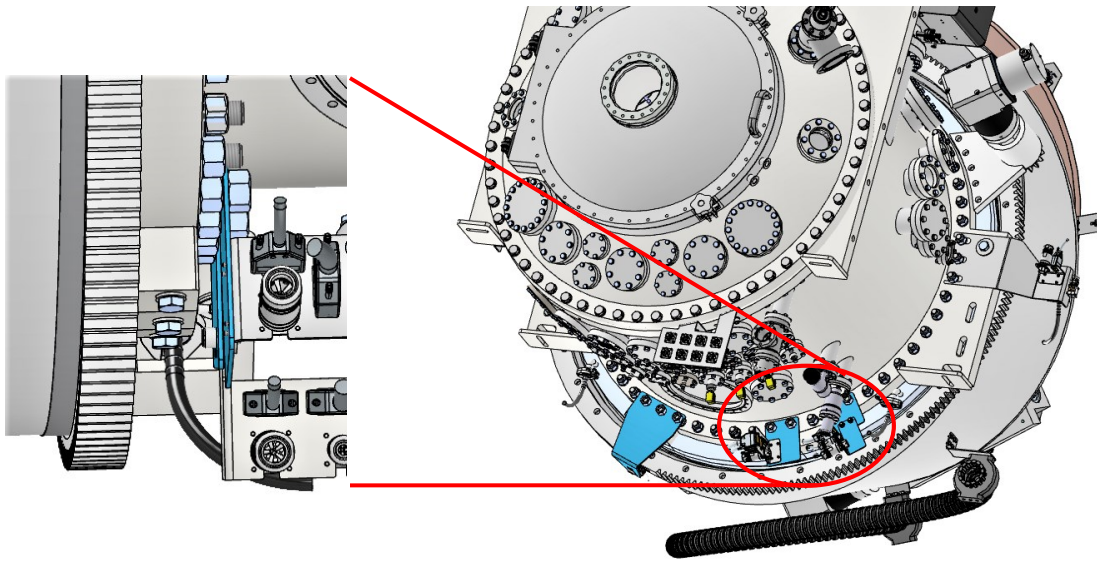


Figure 30: Figure 31: Angle brackets at the sample chamber

16. At this point all nuts can be tightened. This should be done first crosswise in order to achieve regular contact of the flange with the copper gasket and then all around the flange with ca. 80 Nm torque. You can go up to 150 Nm torque when required, e.g. at a specific position at the flange, after leak test.
17. In order to fix the angle bracket shown in Figure 32 for the cable drag chain, four tapped holes M4 x 8 must be drilled into the orange support frame.

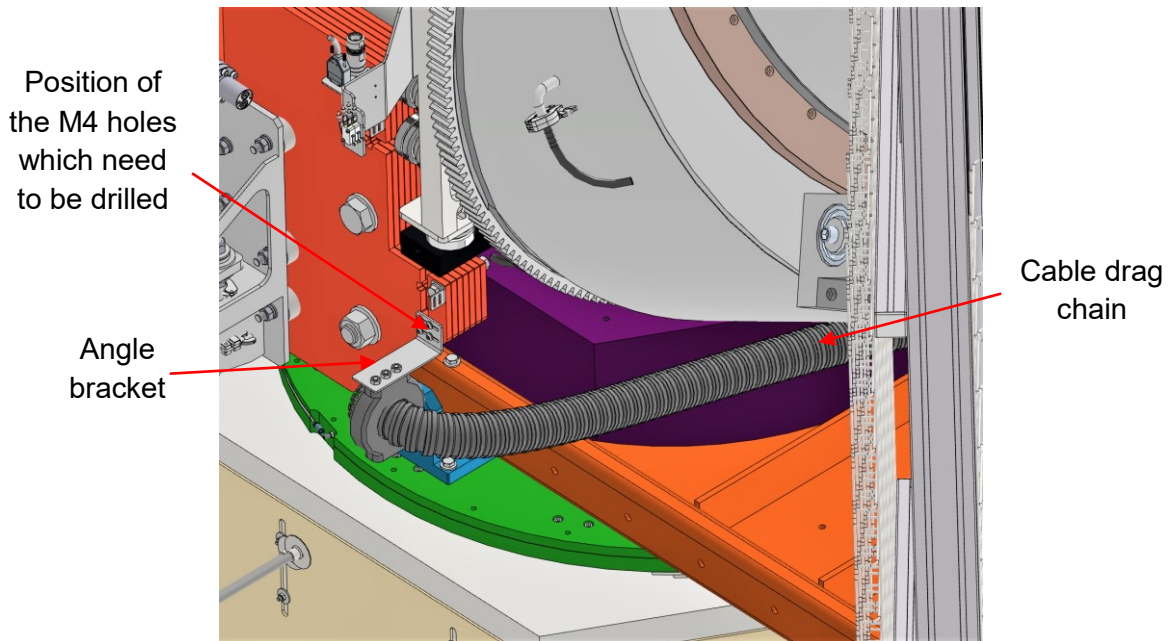


Figure 32: Angle bracket holding cable drag chain

18. At this point all encoder readheads and limit switches can be adjusted. The TRF1 motion axis has two encoder readheads, see left hand side of Figure 33. At both readheads there is a wheel (bearing) available, that keeps the distance to the encoder scale constant during axis movement. The wheel must be mounted slightly pre-tensioned.

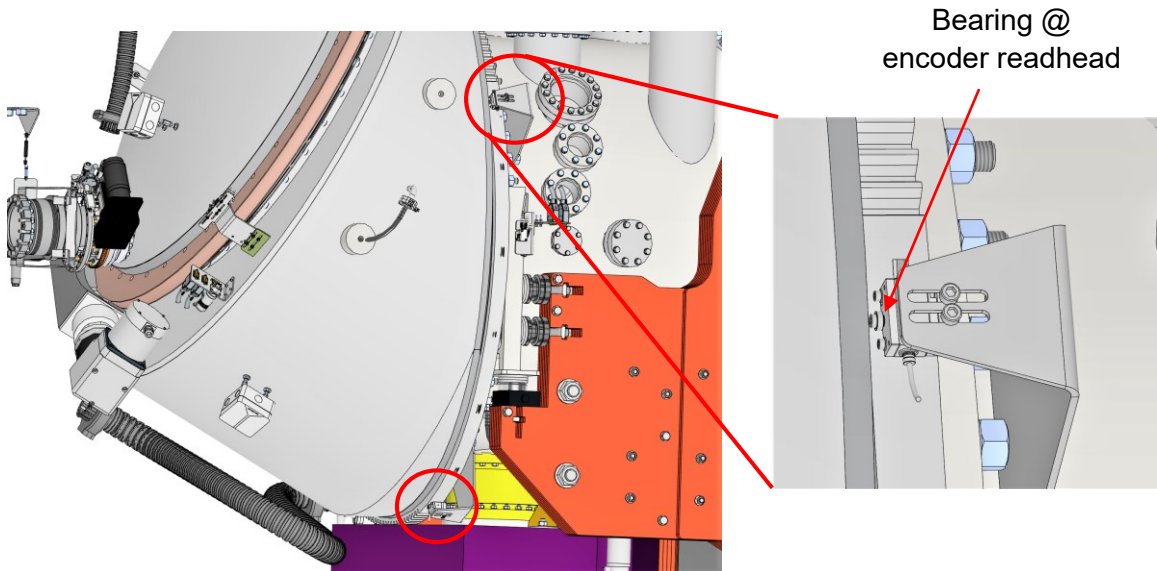


Figure 33: Encoder readheads at the TRF1 motion axis

19. Mount the gate valve at the TRF3 motion axis. The encoder ring does not have to be removed, the nuts can be tightened through the spaces between the posts. In order to reach all the nuts, the TRF3 axis must be moved.

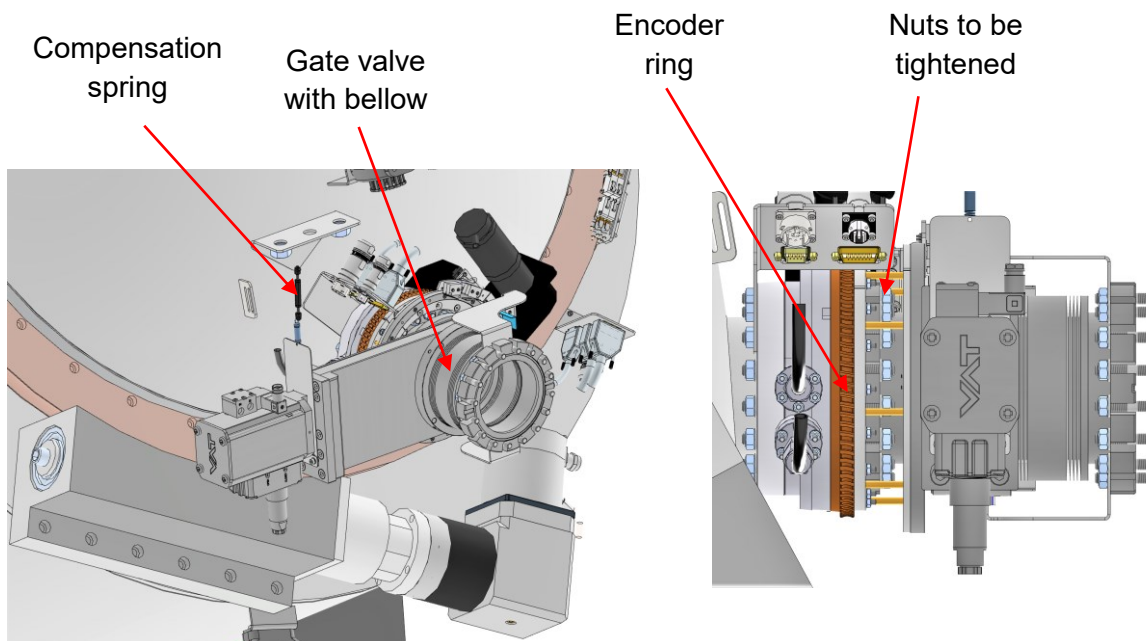


Figure 34: Gate valve at the TRF

20. The metal sheet that activates the limit switches at the TRF3 motion axis (turquoise in Figure 35) has to be glued at the gate valve. Please use for this purpose a heat resistant two component epoxy glue.

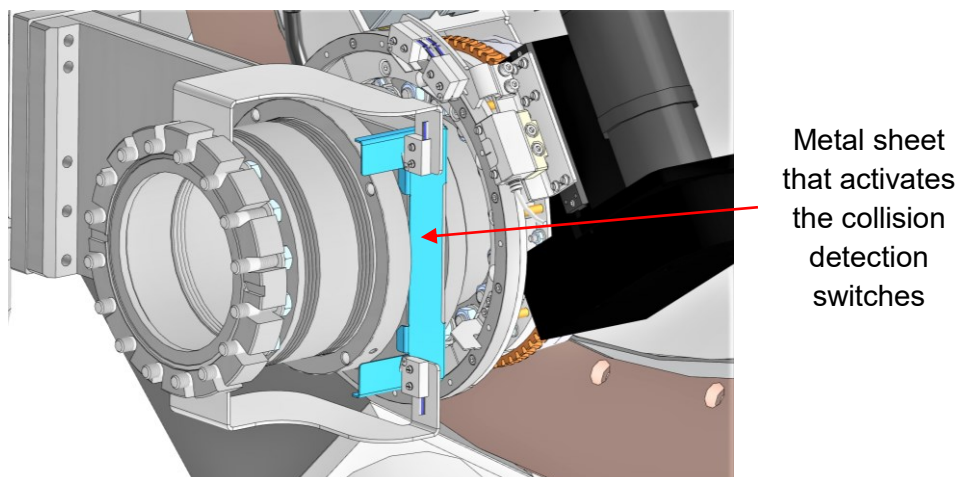


Figure 35: Metal sheet that activates the limit switches at the TRF3 motion axis

21. The next step is to connect the TRF to the 3AA optics tank / spectrometer arm. For this, you have to move the spectrometer arm to an appropriate position.

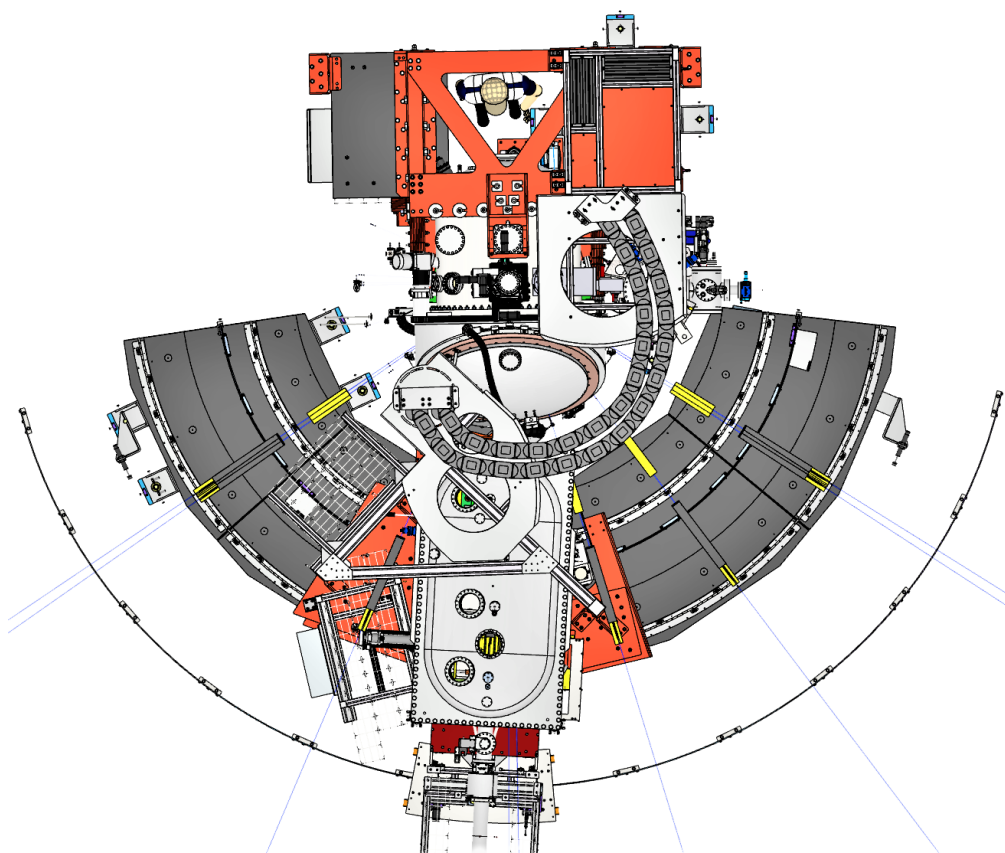


Figure 36: Top view of the sample chamber and 3AA optics tank

22. A gravity compensation spring is available at the gate valve. Please attach /mount it at the 3AA chamber lid as shown in Figure 37.

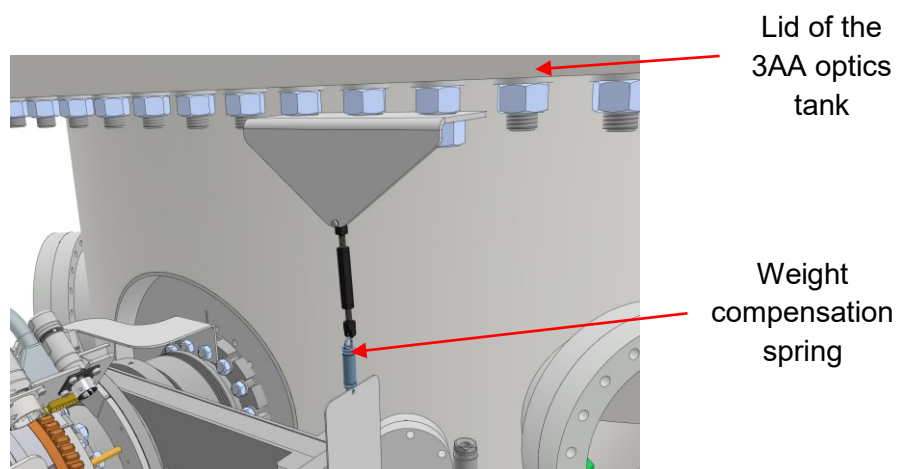


Figure 37: Gravity compensation spring at the gate valve

23. Mount the two metal sheets that hold the collision detection switches (turquoise in Figure 38) using four of the nuts fixing the gate valve bellow to the 3AA chamber, see Figure 39.

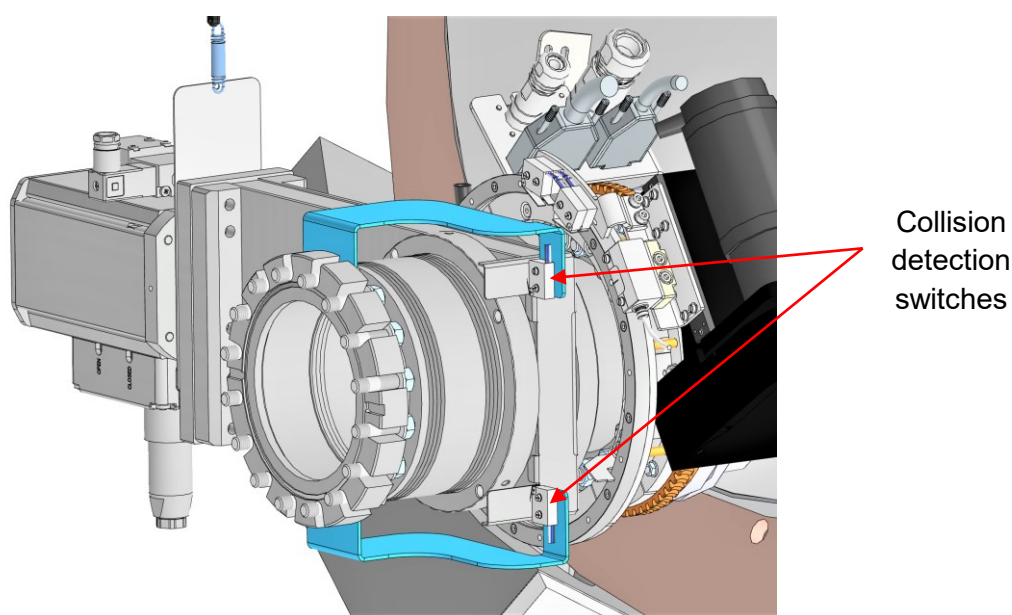


Figure 38: Collision detection switches at the TRF3 motion axis

The lid of the 3AA vacuum chamber does not have to be removed for mounting of the spring

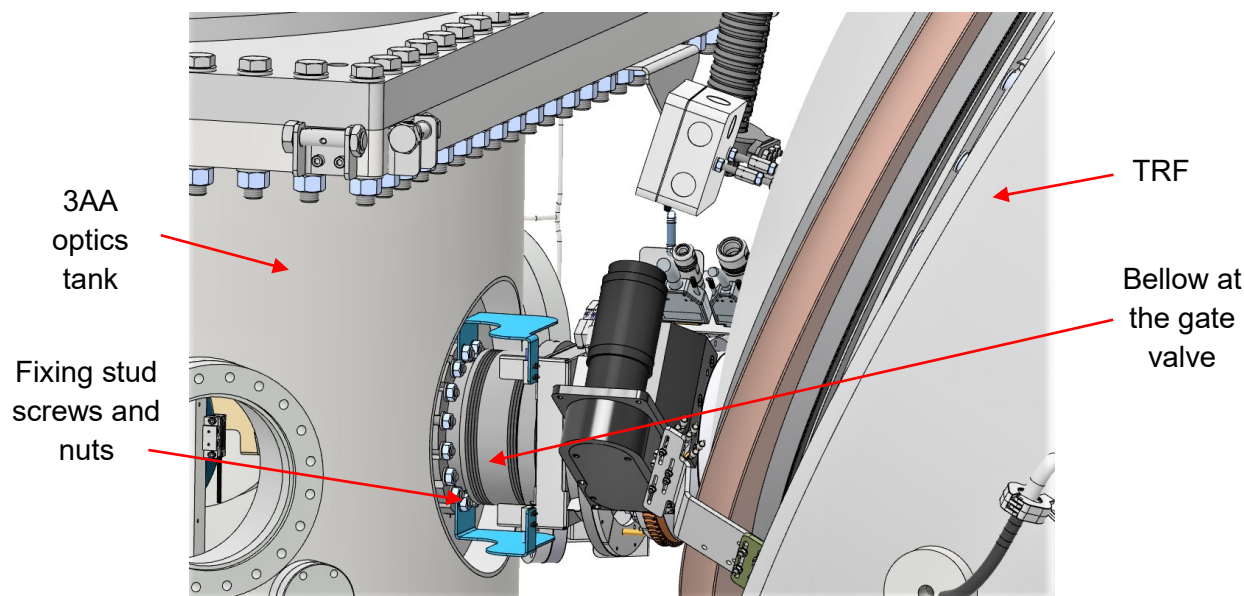


Figure 39: Connection between the TRF and 3AA optics tank

24. Note, that the collision detection switches (CSW1-TRF3 and CSW2-TFR3) have to be electrically connected as shown in Figure 40.

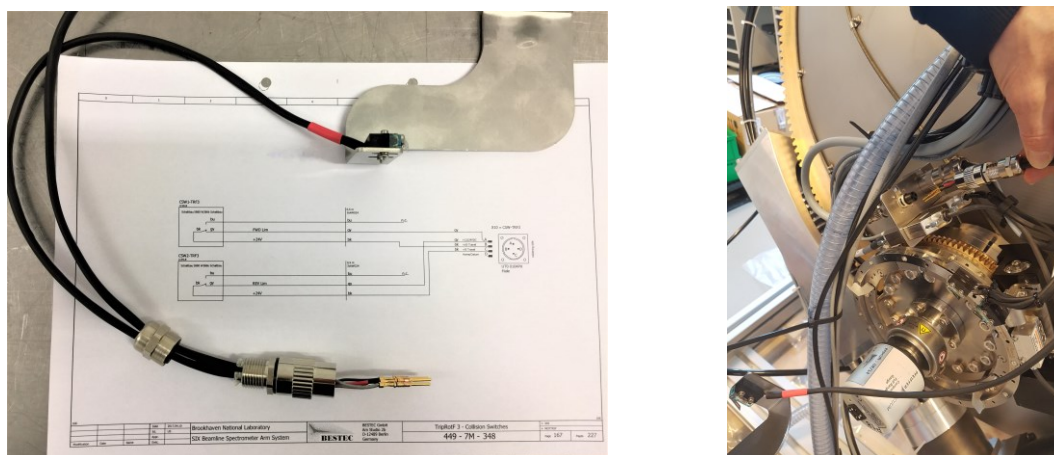


Figure 40: Collision detection switches

CAUTION

Please test the motion of the TRF while the screws fixing the bellow to the 3AA chamber are still loose. A parasitic rotation of the TRF3 relative to the 3AA can cause damage of the bellow by torsion.

7.4 Maintenance

7.4.1 Bakeout procedure

⚠ DANGER Hot surfaces: system bakeout



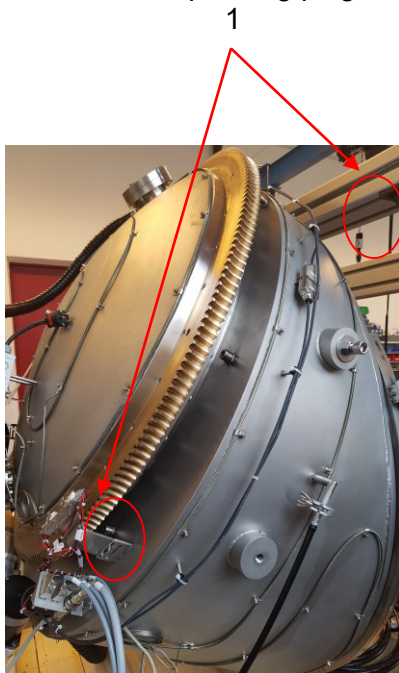
1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system.
2. Wear protective gloves when touching the system while bakeout and cooling down of the system

⚠ WARNING Hot surface: cables at the triple rotation flange

Before starting the bakeout procedure make sure that all cables that are led over the triple rotation flange have no contact with it and are found at a safe distance from the hot surface.

Please refer to Section 6.3.1 for a detailed description of the sample chamber bakeout.

The encoder cables of the readheads ENC1-TRF2 and ENC2-TRF2 have to be disconnected from the corresponding plugs and placed away from the hot chamber wall.



- 1 ENC2-TRF2, ENC1-TRF1 cables to be disconnected
- 2 Encoder readhead (ENC2-TRF2)
- 3 Encoder readhead (ENC1-TRF2)
- 4 Encoder cables placed away from the chamber wall

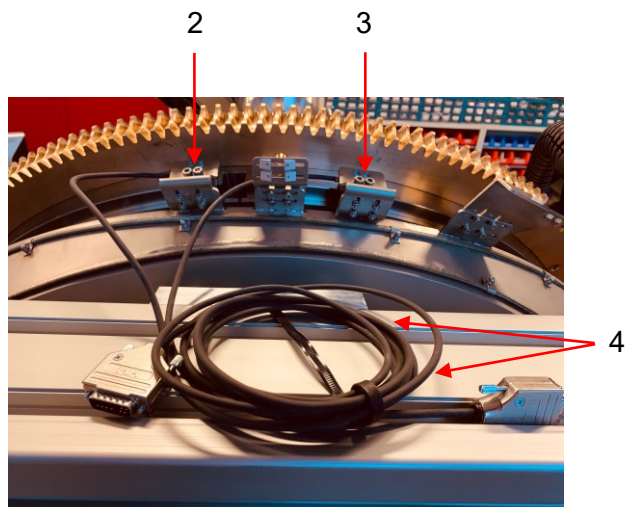


Figure 41: Encoder cables at the triple flange

7.5 Appendix

7.5.1 Overview of technical specification

Triple rotation flange 1 (TRF1)	
Range	-90° to 90°
Motor	Phytron ZSH107/1.200.7-WPLE120-256-SSA3AF-Y
Step resolution (full step)	$4.394 \times 10^{-4} \text{ °/step}$
Encoder	Renishaw RESOLUTE Encoder scale: RTLA-S Readhead (2x): RL32BAT050B30A
Encoder resolution	0.05 μm /count $4.22373 \times 10^{-6} \text{ °/count}$
Limit switches	2 x Schaltbau S880-W3B6k
Kill switches	2 x Schaltbau S880-W3B6k
Triple rotation flange 2 (TRF2)	
Range	-180° to 180°
Motor	Phytron ZSH107/1.200.7-WPLE120-20-SSA3AF-Y
Step resolution (full step)	$3.75 \times 10^{-4} \text{ ° /step}$
Encoder	Renishaw RESOLUTE Encoder scale: RTLA-S Readhead (2x): RL32BAT050B30A
Encoder resolution	0.05 μm /count $5.15936 \times 10^{-6} \text{ °/count}$
Limit switches	2x Cherry D45U-V3RA
Kill switches	2x Cherry D45U-V3RA
Triple rotation flange 3 (TRF3)	
Range	-90° to 90°
Motor	Phytron ZSS57.200.2,5-GPL52/8
Step resolution (full step)	$6.25 \times 10^{-4} \text{ ° /step}$
Encoder	Renishaw RESOLUTE Encoder ring: RESA30USA206B Encoder readhead: RA32BAA206B15A
Encoder resolution	$8.3819 \times 10^{-8} \text{ °/count}$
Limit switches	2 x Schaltbau S880-W3B6k
Kill switches	2 x Schaltbau S880-W3B6k

7.5.2 Rotation angles of the TRF flange

The rotation angles ($\alpha_1, \alpha_2, \alpha_3$) of the TRF axes (TRF1, TRF2, TRF3) as a function of the swivel angle Ω are given below:

$$\text{TRF1:} \quad \alpha_1 = -\arctan\left(\frac{\tan(0.5\Omega)}{\sqrt{\tan^2(28.5^\circ) - \tan^2(0.5\Omega)}}\right)$$

$$\text{TRF2:} \quad \alpha_2 = 2\arcsin\left(\frac{\sin(0.5\Omega)}{\sin(28.5^\circ)}\right)$$

$$\text{TRF3:} \quad \alpha_3 = \alpha_1$$

The swivel angle range is: $-57^\circ \leq \Omega \leq +57^\circ$ see Figure 42.

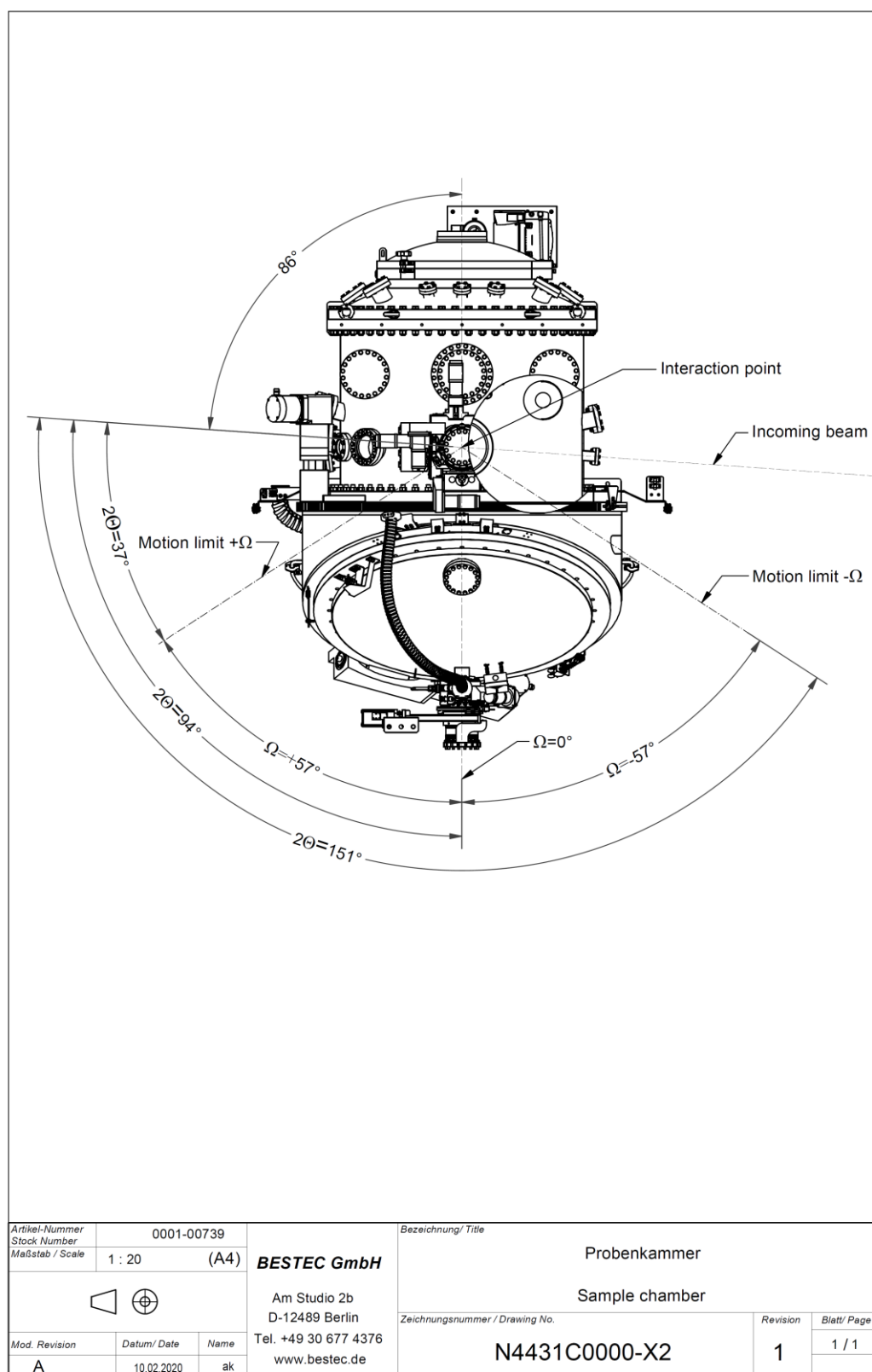


Figure 42: 2Theta and TRF swivel angles ranges

7.5.3 Possible collision regions

Figure 43, Figure 44 and Figure 45 indicate possible collision positions of the TRF with the neighboring parts. The critical positions are marked in yellow.

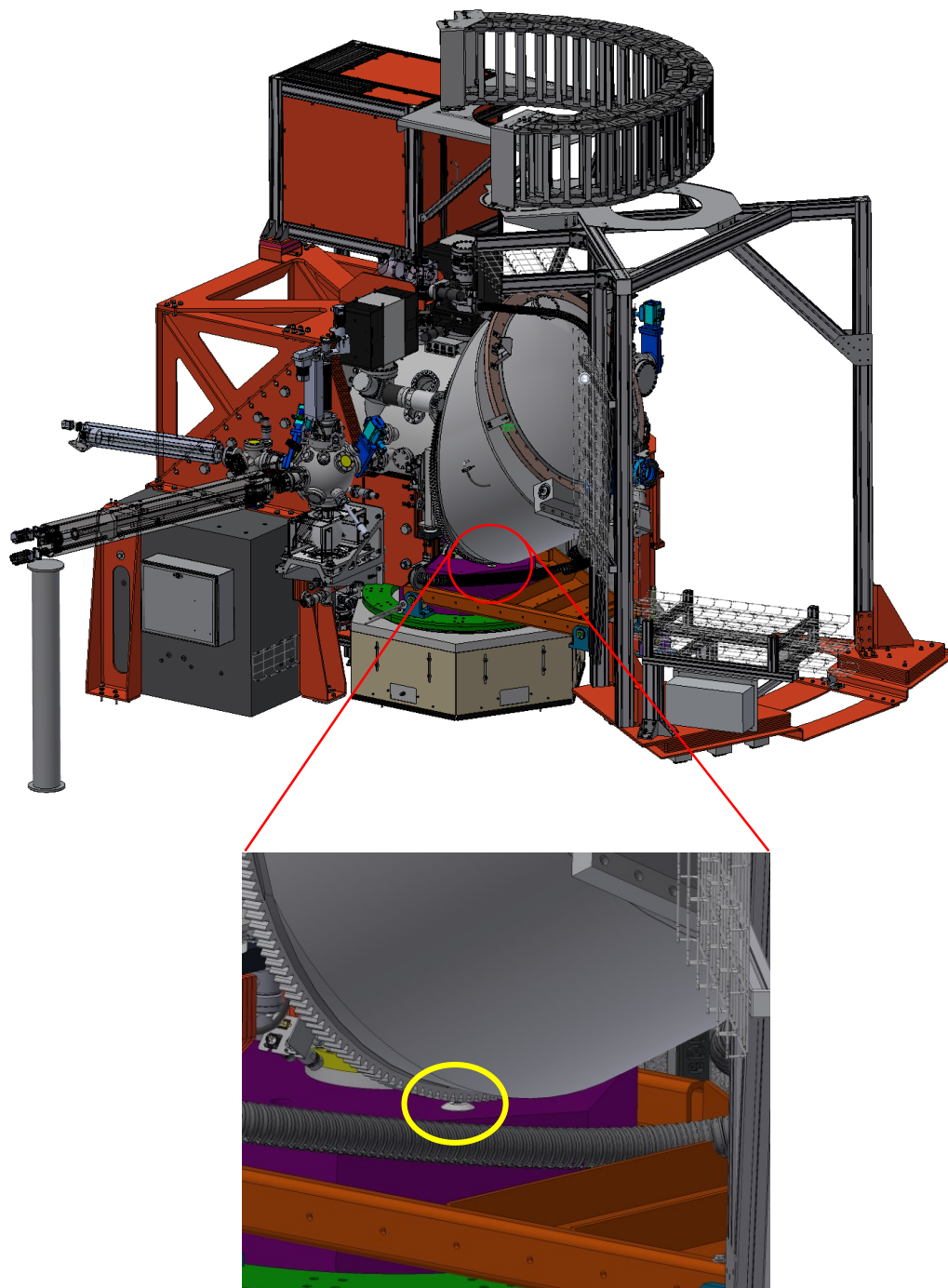


Figure 43: Possible collision region

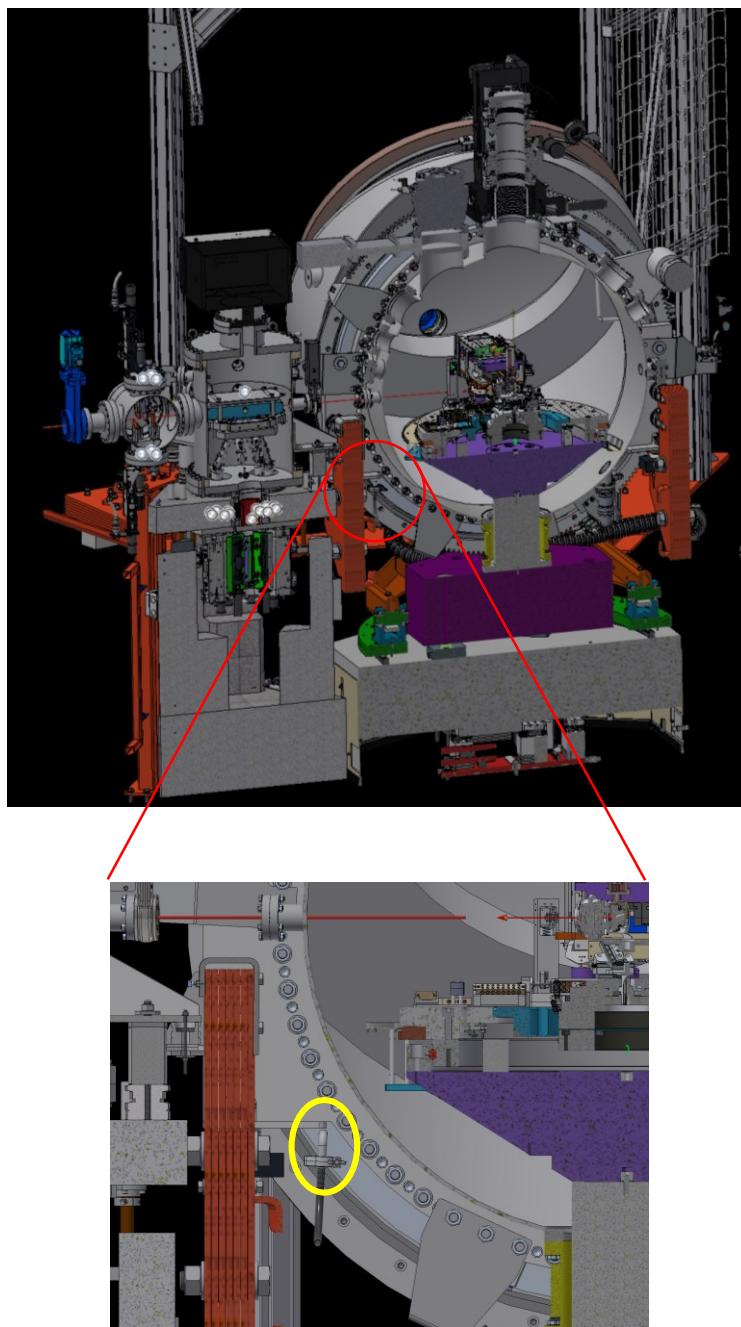


Figure 44: Possible collision position

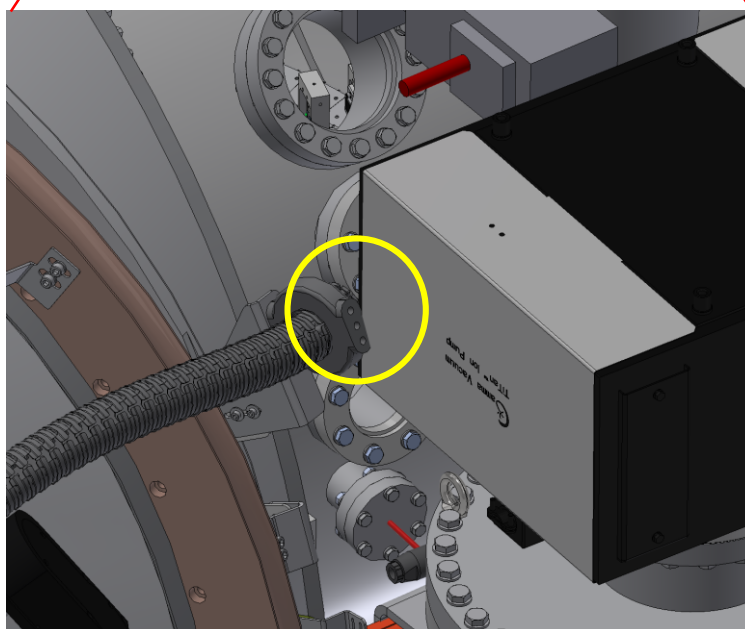
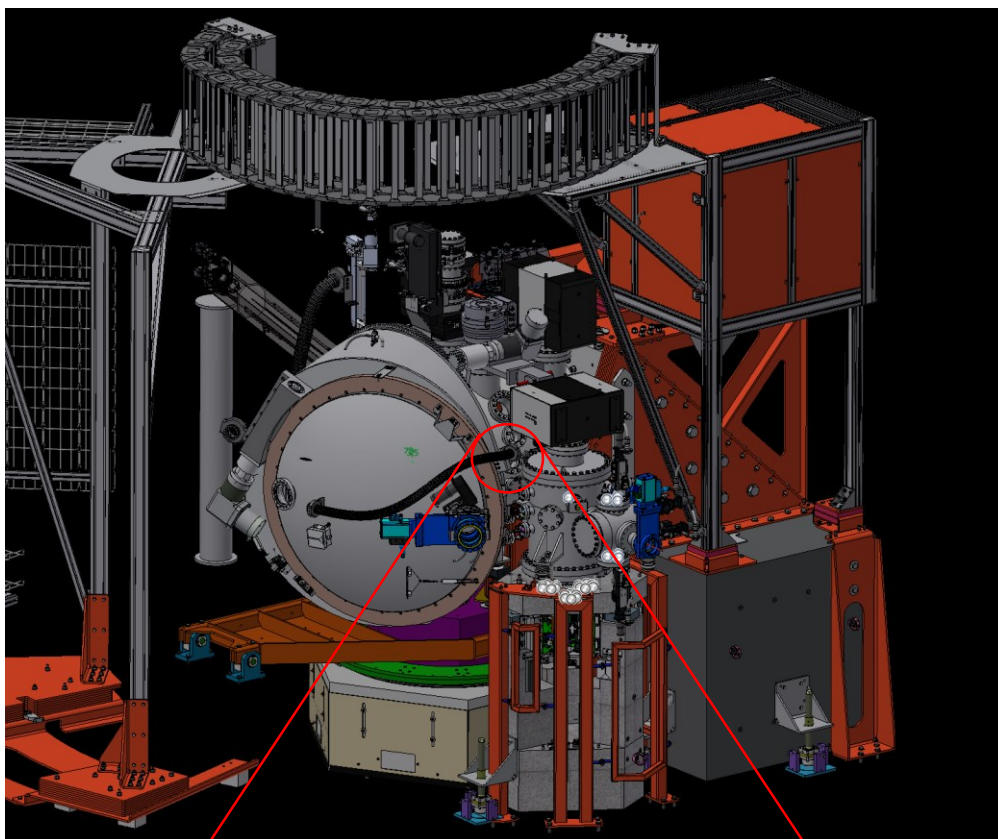


Figure 45: Possible collision position

7.5.4 List of supplied wiring schemes

Table 9 contains a list of the provided wiring schemes. For detailed schemes, please refer to Customer Documentation folder.

Table 10: List of wiring schemes for the triple rotation flange

Scheme number	Description
449 - 7M - 500a	Motion Axes - Overview
449 - 7M - 560a	TripRotF 1 - Motor + Limit switches
449 - 7M - 570a	TripRotF 1 - Kill switches + Encoder
449 - 7M - 560b	TripRotF 2 - Motor + Limit switches
449 - 7M - 570b	TripRotF 2 - Kill switches + Encoder
449 - 7M - 560c	TripRotF 3 - Motor + Limit switches
449 - 7M - 570c	TripRotF 3 - Kill switches + Encoder
449 - 7M - 348	TripRotF 3 - Collision Switches
449 - 7M - 582	TripRotF 2+3 - Motor Cable
449 - 7M - 584	TripRotF 2+3 - Kill Switch Cable
449 - 7M - 591	TripRotF 2+3 - Absolute Encoder Cable
449 - 7M - 589	TripRotF 3 - Collision Switch Cable
449 - 7M - 401a	Bakeout TRF

7.5.5 List of supplied assembly drawings

Table 10 contains a list of the provided assembly drawings. For detailed drawings, please refer to Customer Documentation folder.

Table 11: List of assembly drawings for the triple rotation flange

Drawing number	Description
N4431C0000-X2	Sample chamber
03-1356-00-X1	Triple rotation flange
03-3020-00-X1	Spacer
03-1368-00-X1	Rotary feedthrough
01-2175-00-X1	Holder complete
03-2046-00-X1	Interface
01-7438-00-X1	Switch assembly
01-7458-00-X1	Holder
01-7445-00-X1	Switch assembly
01-7455-00-X1	Holder
01-7434-00-X1	Switch cam assembly
01-3100-00-X1	Holder complete
01-6629-00-X1	Read head holder
01-3071-00-X1	Switch cam complete
01-7472-00-X1	Connector assembly

7.5.6 List of recommended spare parts

If replacement of system components is necessary, please contact Bestec GmbH. Please note that self-repair of system components without Bestec GmbH consultation can lead to loss of warranty claim.

Table 12: List of recommended spare parts

Spare part	Manufacturer	Part number
Encoder readhead	Renishaw	RL32BAT050B30A
		RA32BAA206B15A
Limit switch	Schaltbau Cherry	S880-W3B6k D45U-V3RA

7.5.7 List of supplied subcontractors documents

Table 13 contains a list of provided subcontractors' documents. For detailed manuals please refer to the provided Customer Documentation folder.

Table 13: List of provided subcontractors documents

Part	Manufacturer	File name
Heating cable	BriskHeat	Heating cable.pdf
Stepper motor	Phytron	ZSS.pdf ZSH.pdf
Limit switch	Schaltbau Cherry	Snap action switches.pdf Snap switches.pdf
Encoder	Renishaw	Resolute L-9517-9448-04-F Resolute L-9517-9418-05-B Resolute L-9517-9399-02-A Resolute M-9553-9735-02-B Resolute M-9553-9433-06-A

8 M5 mirror system

8.1 Safety information

8.1.1 Used symbols

The following symbols found in the present document are used to identify hazards. An additional purpose of them is to draw the user's attention to safety instructions or other useful information.

	Information on preventing any kind of physical injury.
	Information on preventing extensive equipment and environmental damage.
	Information on correct handling or use. Disregard can lead to malfunctions or equipment damage.
	General warning
	Pinch point hazard warning
	High voltage hazard warning
	Hot surface hazard warning
	Watch your head hazard warning
	Pinch point/entanglement hazard warning
	Gloves required
	Eye protection required
	Hard hat required

8.1.2 Safety instructions

General

In normal operation, there is no danger when working with the M5 mirror system. The operator of the mirror unit should however be aware of potential dangers that can occur by improper handling of the system and that can cause personnel injury or damage to the equipment.

When the system is operated without fulfilling the recommendations described in this document and the legal health and safety regulation, safety of personnel and equipment can be affected. Potential dangers and measures to avoid them are described in this section.

Electrical safety

In normal operation no hazards can occur associated with electrical voltage and current used by the equipment. Following accidents may happen if the electrical installation of the mirror unit is not properly operated: electric shock and burns, fire and electric arcing.

⚠ DANGER Electrical voltage



1. Voltage and high voltage that may cause injury or death are used in the setup. Service or maintaining work on the electrical installation must only be performed by qualified trained personnel for this type of work.
2. The statutory regulations regarding electrical installations must be respected.
3. The entire electrical equipment supplied as part of the system is provided with a protective ground. Grounding cables must not be removed or manipulated. It is very important that the vacuum system is properly grounded at all times.
4. Fuses may not be bridged, electrical connections must not be separated or otherwise manipulated, protective covers and other electrical insulating material may not be removed or modified.
5. Do not attempt to administer first aid to someone who has suffered electric shock until the source of the shock has been isolated.

Safe handling of gases

In normal operation no dangers through use of gases may appear as long as the recommended precautions are followed. In case of gas handling contrary to the instructions, unconsciousness and associated injuries may occur.

⚠ DANGER Gases



1. If pure nitrogen is used for venting the vacuum system one has to be aware that its inhalation can induce an oxygen deficiency after only few breaths, which may lead to unconsciousness.
2. Areas in which the oxygen concentration is less than 18% must be abandoned immediately.
3. For venting the vacuum system it is recommended to use dry air with normal oxygen concentration instead of pure nitrogen.

Safe handling of mechanical components

In normal operation only few mechanical works on the M5 mirror system is required. For service and maintenance extensive mechanical work may be necessary. Only qualified personnel should attempt to dismantle, modify or repair equipment. The following hazards may arise when dealing with mechanical components: injury when transporting heavy loads (mirror unit, chamber lid), danger of crushing by moving parts (motors at the motion axes), injury by protruding parts at the mirror unit, severe damage of internal mechanics and optical elements (when changing motors at hexapod motion axes).

⚠ DANGER

Heavy loads



1. Only qualified personnel should lift and carry heavy loads.
2. Use a hard hat when working with heavy loads.
3. Lifting eyes are available at the individual components of the system and they must be used for transportation.
4. Parts of the equipment that have to be lifted may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

⚠ DANGER

Moving parts



1. At the mirror unit there are a number of moving parts, which are powered by electrical motors. Make sure that your way of working does not include any safety risk. Do not leave long hair untied, do not wear loose clothing or jewelry as this may involve the danger of becoming caught in the moving mechanisms.
2. Do not remove or modify covers of the moving parts.

⚠ CAUTION

Components in UHV



1. In-UHV components can only be touched with powder/lint-free gloves.
2. Tools use for work on UHV components must be clean and free of grease.
3. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

⚠ CAUTION

Optical elements



1. Use protective gloves and clothing when working with optical elements.
2. Be careful not to drop anything on/touch the optical surface.
3. A suitable cover for the optical surface is recommended.

Safe handling of hot surfaces

During bakeout procedure the system or parts of it become hot ($>100^{\circ}\text{C}$) and can cause burns when touched.

⚠ DANGER Hot surfaces	
 	1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system. 2. Wear protective gloves when touching the system while bakeout and cooling down of the system.

Safe handling of vacuum systems

By appropriate use of and compliance with the safety information during normal operation, no dangers occur from the vacuum system of the mirror unit. The following hazards may arise when dealing with vacuum systems: injury from “flying” pieces caused by exploding or imploding parts of the vacuum system; if the vacuum chamber is over pressurized, viewports may shatter and cause person injury. Use viewport covers and wear eye protection glasses.

⚠ DANGER Vacuum system	
 	1. The walls of the vacuum system can withstand a pressure difference from outside to inside of at least 1 bar, from inside to outside only a pressure difference of about 0.1 bar. An increase in internal pressure of 1.1 bar can lead to the damage of the view ports and must therefore be avoided. Never exceed an overpressure of 0.1 bar inside the vacuum system. 2. The viewports destruction of a chamber under vacuum leads to implosion. There is a risk of injury from flying glass shards. Use viewport and eye protection. 3. Areas in which the oxygen concentration is less than 18% must be immediately abandoned.

8.1.3 Modifications and service

Safe operation, reliability and performance of the equipment may be impaired if assembly operations, extensions, re-adjustments, modifications or repairs are not carried out in accordance with the instructions provided in this document and with any other instructions issued by the manufacturer. If you wish to modify the equipment please contact Bestec GmbH for future advice.

8.2 Design overview of the M5 mirror system

- | | |
|--|---|
| 1 Diagnostic slit and photodiode | 7 Diagnostic slit and photodiode |
| 2 Diagnostic slit and photodiode | 8 Sample manipulator SmarPod (for reference only) |
| 3 Customized cable chain
(allowing $\pm 180^\circ$ rotation) | 9 3AA 2Theta radial guidance |
| 4 M5 2Theta tube
with support pillar | 10 Upper granite part (support of the
sample chamber mechanics and
alignment of the sample chamber pivot) |
| 5 Wedge leveler (3x) for
alignment of the upper
granite part #10 | 11 Rotating metal ring on rails (3AA 2 Theta) |
| 6 M5 mirror assembly | 12 Lower granite part (defines the 3AA pivot) |

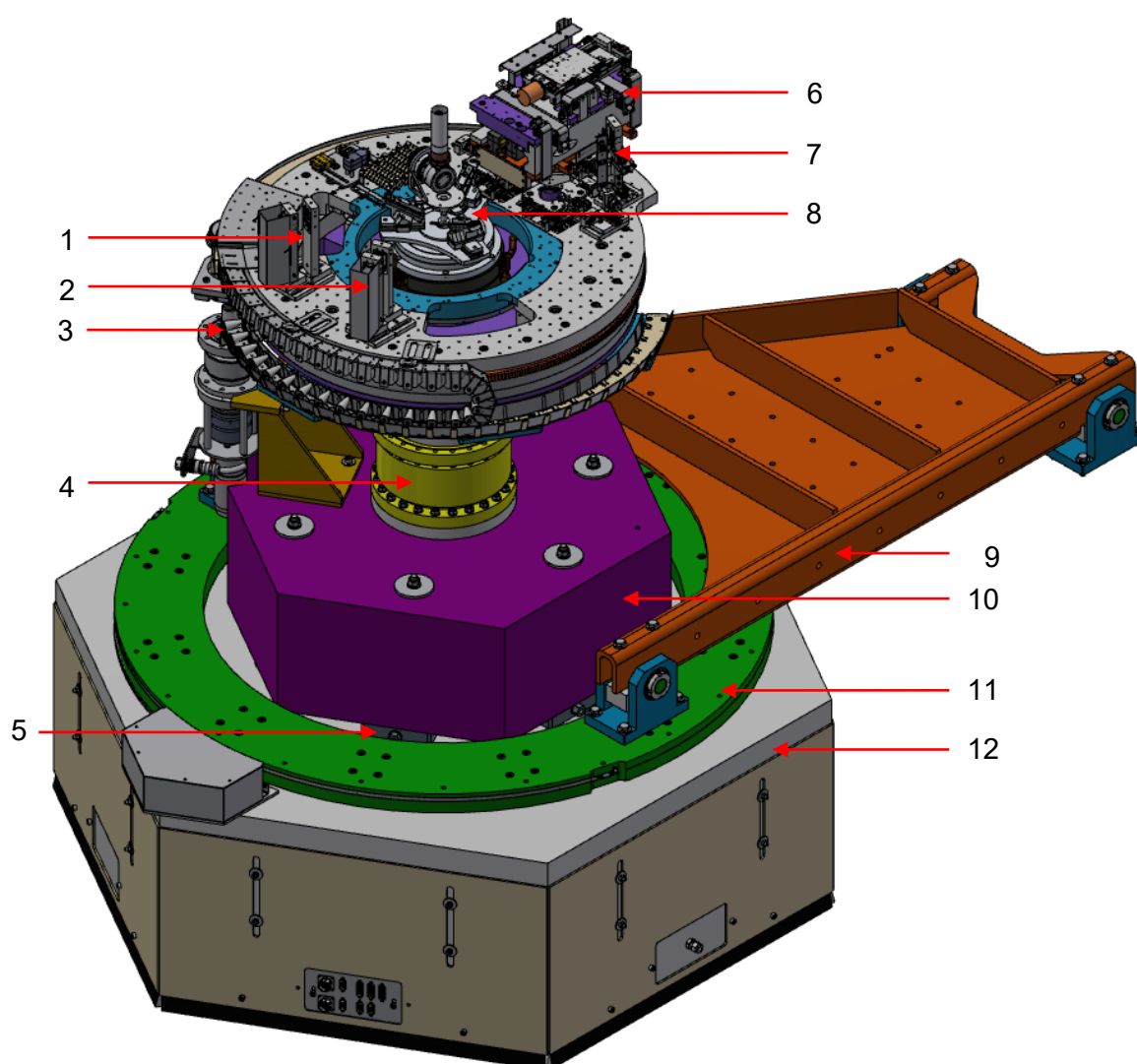


Figure 46: Sample chamber internal mechanics and support structure overview

- 1 M5 2Theta motor
- 2 3AA 2Theta encoder readhead
- 3 Wedge leveler (3x) for alignment of the lower granite #12 in Figure 46

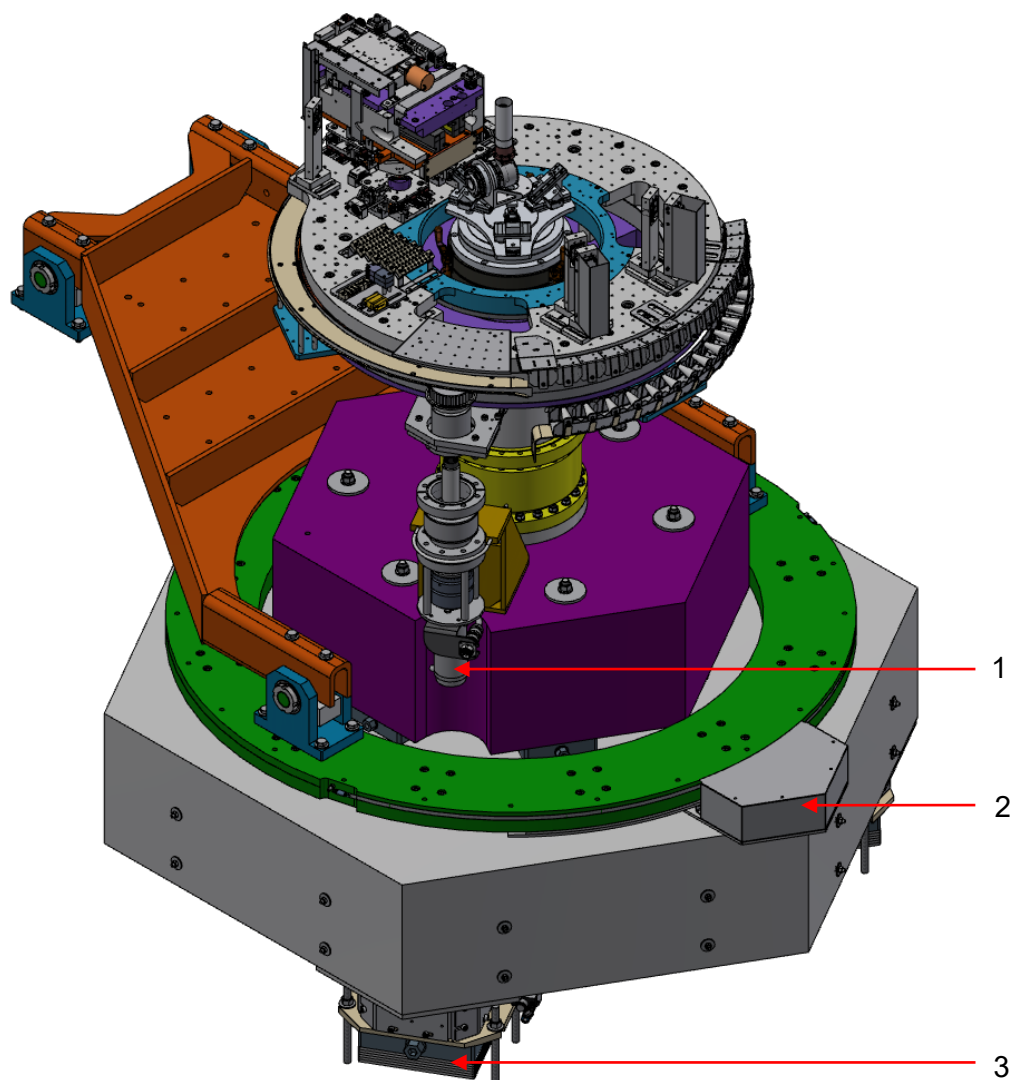


Figure 47: Sample chamber internal mechanics and support structure overview

- | | |
|---|--|
| 1 Wire-race bearing for M5 2Theta rotation | 5 Granite |
| 2 Encoder for M5 2Theta rotation | 6 M5 2Theta encoder ring on detachable metal piece |
| 3 Upper part of the support pillar made of aluminum | 7 Gearwheel |
| 4 Support pillar made of stainless steel | 8 Stainless steel tube |

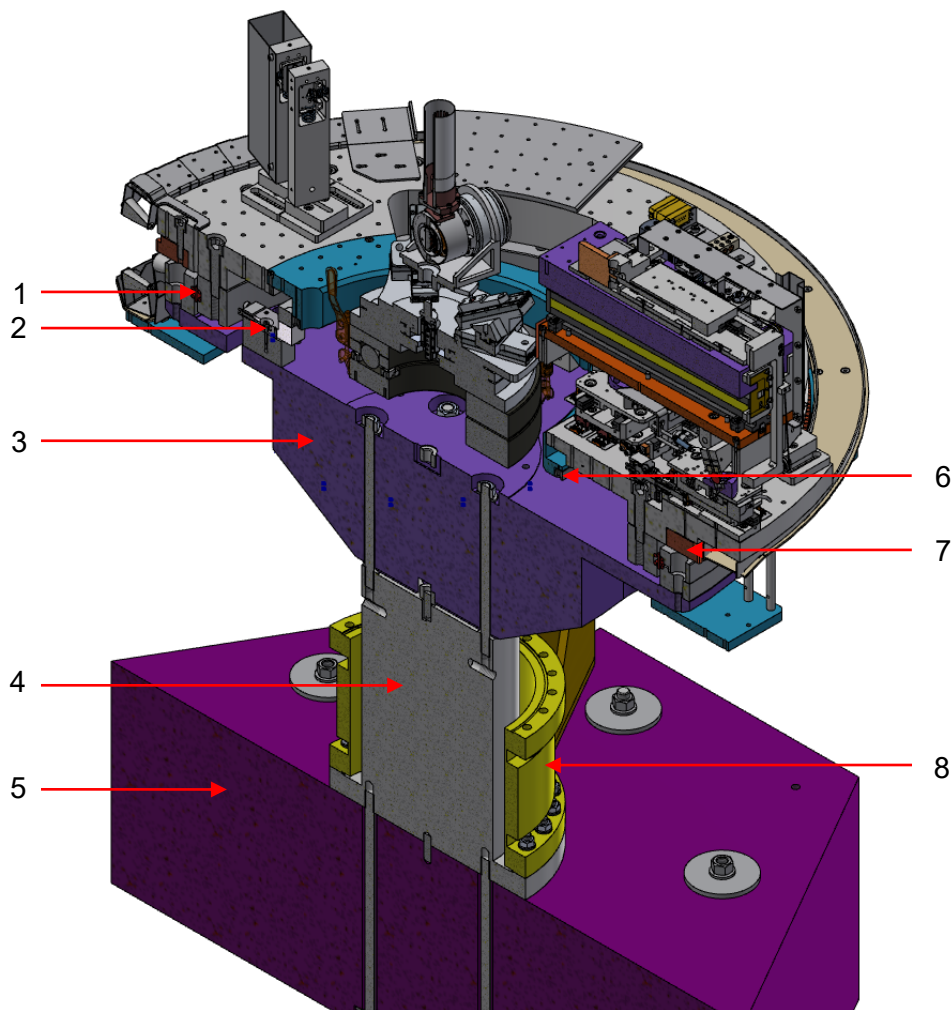


Figure 48: Sample chamber internal mechanics and support structure cross section

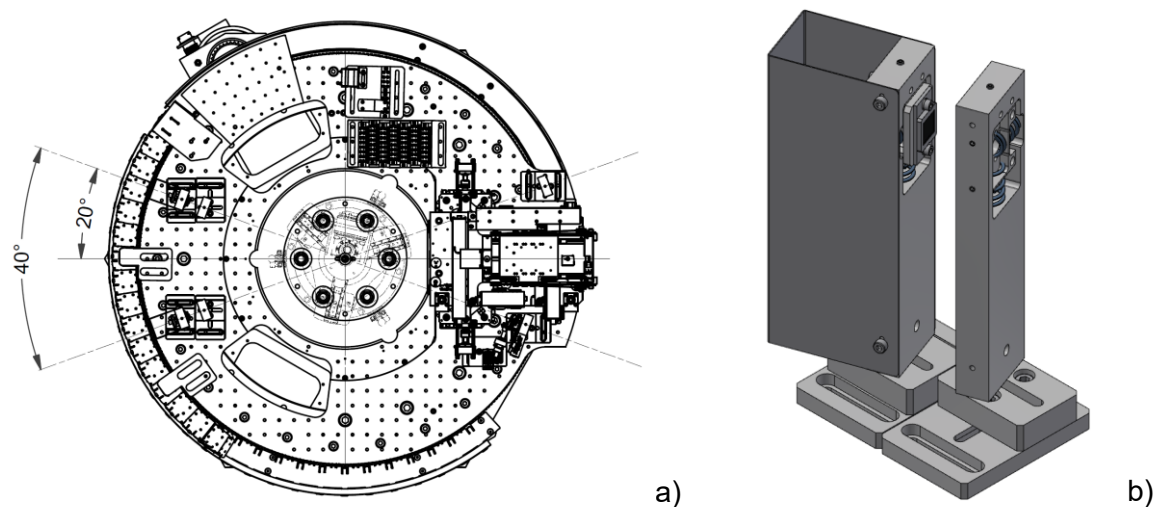


Figure 49: Top view on sample chamber internal mechanics (left)
Diagnostic slit-photodiode assembly (right)

- 1 3AA pivot translation assembly
- 2 Lower granite part (defines 3AA pivot)

- 3 Air pad feet on wedge levelers

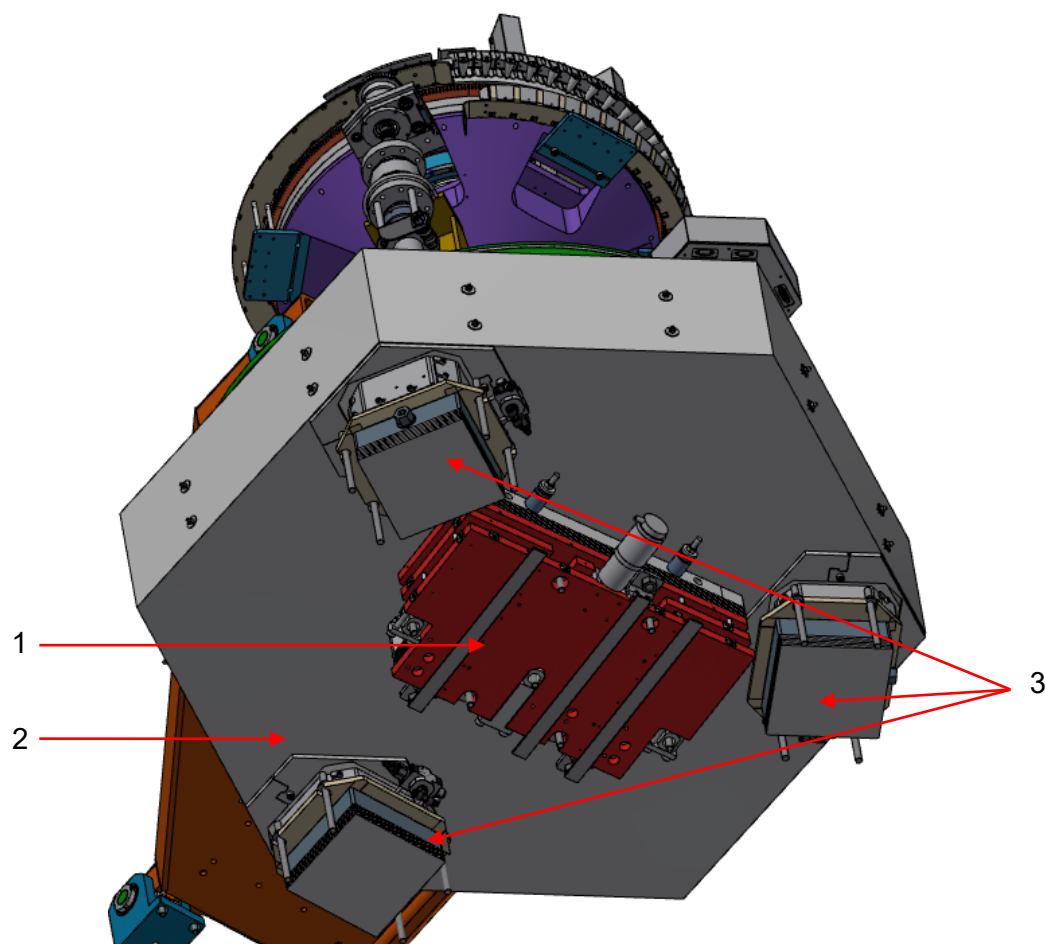


Figure 50: Pivot translation assembly

8.3 M5 mirrors assembly: overview and installation of optical elements

Tripod overview

- | | |
|---|------------------------------|
| 1 Motorized vertical slits | 4 Stacked translation slides |
| 2 Stepper motor driving slits
vertical translation | 5 Piezo linear actuators |
| 3 Tripod joint | |

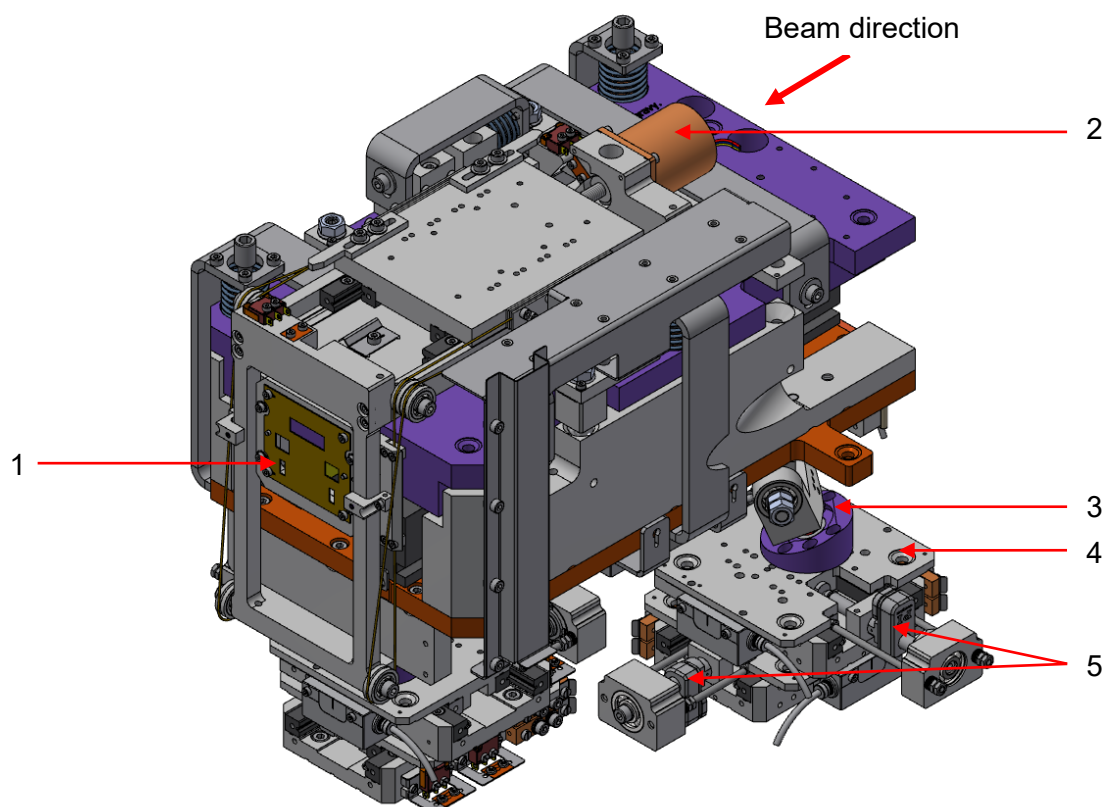


Figure 51: M5 mirrors assembly

- | | |
|--------------------|---|
| 1 Base plate M5B | 4 Collision switch
(Kapton foil with copper layer) |
| 2 Mirror dummy M5B | 5 Base plate M5A |
| 3 Mirror dummy M5A | |

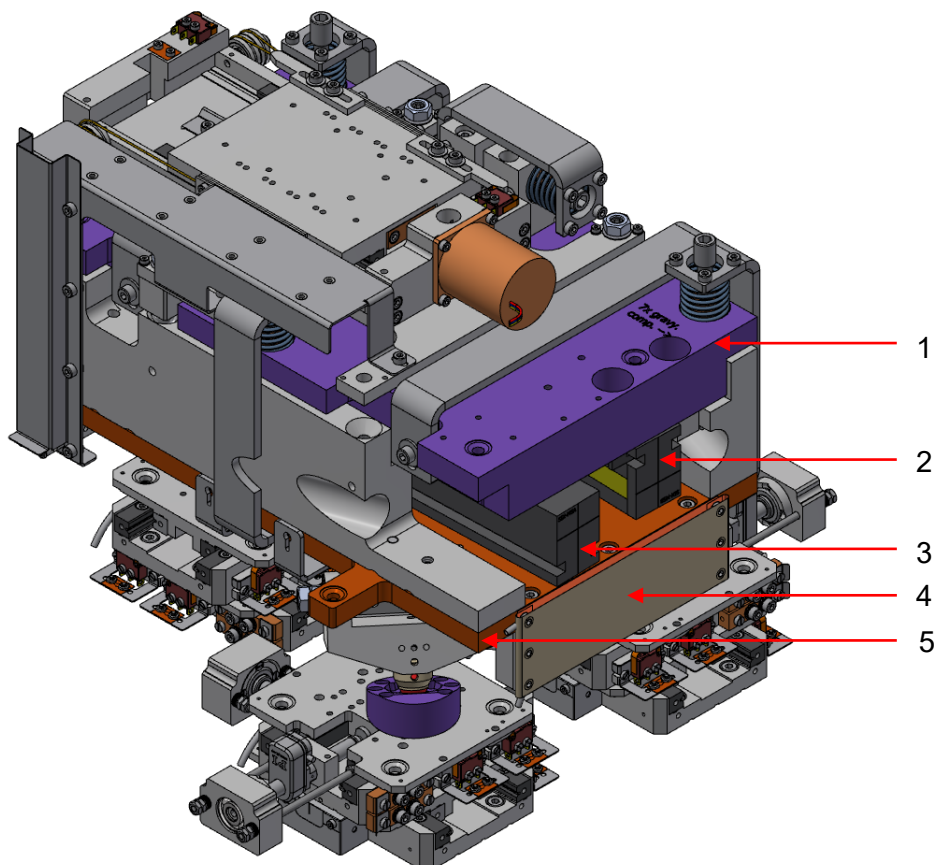


Figure 52: M5 mirrors assembly

A set of vertical slits (M5 mask) are mounted at the downstream end of the M5 mirror assembly are used for defining the horizontal aperture of the beam. The M5 mask is translatable in the vertical direction and is mounted such as to track the M5 translational and angular motions. The positions of the three slits are given in Table 14.

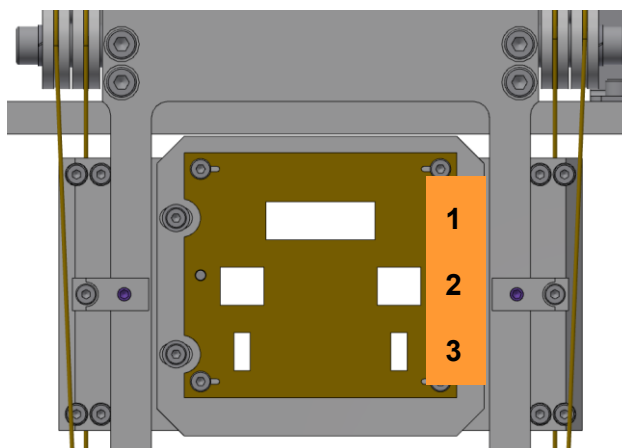


Figure 53: M5 mask assembly (upstream view)

Installation of the M5 mirrors

In order to mount the two M5 optical mirrors, the mirror holder assembly has to be detached from the bottom joints of the tripod.

Please follow the steps described below:

- A. Disconnect the type K thermocouples from the corresponding plugs; see Figure 54, left side.
- B. Disconnect motor/limit switches from corresponding plug; see Figure 54, right side.

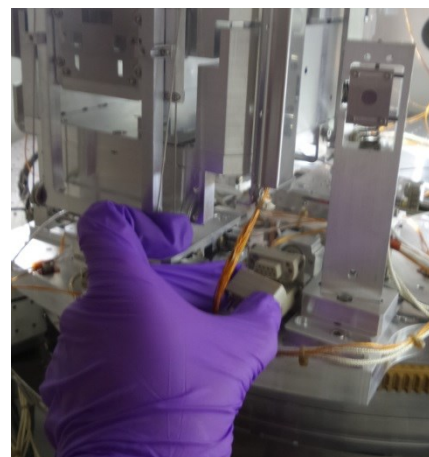
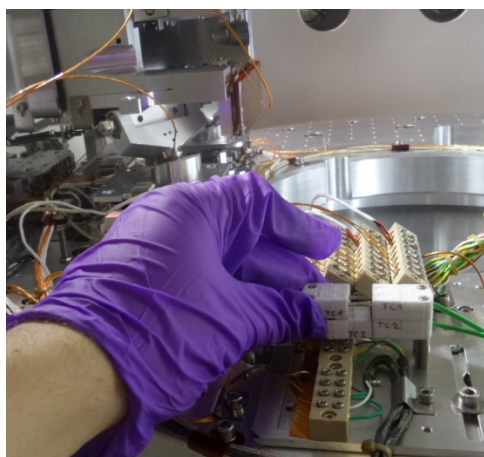


Figure 54: Thermocouple plug (left), motor/limit switches plug (right)

- C. Carefully lift up the upper part of the tripod, as illustrated in Figure 55, left side.
- D. Place the assembly on posts; see as example Figure 53, right side.

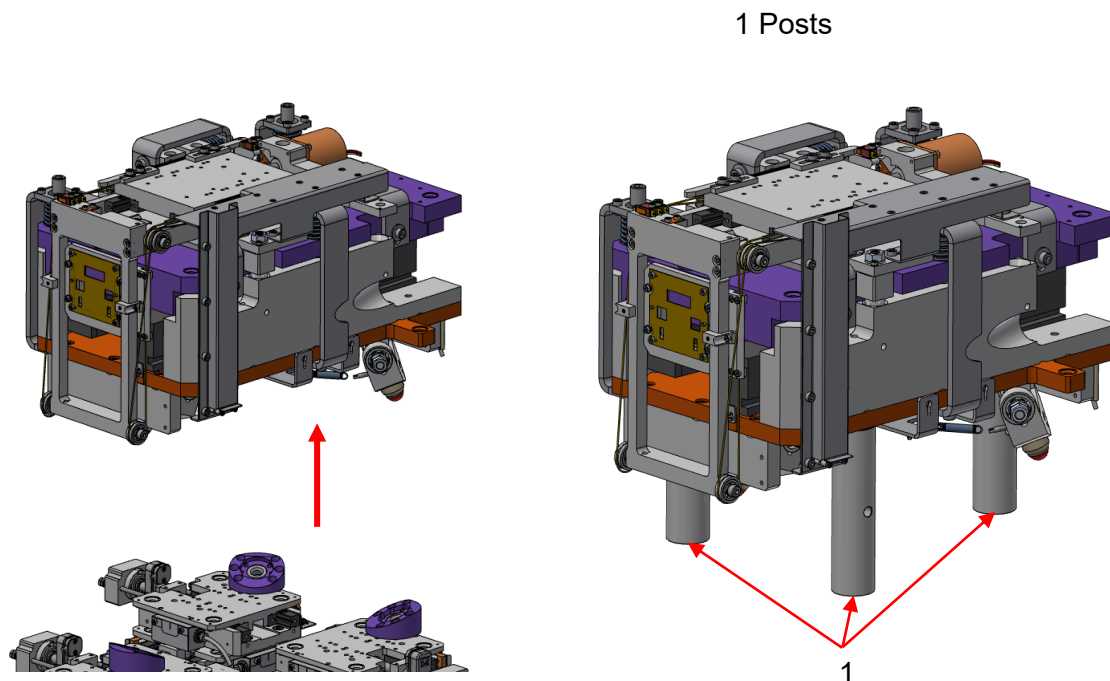


Figure 55: Upper part of the tripod

- E. Remove the collision switch (#4 in Figure 52) cables after loosening the corresponding fixing Allen screws, see Figure 56.



Figure 56: Collision switch at the M5 mirror assembly

- F. Loose and remove the three M6 nuts indicated in the left side of Figure 57.
G. Carefully remove the complete mask assembly as illustrated in the left side of Figure 57.

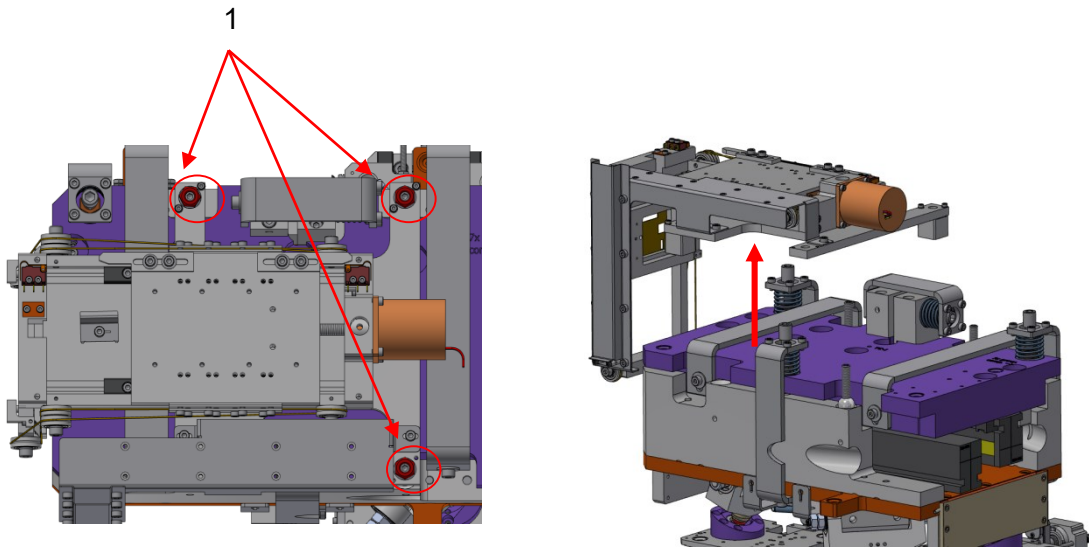


Figure 57: M5 mirror holder assembly top view (left) and side view (right)

H. There are six spring loaded clamps holding both parts of the mirror holder together. They have to be removed one after the other in order to detach the upper mirror holder from the lower one.

- | | |
|------------------------|--------------------|
| 1 Spring loaded clamps | 3 Mirror dummy M5A |
| 2 Mirror dummy M5B | |

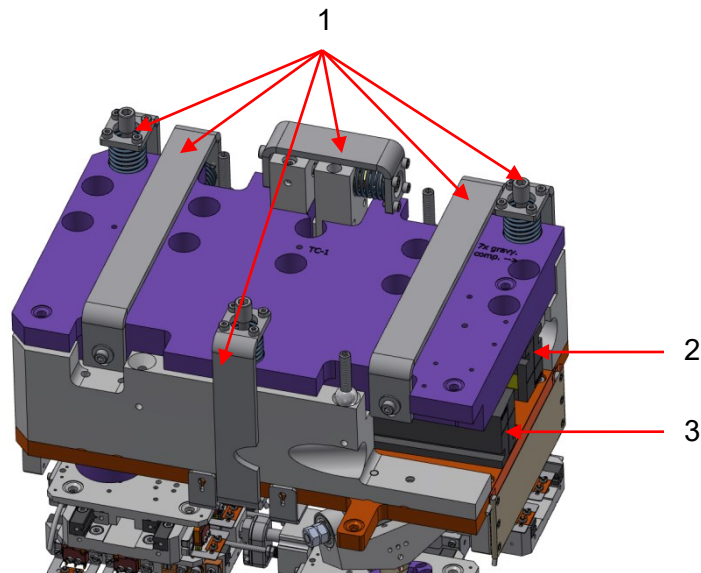


Figure 58: Mirror holder assembly

I. Figure 59 shows how to remove the clamps: press the clamp together such that the spring is slightly compressed, the clamp is then loose and can be removed.

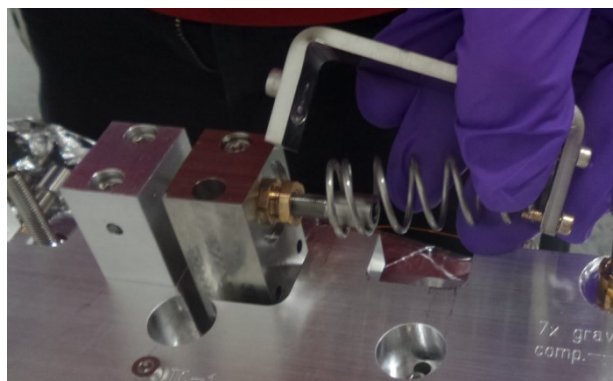
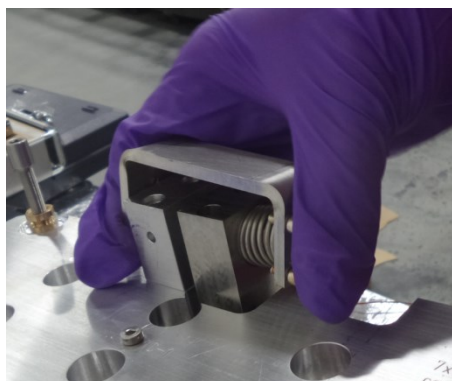


Figure 59: Removal of the fixing clamps at the mirror holder assembly

The upper mirror dummy is fixed on its base plate by three fixing clamps. The right side of Figure 60 shows the cross section of the fixing clamp.

- | | |
|---------------------|------------------------------|
| 1 Fixing clamp (3x) | 4 Gravity compensation screw |
| 2 Holder base plate | 5 Mirror support (fixed) |
| 3 Mirror dummy | 6 Fixing clamp screw |

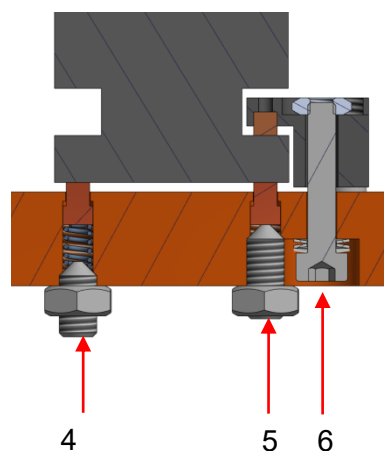
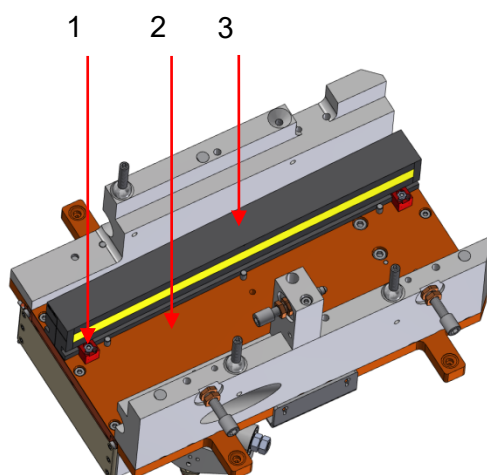


Figure 60: Upper mirror holder (left), cross section of a fixing clamp (right)

The mirror dummy can be removed by carefully sliding it through the fixing clamps after they have been loose (see Figure 62). The clamps can be loose from below the base plate, as shown in Figure 61. Do not remove the fixing clamps/screws!

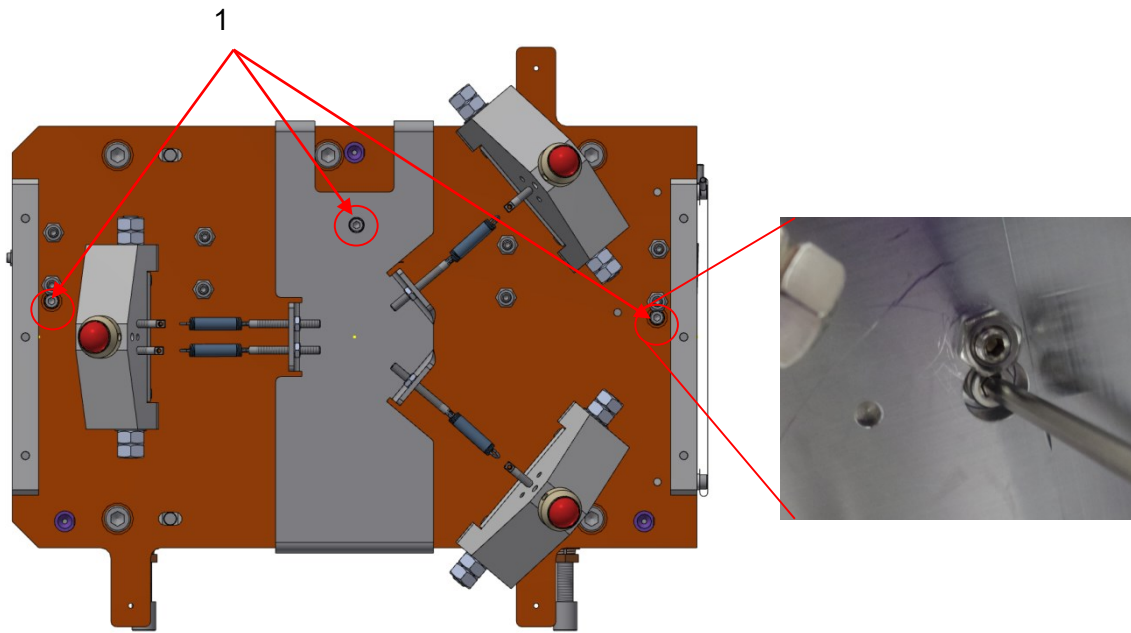


Figure 61: Fixing screws which have to be loosened

The mirror dummy can be replaced by the optical element as illustrated in Figure 62 with a piece of metal sheet.

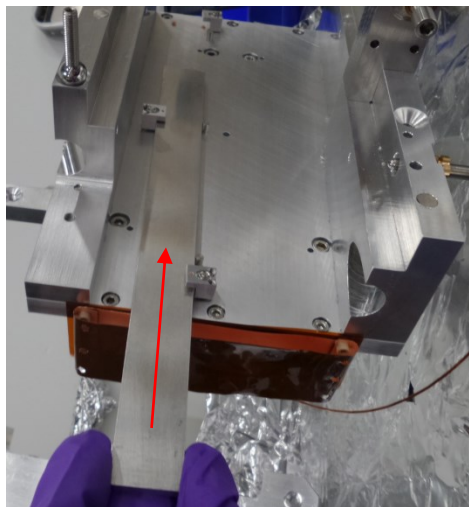


Figure 62: Sliding procedure of the optical mirror through the fixing claps

- J. Carefully tighten the fixing screws at the fixing clamps (#1 in Figure 60) without twisting the clamp on the holder base plate.

The lower mirror dummy is fixed on its base plate by three fixing clamps. The right side of Figure 60 shows the cross section of the fixing clamp.

- | | |
|---------------------|------------------------------|
| 1 Fixing clamp (3x) | 4 Gravity compensation screw |
| 2 Holder base plate | 5 Mirror support (fixed) |
| 3 Mirror dummy | 6 Fixing clamp screw |

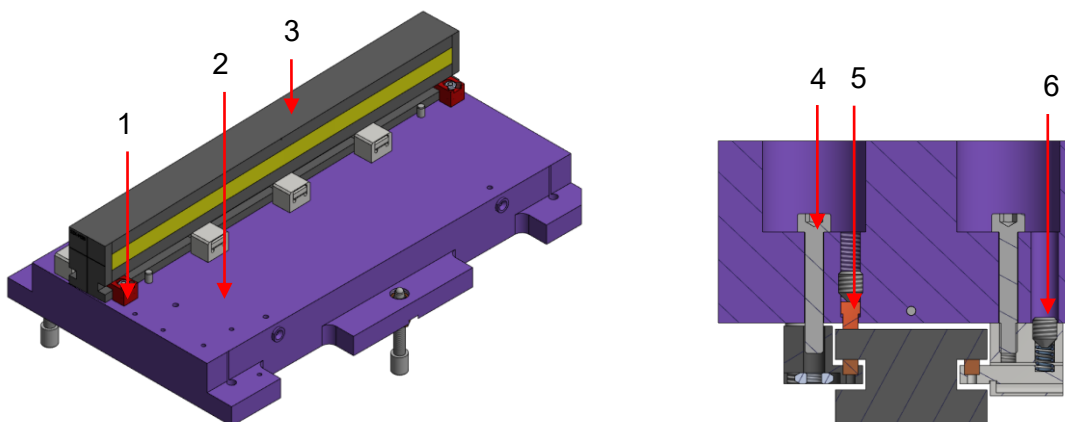


Figure 63: Lower mirror holder (left), cross section of a fixing clamp (right)

The mirror dummy can be removed by carefully sliding it through the fixing clamps after they have been loose (see Figure 64). The clamps can be loose from below the base plate, as shown in Figure 64. Do not remove the fixing clamps/screws!

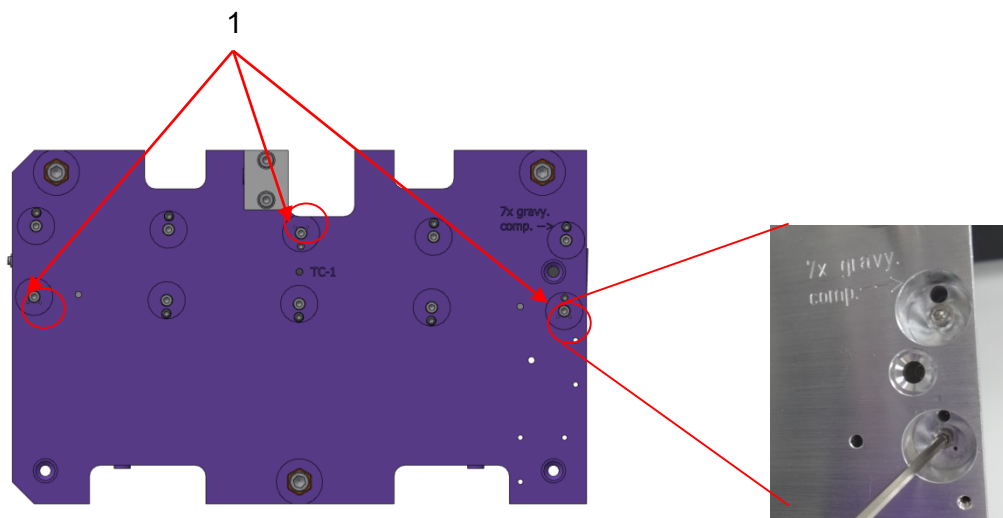


Figure 64: Fixing screws which have to be loosened

The mirror dummy can be replaced by the optical element as illustrated in Figure 62 with a piece of metal sheet.

- K. Carefully tighten the fixing screws at the fixing clamps (#1 in Figure 64) without twisting the clamp on the holder base plate.
- L. Proceed by assembling the complete mirrors holder by following the steps A -> I in reverse order.

Alignment of the M5 mirrors

The alignment of the M5 mirrors was performed with a laser tracker according to the BNL standards. The alignment is limited by the precision of the laser tracker.

8.4 Appendix

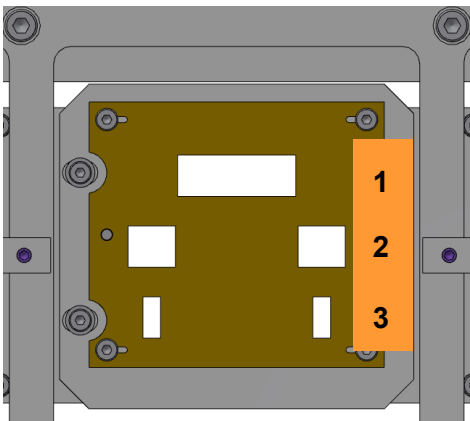
8.4.1 Overview of technical specifications

Table 14: Overview of technical specifications (M5)

General	
Mirror unit	Mirror mechanics (tripod) included in the sample chamber
Beam height	1419.25 mm
Number of mirrors	2 collimating mirrors (M5A, M5B)
Substrate dimensions (L x W x H) Active optical surface (L x W)	260 mm x 50 mm x 72.5 mm 250 mm x 10 mm
Grazing angle	88.5°
M5 mirror rotation 2-theta (M5R2T)	
Total range	365°
Motor	Phytron ZSS57.200.2,5-GPL52/50
Step resolution (full step)	3.6×10^{-3} °/step
Encoder	Renishaw Resolute Encoder scale: RESA30USA350B Readhead: RA26BVA350B15V
Encoder resolution	5.36441×10^{-6} °/count
Limit switches	Huber LS01
FWD LSW-M5R2T	301710645 counts
REV LSW-M5R2T	- 4026308555 counts
M5 2Theta circle Offset from origin point	x = 29 µm z = 80 µm
Circle of minimum confusion	Max. radius = 1.86 µm RMS radius = 1.22 µm
Tangential wobble of the system	~8.7 µrad
M5 mirror pitch	
Range	± 15 mrad
Resolution	1 µrad

Repeatability of mirror rotation upon pitch variation (CLC) - FAT	Pitch: 0.42 μ rad Roll: 0.2 μ rad
Stability	1 μ rad
M5 mirror roll	
Range	± 15 mrad
Resolution	2 μ rad
Repeatability	10 μ rad
Stability	2 μ rad
M5 mirror yaw	
Range	± 20 mrad
Resolution	100 μ rad
Repeatability	500 μ rad
Stability	500 μ rad
M5 mirror lateral translation	
Range	± 5 mm
Resolution	0.1 μ m
Repeatability	0.5 μ m
Stability	1 μ m
M5 mirror longitudinal translation	
Range	± 5 mm
Resolution	0.1 μ m
Repeatability	0.5 μ m
Stability	1 μ m
Actuator drives for all six tripod axes (M5T1 -> M5T6)	
Motor	PI PiezoMike N-470.42U
Offset M5T1 (middle position)	101019622 counts
Offset M5T2 (middle position)	87469900 counts
Offset M5T3 (middle position)	139007368 counts
Offset M5T4 (middle position)	126941347 counts
Offset M5T5 (middle position)	60067971 counts
Offset M5T6 (middle position)	47245379 counts
Travel range	26 mm
Typical step size	20 nm (max. 30 nm)
Encoder	Renishaw Resolute

	Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B50V
Encoder resolution	0.005 μm /count
Limit switches	2 x Huber LS01
FWD LSW-M5T1	103111142 counts
REV LSW-M5T1	98329867 counts
FWD LSW-M5T2	89759054 counts
REV LSW-M5T2	85897896 counts
FWD LSW-M5T3	141295076 counts
REV LSW-M5T3	136728375 counts
FWD LSW-M5T4	129236360 counts
REV LSW-M5T4	124391070 counts
FWD LSW-M5T5	62073690 counts
REV LSW-M5T5	57963734 counts
FWD LSW-M5T6	49482054 counts
REV LSW-M5T6	44979978 counts
Pivot translation X (PTX)	
Range	± 5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Pivot translation Z (PTZ)	
Range	± 5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S

	Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
M5 masks (M5M)	
Total translation range	0 – 55.2 mm
Motor	Phytron VSS32.200.1,2-Pt100-BESTEC
Spindle pitch	1 mm
Step resolution (full step)	5 μm /step
Microstepping 1/64	0.078 μm /step
Limit switches	2 x Huber LS01
REV LSW-M5M	0
FWD LSW-M5M	707418 steps
	Offset (1) = 87880 steps Offset (2) = 241410 steps Offset (3) = 395630 steps (middle positions of the apertures)
Motor retractable photodiode (MPD3)	
Motor	PI PiezoMike N-470.42U
Travel range	26 mm
Typical step size	20 nm (max. 30 nm)
Limit switches	2 x Huber LS01
Diagnostic slits	
Alignment accuracy (factory-aligned)	3 μm

8.4.2 Location of in-vacuum components and sensors

Table 15: Location of in-vacuum components and sensors

In air (feedthrough)	In vacuum	Wiring scheme
TC1	Thermocouple at the base plate of the upper mirror	443 - 7M - 346a
TC2	Thermocouple at the base plate of the lower mirror	443 - 7M - 346a
TC3	Thermocouple at 2Theta motor holder plate	443 - 7M - 346b
TC4	Thermocouple at one of the 2Theta trailing chain support plates	443 - 7M - 346b
TC5	Thermocouple at the inboard side of 2Theta support pillar	443 - 7M - 346b
TC6	Thermocouple at the outboard side of 2Theta support pillar	443 - 7M - 346b
CSW-M5T1	Collision switch at the M5 mirror holder assembly	443 - 7M - 348a

8.4.3 List of supplied wiring schemes

Table 9 contains a list of the provided wiring schemes. For detailed schemes, please refer to Customer Documentation folder.

Table 16: List of wiring schemes for the complete spectrometer arm system

Scheme number	Description
443 - 7M - 500a	Motion Axes - Overview
443 - 7M - 500b	Motion Axes - Overview
443 - 7M - 560a	PTX - Motor + Limit switches
443 - 7M - 562a	PTX - Kill switches
443 - 7M - 571a	PTX - Encoder
443 - 7M - 560b	PTZ - Motor + Limit switches
443 - 7M - 562b	PTZ - Kill switches
443 - 7M - 571b	PTZ - Encoder
443 - 7M - 560c	M5R2T - Motor + Limit switches
443 - 7M - 571c	M5R2T - Encoder

443 - 7M - 562a	M5M - Motor + Limit switches
443 - 7M - 583a	M5M - Motor + Limit switches
443 - 7M - 561a	M5T1 - Motor + Limit switches
443 - 7M - 586a	M5T1 - Limit switches
443 - 7M - 561b	M5T2 - Motor + Limit switches
443 - 7M - 586b	M5T2 - Limit switches
443 - 7M - 561c	M5T3 - Motor + Limit switches
443 - 7M - 586c	M5T3 - Limit switches
443 - 7M - 561d	M5T4 - Motor + Limit switches
443 - 7M - 586d	M5T4 - Limit switches
443 - 7M - 561e	M5T5 - Motor + Limit switches
443 - 7M - 586e	M5T5 - Limit switches
443 - 7M - 561f	M5T6 - Motor + Limit switches
443 - 7M - 586f	M5T6 - Limit switches
443 - 7M - 571d	M5T1 - Encoder
443 - 7M - 571e	M5T2 - Encoder
443 - 7M - 571f	M5T3 - Encoder
443 - 7M - 571g	M5T4 - Encoder
443 - 7M - 571h	M5T5 - Encoder
443 - 7M - 571i	M5T6 - Encoder
443 - 7M - 561g	PD3 - Motor + Limit switches
443 - 7M - 586g	MPD3 - Limit switches
443 - 7M - 345a	Air Bearing M5R2T
443 - 7M - 346a	Thermocouples
443 - 7M - 346b	Thermocouples
443 - 7M - 348a	Collision Switch M5<>SmarPod
443 - 7M - 585a	Collision Switch M5<>SmarPod

443 - 7M - 349a	Photodiodes
443 - 7M - 401a	Bakeout SC
443 - 7M - 401b	Bakeout SC
443 - 7M - 300	PLC Box 1 Layout
443 - 7M - 347e	Air Bearing
443 - 7M - 346c	PT100 Motor M5 Masks

8.4.4 List of supplied assembly drawings

Table 10 contains a list of the provided assembly drawings. For detailed drawings, please refer to Customer Documentation folder.

Table 17: List of assembly drawings for the M5 mirror system

Drawing number	Description
N4430C0002	M5 mirror system mechanics
N4434C0000	Granit support structure
N4432C0002	Cross table
N4432C0010	Manipulator
N4433C0000	Manipulator
N4433C0031	Photodiode assembly
N4433C0065	Diagnostic slit with retractable photodiode
N4433C0058	Diagnostic slit holder
N4433C0066	Diagnostic slit holder
N4433C0001	Rotation platform
N4434C0002	2Theta radial guidance
N4433C0007	Manipulator
N4433C0012	Mask assembly
N4433C0019	Mask mechanism
N4433C0040	Mask actuator
N4433C0032	Mirrors holder assembly
N4433C0048	Mirror holder-top mirror
N4433C0041	Mirror holder-bottom mirror

N4433C0049	Mirror holder- bottom mirror
------------	------------------------------

8.4.5 List of recommended spare parts

If replacement of system components is necessary, please contact Bestec GmbH. Please note that self-repair of system components without Bestec GmbH consultation can lead to loss of warranty claim.

Table 18: List of recommended spare parts

Spare part	Manufacturer	Part number
Encoder readhead	Renishaw	RA26BVA350B15V
		RL32BVT005B50V
		RL32BAT050B30A
Limit switch	Schaltbau	S880
Kill switch	Huber	LS01

8.4.6 List of supplied subcontractor documents

Table 13 contains a list of provided subcontractor documents. For detailed manuals please refer to the provided Customer Documentation folder.

Table 19: List of provided subcontractors documents

Part	Manufacturer	File name
Heating cable	BriskHeat	Heating cable.pdf
Stepper motor	Phytron	ZSS.pdf VSS.pdf
Piezo motor	PI	PiezoMike linear actuator.pdf
Limit / kill switch	Schaltbau	Snap action switches.pdf
	Huber	Limit switches.pdf
Encoder	Renishaw	Resolute L-9517-9530-01-K Resolute L-9517-9399-03-A Resolute M-9553-9735-02-B Resolute M-9553-9433-06-C

9 3AA - 3 m arm and optics tank assembly

9.1 Safety information

9.1.1 Used symbols

The following symbols found in the present document are used to identify hazards. An additional purpose of them is to draw the user's attention to safety instructions or other useful information.

	Information on preventing any kind of physical injury.
	Information on preventing extensive equipment and environmental damage.
	Information on correct handling or use. Disregard can lead to malfunctions or equipment damage.
	General warning
	Pinch point hazard warning
	High voltage hazard warning
	Hot surface hazard warning
	Watch your head hazard warning
	Pinch point/entanglement hazard warning
	Gloves required
	Eye protection required
	Hard hat required

9.1.2 Safety instructions

General

In normal operation, there is no danger when working with the PGM system. The operator of the PGM should however be aware of potential dangers that can occur by improper handling of the system and that can cause personnel injury or damage to the equipment.

When the system is operated without fulfilling the recommendations described in this document and the legal health and safety regulation, safety of personnel and equipment can be affected. Potential dangers and measures to avoid them are described in this section.

Electrical safety

In normal operation no hazards can occur associated with electrical voltage and current used by the equipment. Following accidents may happen if the electrical installation of the PGM unit is not properly operated: electric shock and burns, fire and electric arcing.

⚠ DANGER Electrical voltage



1. Voltage and high voltage that may cause injury or death are used in the setup. Service or maintaining work on the electrical installation must only be performed by qualified trained personnel for this type of work.
2. The statutory regulations regarding electrical installations must be respected.
3. The entire electrical equipment supplied as part of the system is provided with a protective ground. Grounding cables must not be removed or manipulated. It is very important that the vacuum system is properly grounded at all times.
4. Fuses may not be bridged, electrical connections must not be separated or otherwise manipulated, protective covers and other electrical insulating material may not be removed or modified.
5. Do not attempt to administer first aid to someone who has suffered electric shock until the source of the shock has been isolated.

Safe handling of gases

In normal operation no dangers through use of gases may appear as long as the recommended precautions are followed. In case of gas handling contrary to the instructions, unconsciousness and associated injuries may occur.

⚠ DANGER Gases



1. If pure nitrogen is used for venting the vacuum system one has to be aware that its inhalation can induce an oxygen deficiency after only few breaths, which may lead to unconsciousness.
2. Areas in which the oxygen concentration is less than 18% must be abandoned immediately.
3. For venting the vacuum system it is recommended to use dry air with normal oxygen concentration instead of pure nitrogen.

Safe handling of mechanical components

In normal operation only few mechanical works on the PGM unit is required. For service and maintenance extensive mechanical work may be necessary. Only qualified personnel should attempt to dismantle, modify or repair equipment. The following hazards may arise when dealing with mechanical components: injury when transporting heavy loads (PGM unit, chamber lid, grating cradle), danger of crushing by moving parts (motors at the motion axes), injury by protruding parts at the PGM unit.

⚠ DANGER

Heavy loads



1. Only qualified personnel should lift and carry heavy loads.
2. Use a hard hat when working with heavy loads.
3. Lifting eyes are available at the individual components of the system and they must be used for transportation.
4. Parts of the equipment that have to be lifted may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

⚠ DANGER

Moving parts



1. At the PGM unit there are a number of moving parts, which are powered by electrical motors. Make sure that your way of working does not include any safety risk. Do not leave long hair untied, do not wear loose clothing or jewelry as this may involve the danger of becoming caught in the moving mechanisms.
2. Do not remove or modify covers of the moving parts.

⚠ CAUTION

Components in UHV



1. In-UHV components can only be touched with powder/lint-free gloves.
2. Tools use for work on UHV components must be clean and free of grease.
3. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

⚠ CAUTION

Optical elements



1. Use protective gloves and clothing when working with optical elements.
2. Be careful not to drop anything on/touch the optical surface.
3. A suitable cover for the optical surface is recommended.

Safe handling of hot surfaces

When baking-out the system or parts of it components become hot ($>100^{\circ}\text{C}$) and can cause burns when touched.

⚠ DANGER Hot surfaces	
 Two circular icons: the top one is yellow with a black border and three wavy lines representing heat; the bottom one is blue with a white border and shows a white hand wearing a protective glove.	1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system. 2. Wear protective gloves when touching the system while bakeout and cooling down of the system.

Safe handling of vacuum systems

By appropriate use of and compliance with the safety information during normal operation, no dangers occur from the vacuum system of the PGM. The following hazards may arise when dealing with vacuum systems: injury from “flying” pieces caused by exploding or imploding parts of the vacuum system; if the vacuum chamber is over pressurized, viewports may shatter and cause person injury. Use viewport covers and wear eye protection glasses.

⚠ DANGER Vacuum system	
 Two circular icons: the top one is yellow with a black border and a black exclamation mark; the bottom one is blue with a white border and shows a white face wearing safety goggles.	1. The walls of the vacuum system can withstand a pressure difference from outside to inside of at least 1 bar, from inside to outside only a pressure difference of about 0.1 bar. An increase in internal pressure of 1.1 bar can lead to the damage of the view ports and must therefore be avoided. Never exceed an overpressure of 0.1 bar inside the vacuum system. 2. The viewports destruction of a chamber under vacuum leads to implosion. There is a risk of injury from flying glass shards. Use viewport and eye protection. 3. Areas in which the oxygen concentration is less than 18% must be immediately abandoned.

9.1.3 Modifications and service

Safe operation, reliability and performance of the equipment may be impaired if assembly operations, extensions, re-adjustments, modifications or repairs are not carried out in accordance with the instructions provided in this document and with any other instructions issued by the manufacturer. If you wish to modify the equipment please contact Bestec GmbH for future advice.

9.2 Design overview of the 3AA system

Figure 65 illustrates the complete 3AA system: the optics tank, its adjustable support structure and the 3AA 2Theta guidance and drive.

- | | |
|--------------------------------------|--------------------------------|
| 1 Support structure of cable harness | 4 Toothed belt |
| 2 Optics tank | 5 3AA 2Theta motor |
| 3 3AA 2Theta guidance | 6 Adjustable support structure |

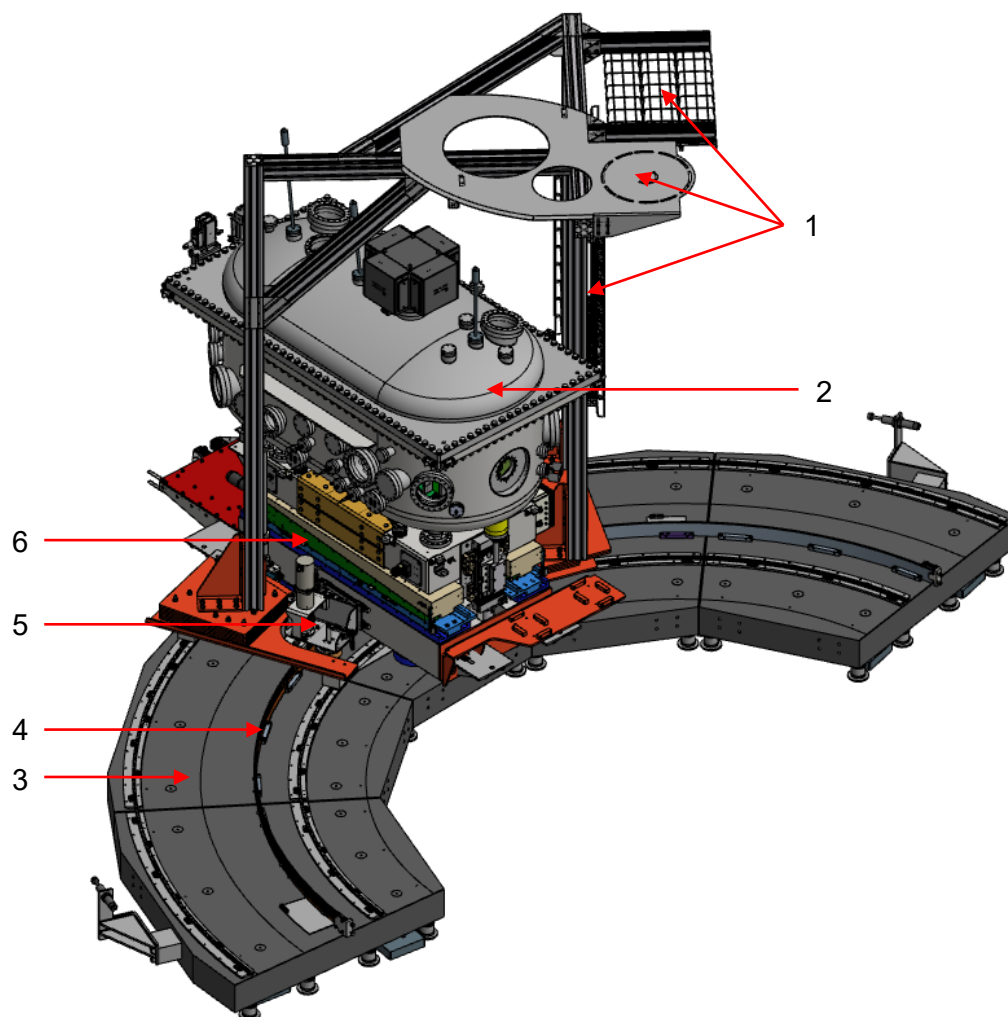


Figure 65: Design overview of the 3AA system

The 3AA 2Theta support consists of five granite base plates fixed to the laboratory floor. They are resting on wedge levelers and WASI feet.

The adjustable support structure consists of a motorized translation slide for longitudinal adjustment of the optics tank and two motorized wedge levelers for the height adjustment of the optics tank.

9.3 Design overview of the optics tank

Figure 66 and Figure 67 show the design overview of the optics tank.

- | | |
|---------------------------------------|---|
| 1 Wobble sticks for in-UHV adjustment | 11 Wobble stick (M6 yaw) |
| 2 Grating mask actuator | 12 Corner valve |
| 3 Ion getter pump | 13 Collision detection box |
| 4 Pirani gauge | 14 M6 rotation actuator |
| 5 Cold cathode | 15 Connector between camber and upper granite plate |
| 6 Gratings translation actuator (GT) | 16 Encoder feedthrough (M6TZ) |
| 7 Interpolator (DSI-GP) | 17 Thermocouple type K feedthrough (TC7-M6) |
| 8 Motorized wedge leveler (IVY) | 18 Beam entrance flange |
| 9 Encoder feedthrough (GT) | 19 Motor feedthrough (M6TZ) |
| 10 Encoder feedthrough (GP) | |

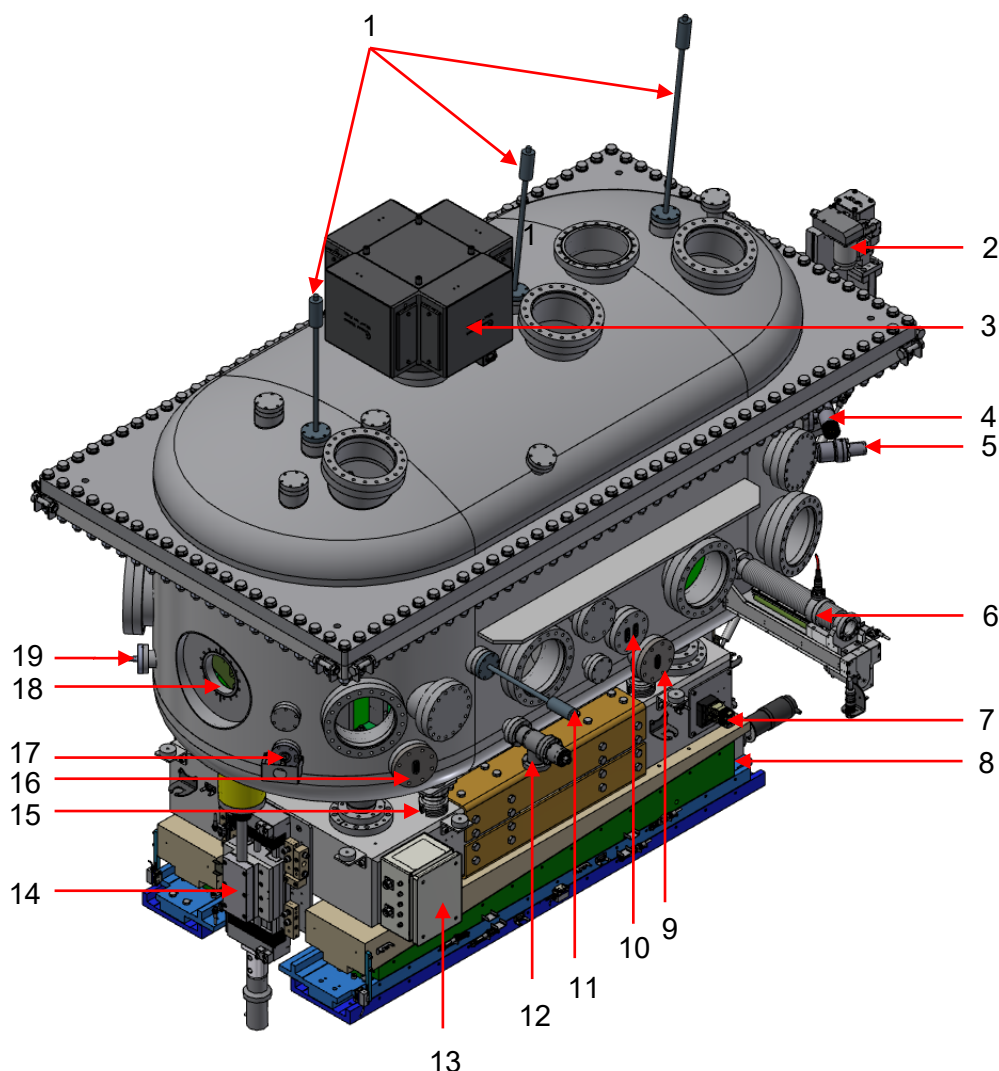


Figure 66: Design overview of the optics tank (inboard)

- | | |
|---|--|
| 1 Encoder feedthrough (M6P) | 8 Thermocouple type K feedthrough (TC8-M7, TC9-GR) |
| 2 Limit switches feedthrough (M6TZ) | 9 Motorized wedge leveler (OVY) |
| 3 Interpolator (DSI-M6P) | 10 Gratings rotation actuator (GP) |
| 4 Motor feedthrough (M6M) | 11 Interpolator (DSI-M7P) |
| 5 Pt100 feedthrough (M6M) | 12 Support for grating mask assembly |
| 6 Collision switch feedthrough (CSW-GP) | 13 Gate valve with aperture |
| 7 Encoder feedthrough (M7P) | 14 Grating mask |

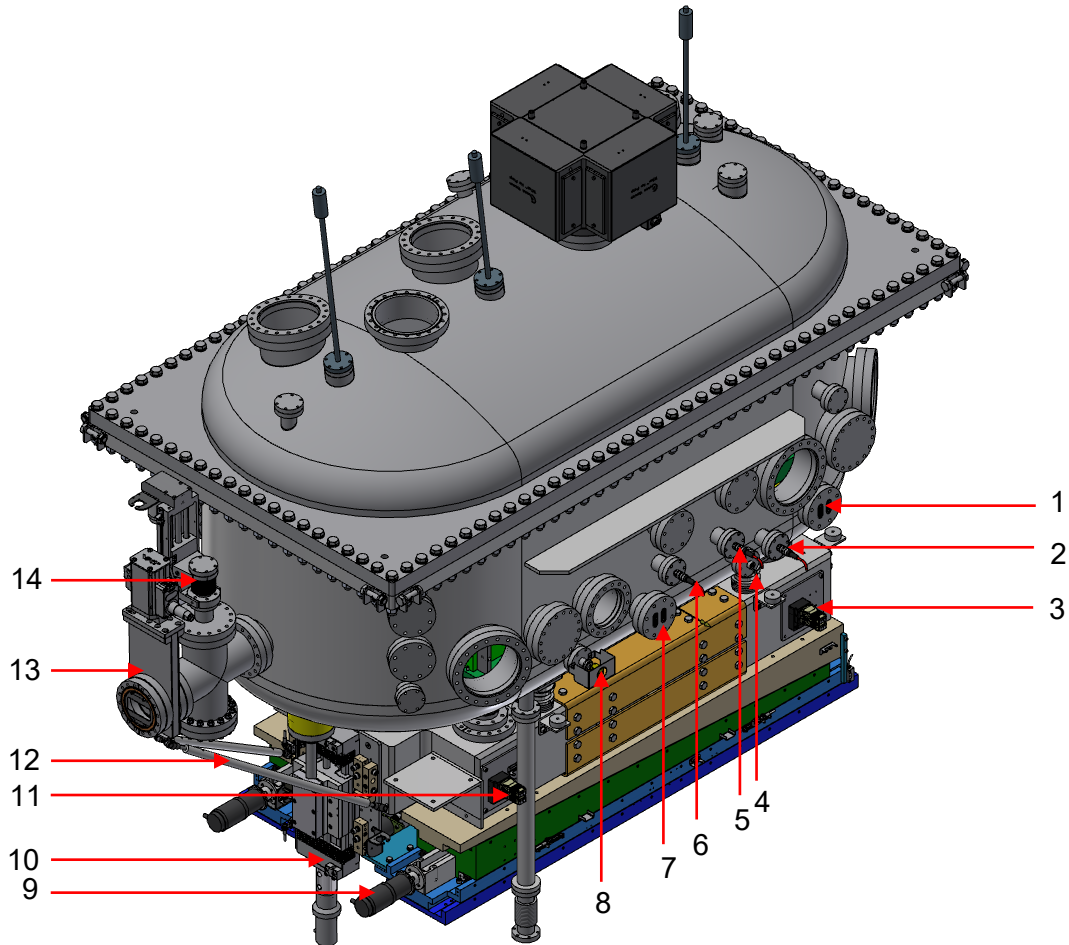


Figure 67: Design overview of the optics tank (outboard)

A set of vertical slits (M7 mask) is available downstream of the gratings, translatable in the vertical direction, for defining the vertical aperture of the beam leaving the grating. For a given selected slit, its center remains on the outgoing fixed optical axis, regardless of the pitch angle of the grating.

Figure 68 shows the gratings slits assembly.

1 Gate valve with aperture

2 Grating slits actuator

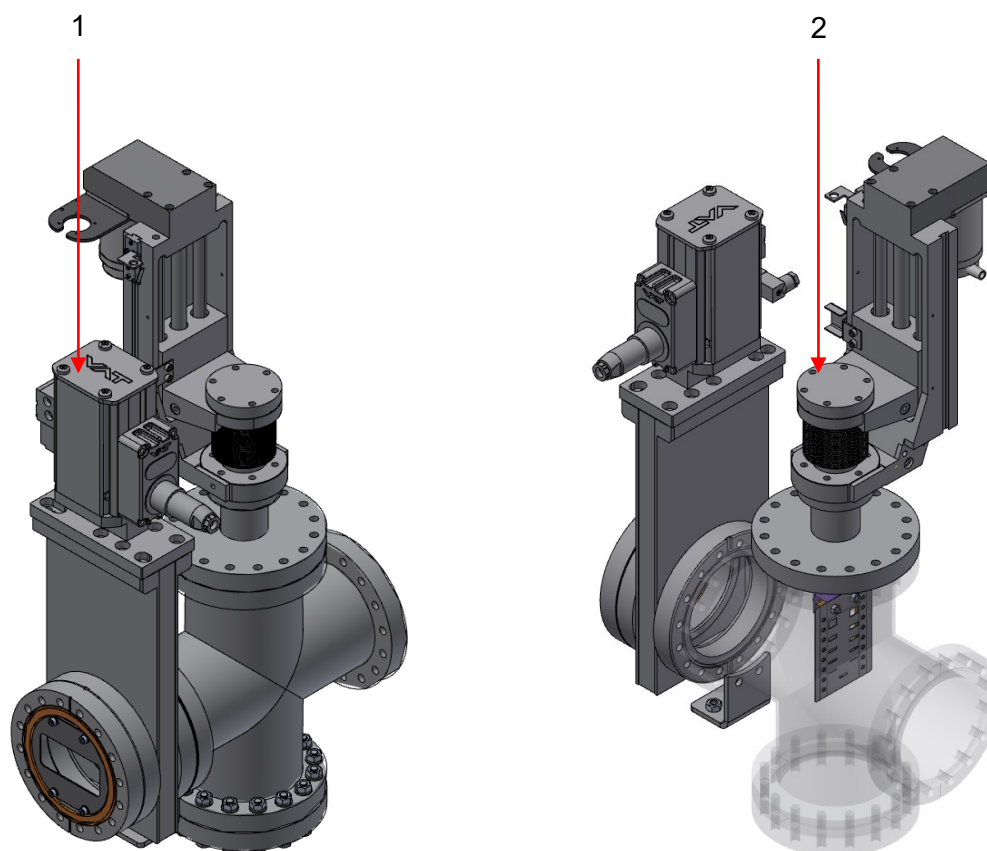
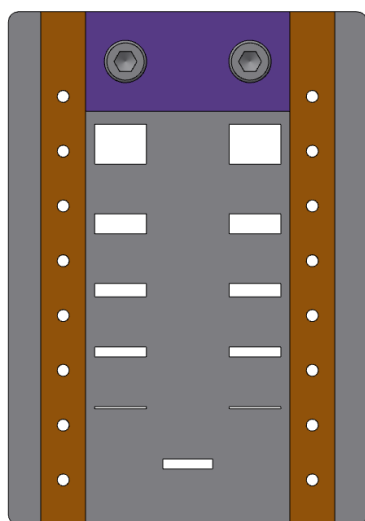


Figure 68: Gratings slits assembly (M7 mask), upstream view (left) and downstream view (right)



The positions of the grating
slits are given in Table 21.

Figure 69: M7 mirror mask assembly (downstream view)

For the 2-theta motion the optics tank unit moves on top of the granite base fixed to the floor. The granite block carrying the optics tank moves on two bended guide rails (#3, 6 in Figure 70), which are fixed onto the granite base. The friction wheel drive assembly is shown in Figure 70.

- | | |
|----------------------------------|---|
| 1 Motor/limit switches connector | 7 Timing belt wheel |
| 2 Limit/kill switches | 8 Tension pulley |
| 3 Guide rail | 9 Guide rail |
| 4 Motor with gear | 10 Timing belt |
| 5 Air bearing (6x) | 11 Granite base fixed to laboratory floor |
| 6 Counter wheel | |

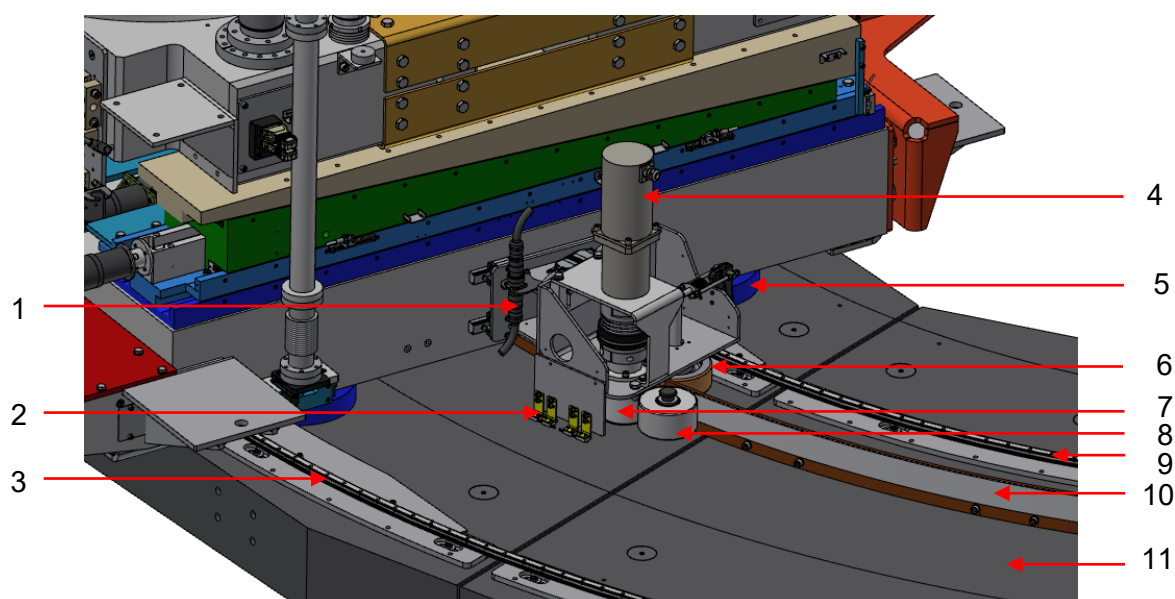


Figure 70: 2-theta rotation stage of the optics tank

The internal mechanics of the optics tank is illustrated in Figure 71.

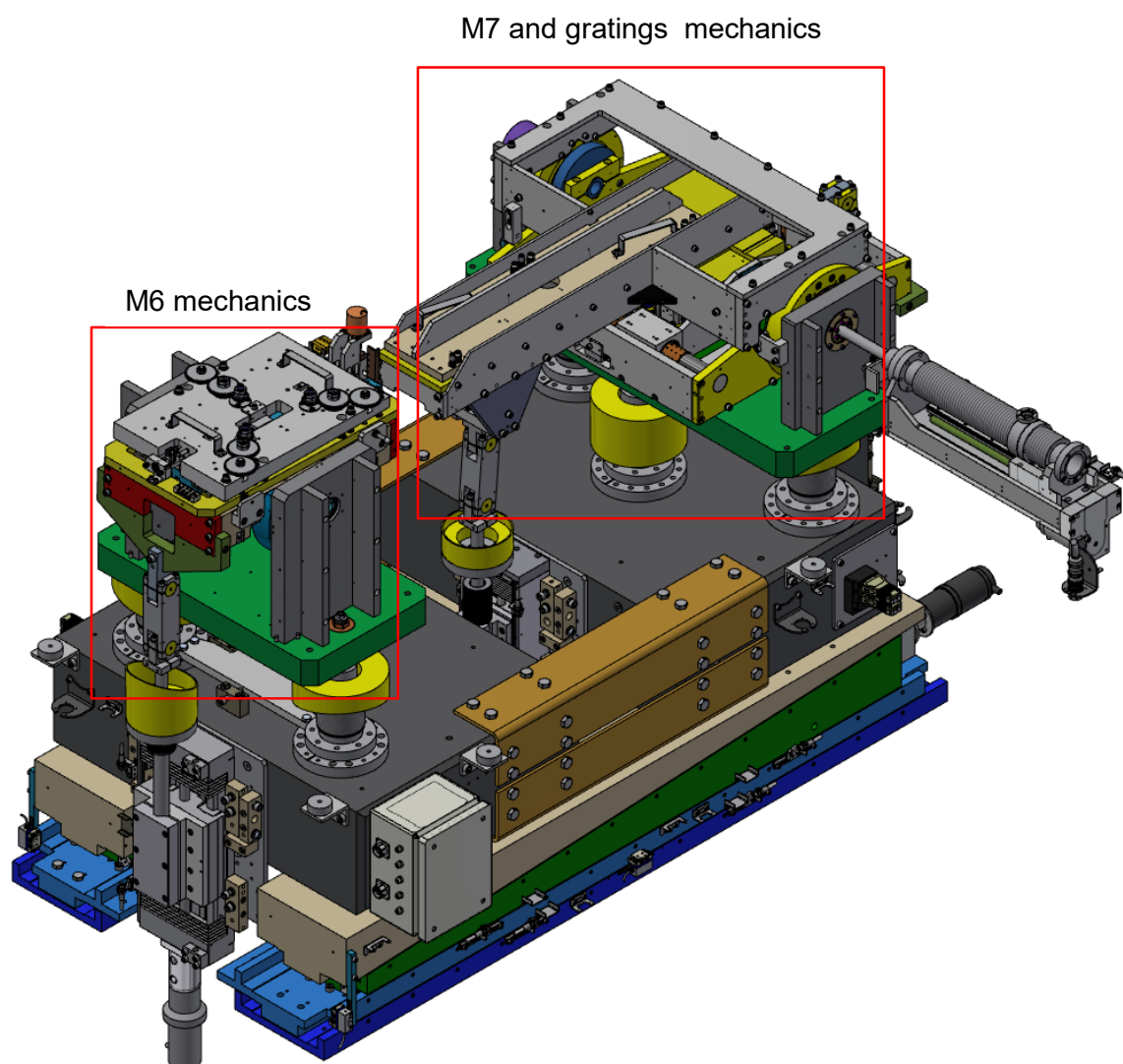


Figure 71: Internal mechanics of the optics tank

The internal mechanics of the optics tank is decoupled from the vacuum chamber by bellows. The two base plates supporting the M6 mechanics and the M7/grating mechanics (#2, 12 in Figure 72) are kinematically attached to their corresponding granite plates. Both granite plates are connected to each other (#10 in Figure 72) leading to the upper granite plate. The vacuum chamber rests on four feet directly connected to the upper granite plate (see #15 in Figure 66). The upper granite plate is kinematically attached to the adjustable support structure (see #6 in Figure 65).

- | | |
|--|---|
| 1 Sine-bar drive - grating pitch | 8 Translation slide (BTZ) |
| 2 M7 & gratings base plate | 9 Kinematic mount of M7/gratings granite plate |
| 3 Sine-bar drive - M7 pitch | 10 Connector between the two parts of the upper granite plate |
| 4 Kinematic mount of M7/gratings mechanics | 11 Kinematic mount of M6 mechanics |
| 5 Upper granite plate (reference) | 12 M6 base plate |
| 6 Kinematic mount of M6 granite plate | 13 Sine-bar driving M6 pitch |
| 7 Wedge leveler (OVY) | |

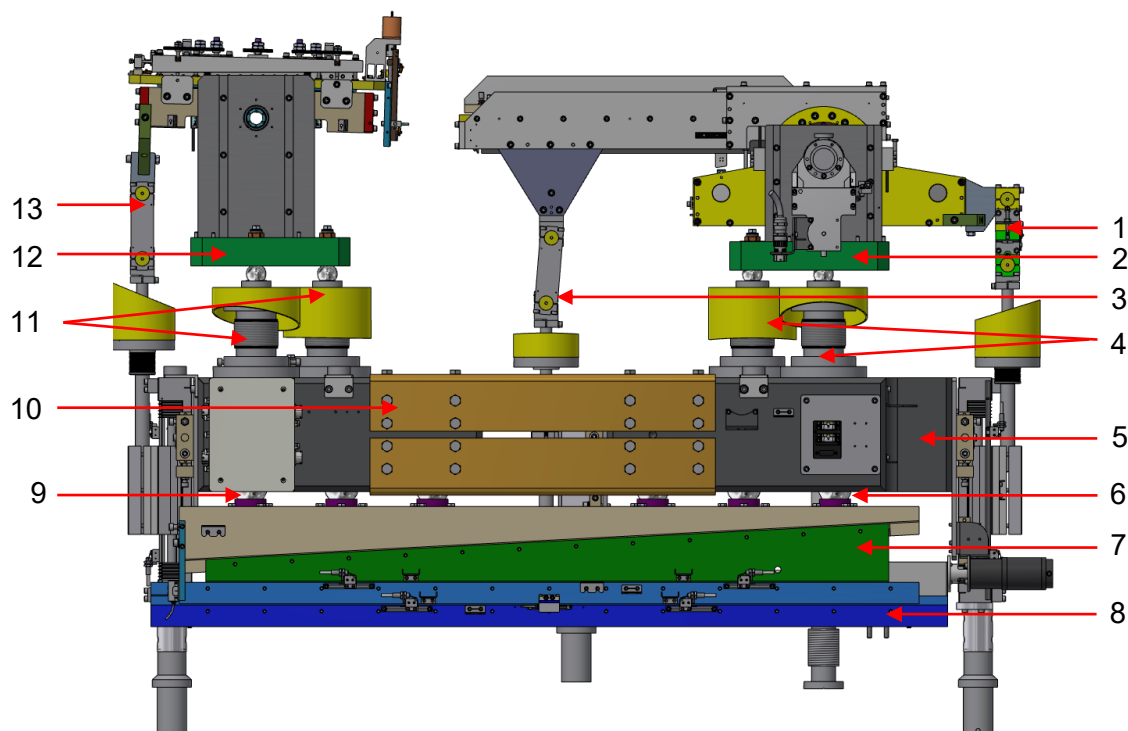


Figure 72: Internal mechanics of the optics tank (side view)

9.3.1 M6 mirror assembly: overview and installation

The M6 mirror is mounted in a downward deflecting geometry, collecting the scattered light from the sample and focusing it to a virtual focus located behind the gratings.

The mirror pitch is driven in air by a high-precision feedthrough equipped with limit switches and hard stops and transferred into UHV through an edge-welded bellow. The linear movement is transformed into rotation using a sine-bar mechanism. The mirror pitch is oriented at 1.5° relative to the horizontal plane. This is achieved by a wedge-shaped mirror holder base plate.

The mirror translation is realized in-UHV by a piezo motor. The mirror holder moves over two linear guideways with balls.

- | | |
|--|---------------------------------|
| 1 Wobble sticks inserts for UHV adjustment | 6 M6 mirror cradle |
| 2 Piezo motor for M6 translation | 7 Bearing block |
| 3 Handle | 8 M6 mechanics base plate |
| 4 Mirror holder base plate | 9 Encoder read head (rotation) |
| 5 Yaw plate | 10 Motorized vertical apertures |

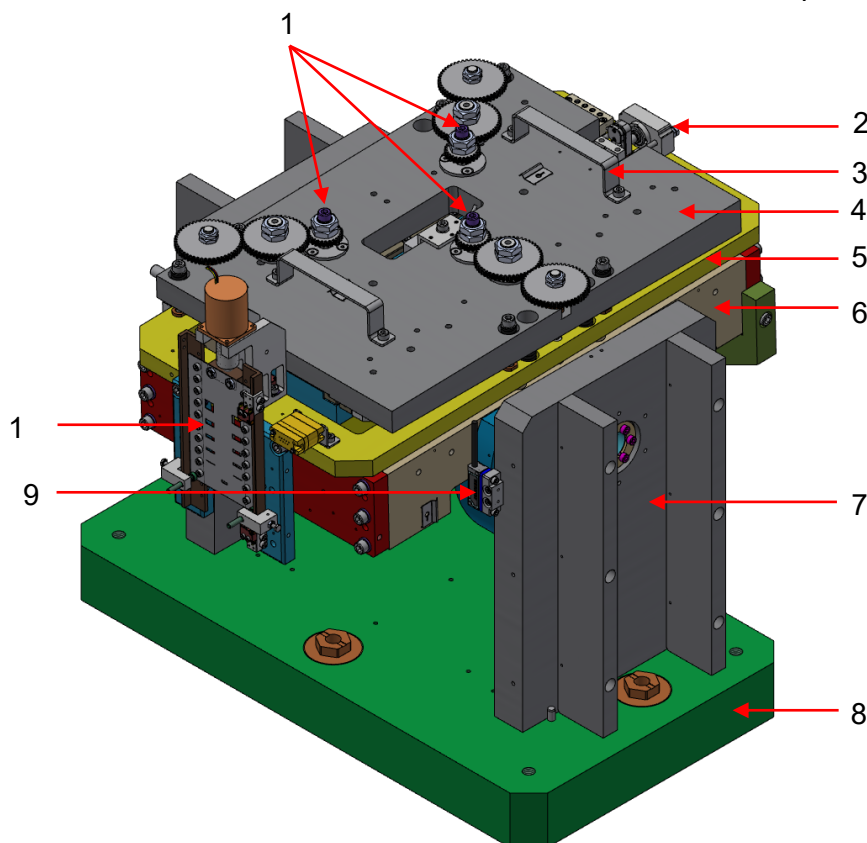


Figure 73: M6 mirror assembly

A set of vertical apertures are available downstream of M6 mirror (M6 mirror mask). They are translatable in the vertical direction and are used for defining the vertical aperture of the beam near the mirror. The apertures are attached to the mirror cradle in order to facilitate a coupled rotation with the mirror pitch. The mask can be moved 15 mm downward and 70 mm upward (center of the top aperture aligned vertically with respect to the optical axis).

- | | |
|-----------------|---------------------------|
| 1 Stepper motor | 3 Stepper motor connector |
| 2 Limit switch | 4 Bearing block |

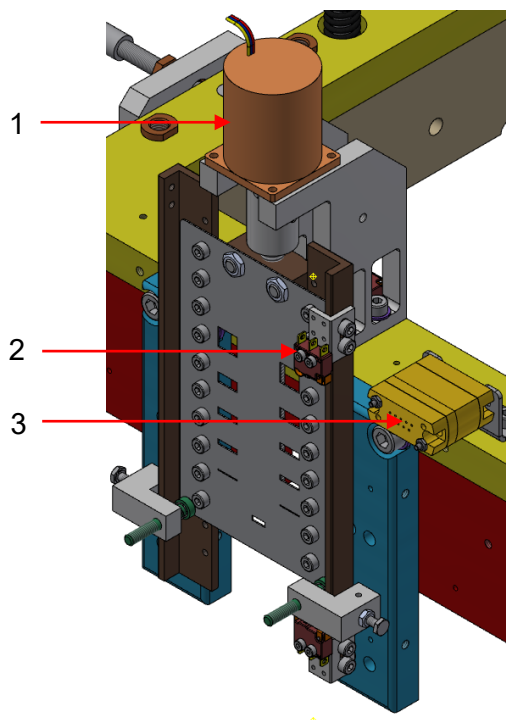
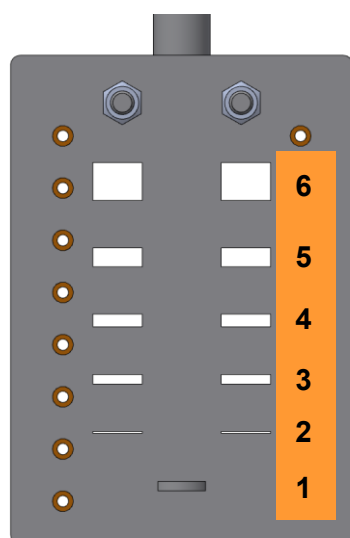


Figure 74: Vertical aperture assembly



The positions of the apertures are given in Table 20.

Figure 75: M6 mirror mask (upstream view)

The mirror holder assembly is kinematically attached to the yaw plate. The mirror holder can be adjusted in UHV by using a wobble stick fixed to the chamber lid.

A distance of 3.5 mm at one short side of the mirror holder plate and a distance of 10.5 mm at the other short side (measured between mirror holder base plate and the mirror cradle) define the nominal position of the mirror surface.

- 1 Encoder readhead (translation) 2 Yaw adjustment screw

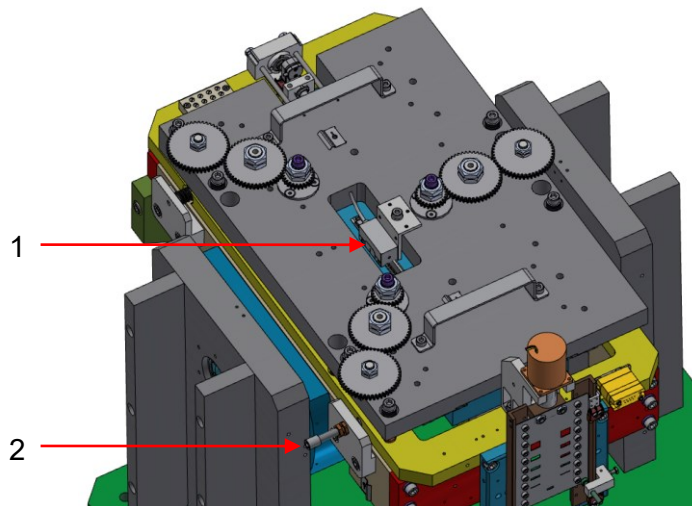


Figure 76: M6 mirror base plate

Integration and alignment of the M6 mirror

1. Chamber lid removal

⚠ DANGER Heavy load: Chamber lid



1. Only qualified personnel should lift/transport the chamber lid.
2. Use a hard hat while lift/transport the chamber lid.
3. Lifting eyes are available at the chamber lid and they must be used for lift/transportation.
4. The chamber lid may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

- Secure the lid of the chamber with the help of a crane by using the provided lifting eyes.
- Loosen and remove all fixing bolts of the lid to the vacuum chamber.
- Carefully lift up the lid with the crane; the ion getter pump can remain connected to the chamber lid during lifting/transportation procedure.
- Place the chamber lid into a safe place onto a clean surface.

2. M6 mirror holder removal

In order to remove the mirror holder please follow the steps listed below:

- Drive mirror and grating cradle in horizontal position.
- Disconnect the piezo motor cable from the luster terminal (# 2 in Figure 77).
- Disconnect the mirror translation encoder cable from its vacuum feedthrough.
- Disconnect the thermocouple wire from its contact point.
- Loosen and remove all six fixing screws (# 6 in Figure 77) of the mirror holder base plate to the yaw plate.
- Remove cable strain reliefs if necessary.
- Carefully remove the mirror holder straight upwards using the provided handles.

- Place the mirror holder on a safe surface with the dummy showing upwards; secure the mirror holder before proceeding with mirror dummy removal.

- | | |
|------------------------------------|---|
| 1 Piezo motor | 4 Mirror holder base plate |
| 2 Luster terminal | 5 Encoder readhead (mirror translation) |
| 3 Fixing screws to be removed (6x) | 6 Handle |

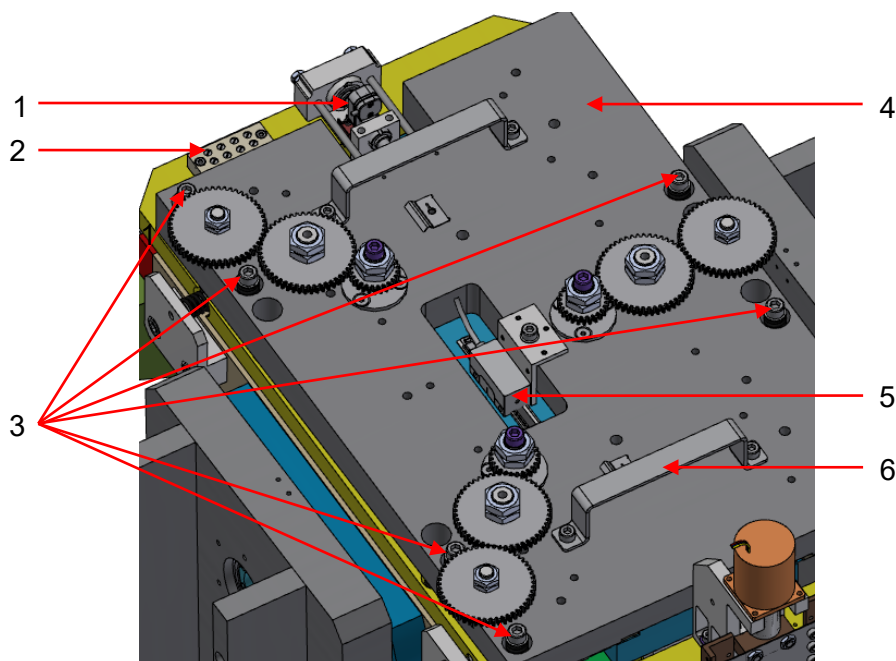


Figure 77: Top view on the M6 mirror assembly

3. M6 mirror dummy removal

Two grooves at the long side of the mirror enable fixing the mirror to its base plate (#4 in Figure 78). Six holding clamps (#3 in Figure 78) are used for this purpose, three of them fixed and the other three spring loaded.

- | | |
|--|----------------------------|
| 1 Mirror dummy with slot for auxiliary test mirror | 4 Mirror base plate |
| 2 Linear guide with balls | 5 Mirror holder base plate |
| 3 Clamp (3x fixed, 3x spring loaded) | |

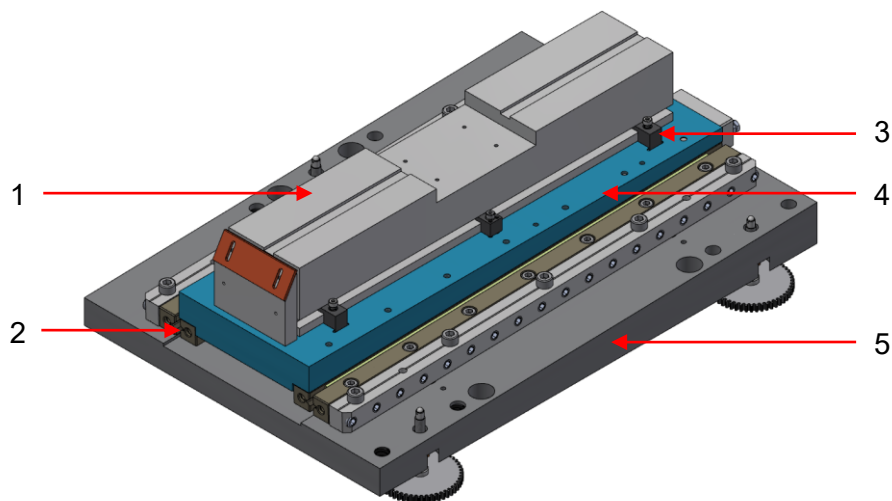


Figure 78: M6 mirror holder assembly

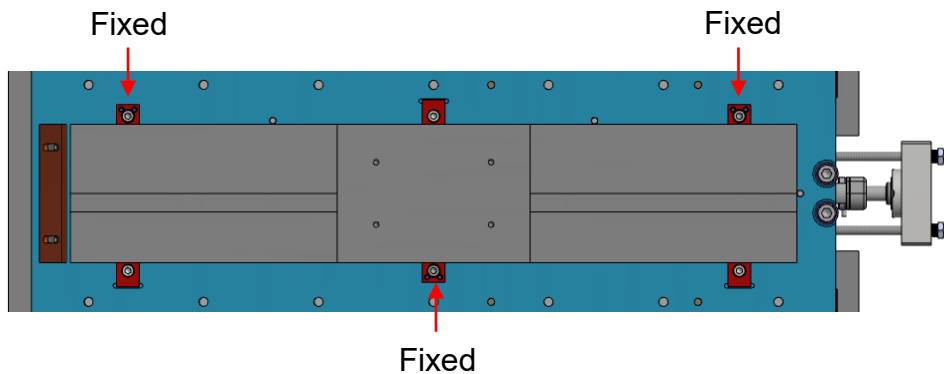


Figure 79: View on the M6 dummy surface

- Loosen and remove all six clamps fixing the mirror to the base plate.
- At this point the grating dummy is free and can be removed.

4. Integration of the M6 mirror

CAUTION

Optical elements: M6 mirror



1. Use protective gloves and clothing when working with mirror.
2. Be careful not to drop anything on/touch the optical surface of the mirror.
3. A suitable cover for the optical surface is recommended.

- Carefully place the mirror on its base plate; three dowel pins are available for the correct positioning of the mirror on the base plate.
- Carefully fix the holding clamps in the side grooves of the mirror substrate and fasten the corresponding screws.
- Re-insert the mirror holder assembly into the vacuum chamber and fix it to the yaw plate with the corresponding fixing screws, # 3 in Figure 77.
- Connect the piezo motor cable, the mirror translation encoder cable and the thermocouple wire to their corresponding position, see Figure 77.
- Dismount the mirror holder handles if necessary, # 6 in Figure 77.

5. Alignment of the M6 mirror

The alignment of the M6 mirror was performed with a laser tracker according to the BNL standards. The alignment is limited by the precision of the laser tracker.

9.3.2 M7 mirror and gratings mechanics: overview and installation

The M7/grating mechanics is shown in Figure 80.

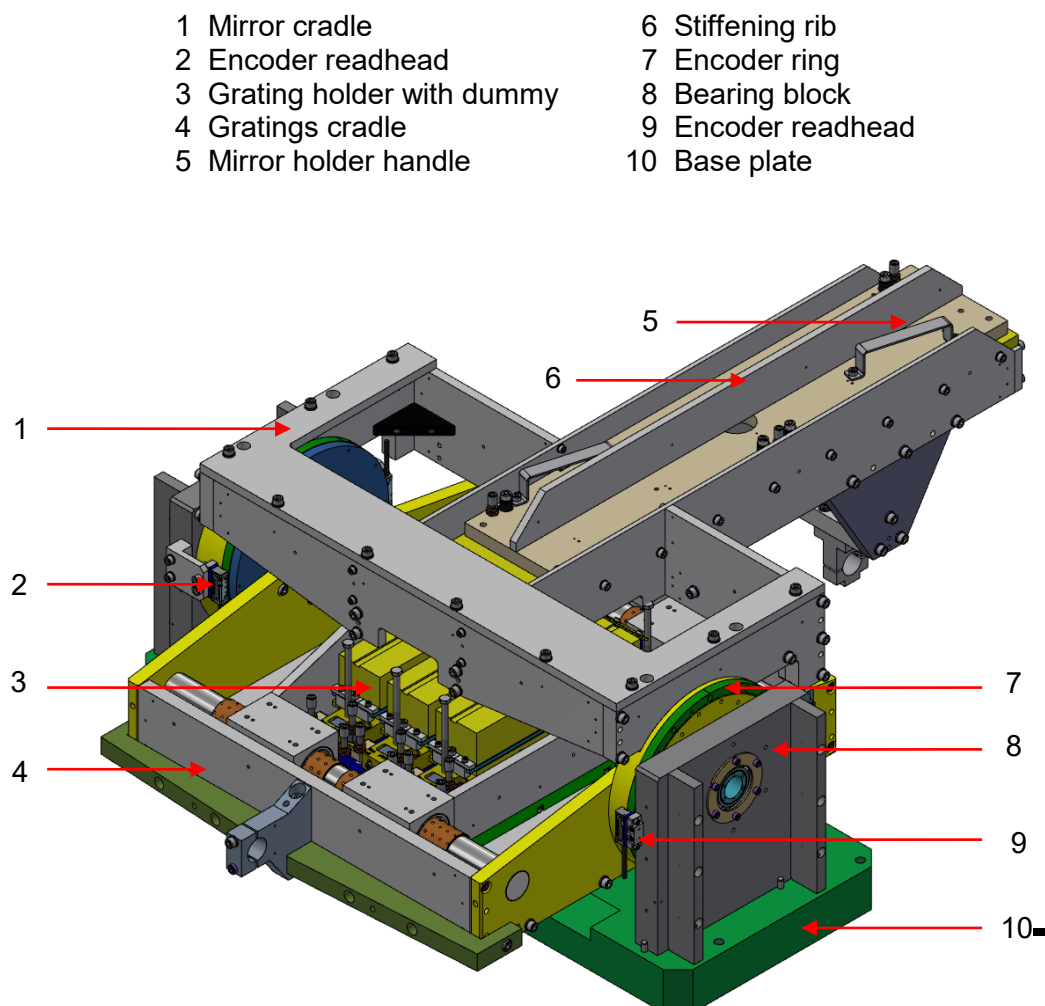


Figure 80: Internal mechanics (M7 and gratings) of the PGM unit

Integration and alignment of the gratings

In order to reach the gratings holders, the mirror cradle has to be detached from its bearing blocks and completely removed from the chamber

1. Mirror holder removal

The mirror is mounted on a holder which is kinematically attached to the mirror cradle. To install the optical mirror, its individual holder has to be detached from the mirror cradle.

After removing all four spring loaded fixing screws (#1, 2, 3, 6 in Figure 81) the mirror holder is free and can be lifted from the vacuum chamber by using the two available handles (#5 in Figure 81).

- | | |
|------------------------------|------------------------------|
| 1 Fixing screw to be removed | 5 Mirror holder handle |
| 2 Fixing screw to be removed | 6 Fixing screw to be removed |
| 3 Fixing screw to be removed | 7 Mirror cradle |
| 4 Stiffening rip | |

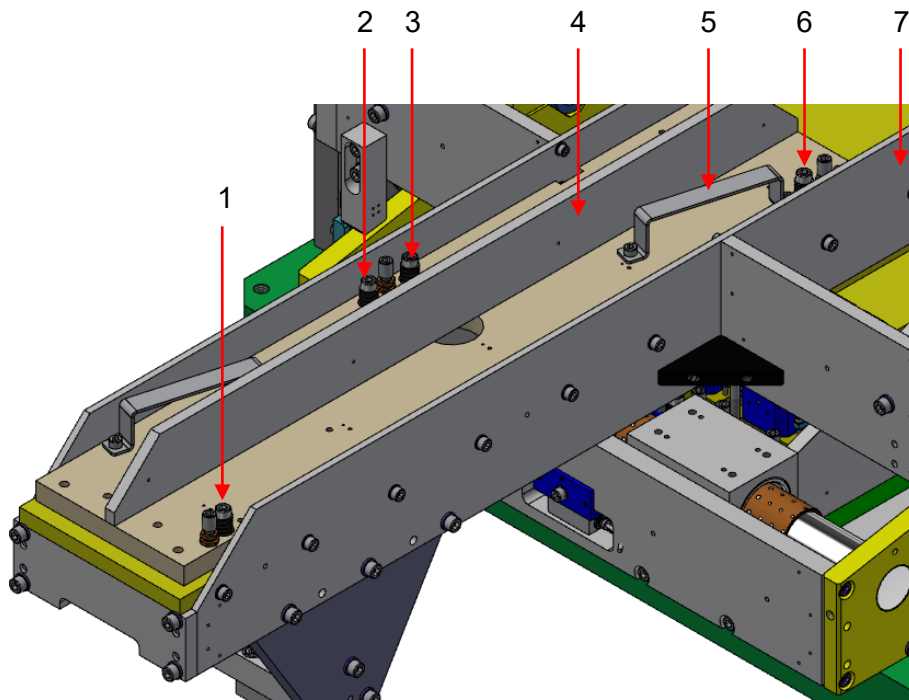


Figure 81: M7 mirror cradle

2. Mirror cradle removal

After mirror holder removal proceed following the next steps:

- A. Drive mirror and gratings cradles in horizontal position.
- B. Secure the mirror cradle with the help of a crane.
- C. Disconnect all cables (thermocouple, collision switch) from their corresponding vacuum feedthroughs; remove cable strain reliefs if necessary.
- D. Loosen and remove all screws (12x) indicated in Figure 82.

1, 2 Screws to be removed (12x)

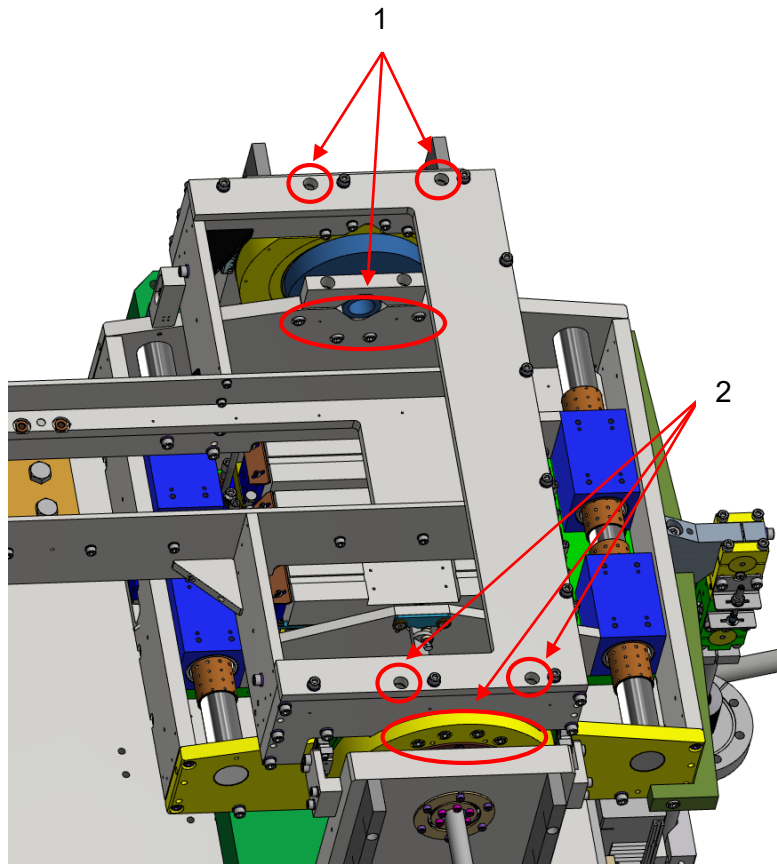


Figure 82: Top view of the mirror cradle

- E. Loosen and remove the two screws (#4 in Figure 84) at the connection rod of the sine-bar drive for the mirror rotation. Remove the clamp indicated in Figure 84, #1.

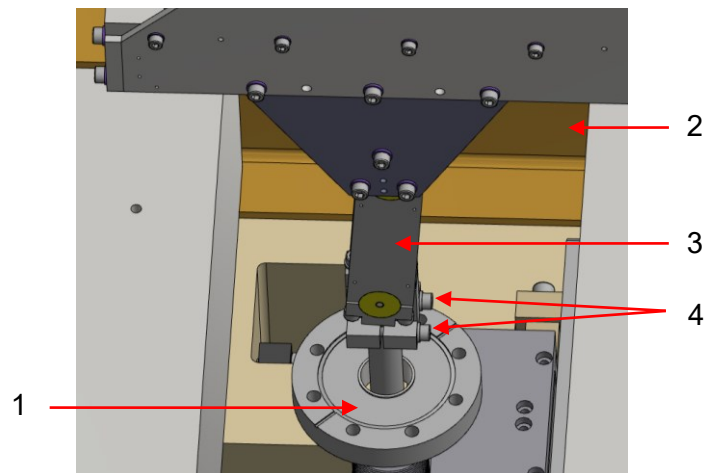


Figure 83: Sine-bar drive for mirror rotation

3. Grating holder removal

Each grating is mounted on an individual grating holder which is kinematically attached to the grating translation slide. For installing a particular grating its individual holder has to be detached from the translation slide.

After removing the two spring loaded fixing screws (# 3, 5 in Figure 84) the grating holder (G3 in Figure 84) is free and can be lifted of the vacuum chamber by using two long screws as lifting aid, as shown in Figure 84, # 1, 4.

- | | |
|--|------------------------------|
| 1 Screw used to lift the grating holder | 5 Fixing screw to be removed |
| 2 Grating dummy with slot for auxiliary mirror | 6 Grating translation slide |
| 3 Fixing screw to be removed | 7 Grating cradle |
| 4 Screw used to lift the grating holder | |

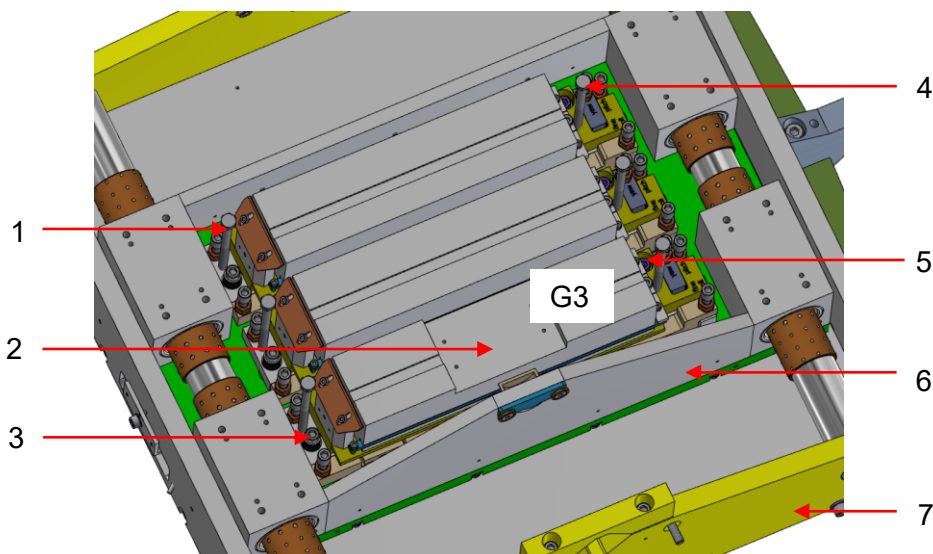


Figure 84: Grating cradle, top view

Figure 85 illustrates the grating holder assembly. Two grooves at the long side of the dummy enable fixing the dummy to the yaw plate. This is realized by using two clamping strips guided through the grooves and fixed at both ends to the yaw plate.

- A. Loose and remove the two fixing screws (#1 in Figure 85) and the two clamps (#4 in Figure 85).
- B. Carefully remove both clamping stripes (#2 in Figure 85).

- | | |
|------------------------------|--|
| 1 Fixing screw to be removed | 3 Grating dummy |
| 2 Clamping strip | 4 Fixing screw and clamp to be removed |

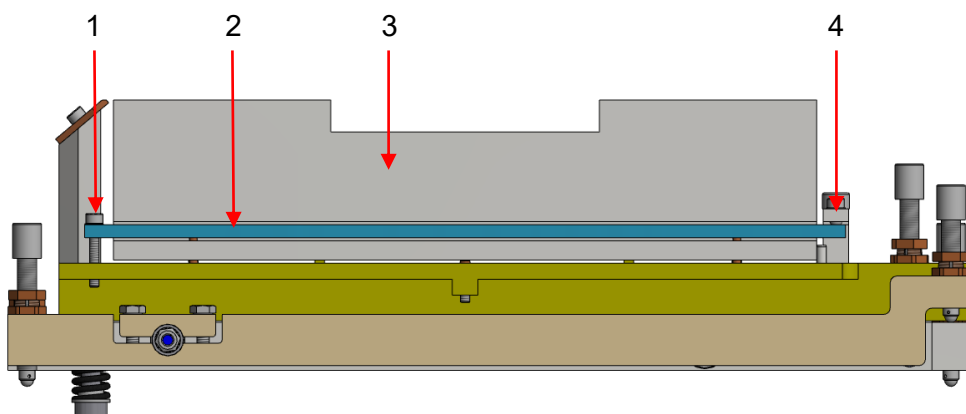


Figure 85: Grating holder assembly (side view)

4. Integration of the optical grating

CAUTION

Optical elements: grating



1. Use protective gloves and clothing when working with grating.
2. Be careful not to drop anything on/touch the optical surface of the grating.
3. A suitable cover for the optical surface is recommended.

- A. Place the grating on its yaw plate.
- B. Carefully slide the two clamping strips through the side grooves of the grating substrate and fasten them with the corresponding screws.
- C. Carefully re-insert the grating holder into the vacuum chamber.
- D. Tighten the spring loaded fixing screws (#3, 5 in Figure 84).
- E. Dismount the two screws used as lifting aid (#4 in Figure 84).

5. Alignment of the optical grating

The alignment of the optical grating was performed with a laser tracker according to the BNL standards. The alignment is limited by the precision of the laser tracker.

Integration and alignment of the mirror

6. Mirror dummy removal

Two grooves at the long side of the mirror enable fixing the mirror to its base plate. Ten holding clamps (#3 in Figure 78) are used for this purpose, three of them are fixed and the other seven are spring loaded.

- A. Loose all ten screws and carefully remove the corresponding clamps (#3, 4 in Figure 86). Be aware of the seven springs at the spring loaded clamps.

- | | |
|----------------------------|-----------------------------------|
| 1 Chin guard | 3 Spring loaded fixing clamp (7x) |
| 2 Mirror holder base plate | 4 Fixed clamps (3x) |

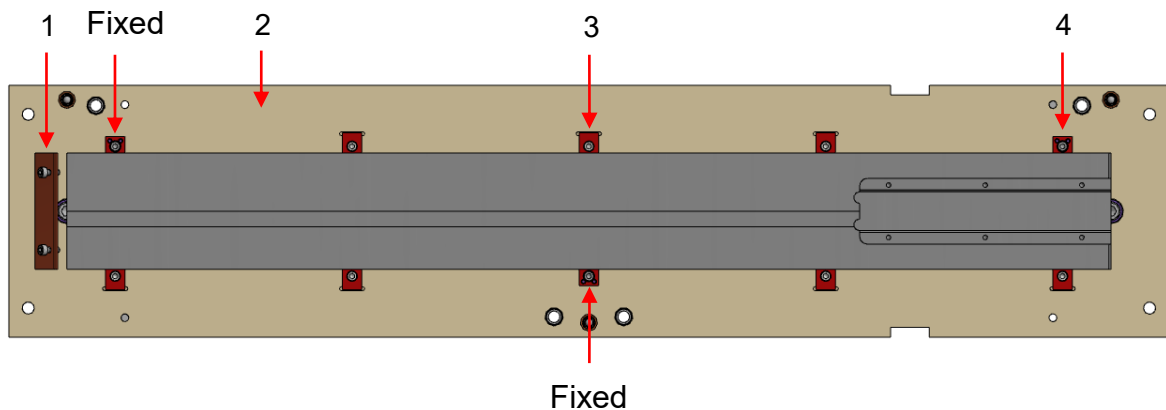


Figure 86: M7 mirror holder assembly

7. Integration of the plane mirror

CAUTION

Optical elements: M7 mirror



1. Use protective gloves and clothing when working with optical mirror.
2. Be careful not to drop anything on/touch the optical surface of the mirror.
3. A suitable cover for the optical surface is recommended.

- Carefully place the mirror on its base plate.
- Carefully fix the holding clamps in the side grooves of the mirror substrate and fasten the corresponding screws. Do not forget to use the corresponding springs at the spring loaded clamps, see Figure 86.
- Insert the mirror holder into the mirror cradle and tighten the fixing screws #1, 2, 3, 6 indicated in Figure 81.
- Re-insert the mirror cradle into the vacuum chamber and fix it with the corresponding screws, in the reverse order as described in the previous paragraph.
- Connect the sine-bar drive for mirror cradle rotation.
- Connect all cables to their corresponding positions.
- Dismount the mirror holder handles if necessary, # 5 in Figure 81.

8. Alignment of the plane mirror

The alignment of the M7 mirror was performed with a laser tracker according to the BNL standards. The alignment is limited by the precision of the laser tracker.

9.4 Maintenance

9.4.1 Bakeout procedure

 Hot surfaces: system bakeout	
 	3. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system. 4. Wear protective gloves when touching the system while bakeout and cooling down of the system.

A successful bakeout procedure requires following the next steps:

- A. Drive the mirror into its horizontal position.
- B. Connect a suitable pumping stand with dry membrane/scroll roughing pump and turbo-molecular pump via an all-metal right angle valve.
- C. Remove all actuators cables.
- D. Connect a RGA to a suitable place.
- E. Wrap edge welded bellows and viewports with thin Aluminum foil.
- F. Wrap the chamber in one or two layers of thicker Aluminum foil.
- G. Start pumping the system. The bakeout procedure can be started when the pressure is lower than 1×10^{-6} mbar.
- H. Switch on the ion getter pump (if present) before starting the bakeout and let it run for ca. 30 minutes. Switch off the pump after this period of time.
- I. Ramp the system temperature up in stages to ensure the heating process is under control. When a certain temperature level is reached, leave the system at that heating level until the entire system is stable and at constant temperature before continuing.
- J. When the system reaches the target bakeout internal temperature (max. 120°C without optical elements, max. 80°C with optical elements), keep the system at that temperature for a period of time of at least 48 hours, or until the total pressure falls below 2×10^{-7} mbar.
- K. Stop the heating of the ion getter pump at least 6 hours before starting to ramp down the system temperature.
- L. Switch on the ion getter pump.

M. Switch on the mass spectrometer.

N. Ramp the system temperature down in stages to ensure the cooling process is under control. At each stage leave the system at that heating level until the entire system is stable and at a uniform temperature distribution before proceeding to the next stage.

9.4.2 Sealing of the vacuum chamber main flange

⚠ DANGER

Heavy load: Chamber lid



9. Only qualified personnel should lift/transport the chamber lid.
10. Use a hard hat while lift/transport the chamber lid.
11. Lifting eyes are available at the chamber lid and they must be used for lift/transportation.
12. The chamber lid may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

⚠ CAUTION

Components in UHV: wire gasket



5. In-UHV components can only be touched with powder/lint-free gloves.
6. Tools used for work on UHV components must be clean and free of grease.
7. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

- | | |
|---|---|
| 1 Sealing wire at the long side of the chamber | 4 Vacuum chamber main flange |
| 2 Turnbuckle with hole | 5 Intersection point of the sealing wires |
| 3 Sealing wire at the short side of the chamber | |

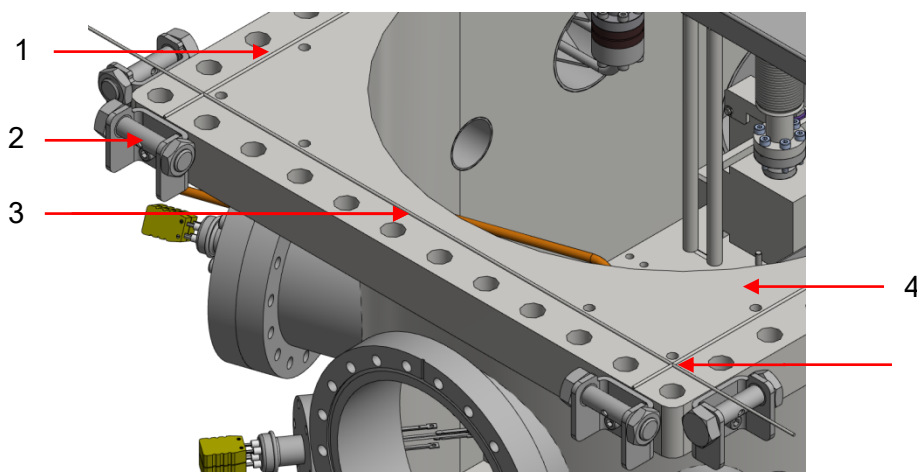


Figure 87: Sealing of the main flange

For the sealing of the vacuum chamber (stadium shaped) Aluminum wire has to be used. Please follow the steps listed below:

- A. Clean the Aluminum wire with a suitable lint-free cloth and alcohol.
- B. Clean the main chamber flange of possible particle remnants with a suitable lint-free cloth and alcohol.
- C. Start with one short side of the chamber:
 - Four turnbuckles are available at the upper corners of the vacuum chamber. Thread the end of the Aluminum wire through the hole available at the corresponding turnbuckle.
 - Carefully rotate the turnbuckle screw such that the wire will be wrapped around it two or three times.
 - Carefully pull the wire along the short side of the main flange to the opposite turnbuckle. Cut the wire to a length about 5-10 cm longer than the length of the chamber short side. Thread the wire through the hole available at the turnbuckle. Start to rotate the turnbuckle screw until a proper tension of the wire is achieved. The wire has to be straight but not too tense.

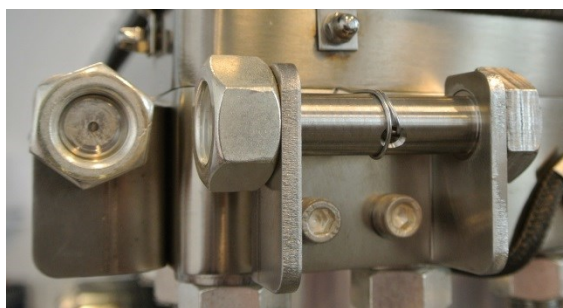


Figure 88: Turnbuckle at the M1 vacuum chamber main flange

- D. Repeat the procedure described above for the other short side of the chamber.
- E. Continue with the long sides of the chamber and tense the long side wires over the short side wires.
 - Thread the end of the Aluminum wire through the hole available at the corresponding turnbuckle.
 - Carefully rotate the turnbuckle screw such that the wire is wrapped around it two or three times.
 - Carefully pull the wire along the long side of the main flange to the opposite turnbuckle. Cut the wire to a length about 5 to 10 cm longer than the length of the chamber long side. Thread the wire through the hole available at the turnbuckle. Start to rotate the turnbuckle screw until a proper tension of the wire is achieved. The wire has to be straight but not too tense.
- F. Repeat the procedure for the other long side of the chamber.
- G. The four intersection points of the Aluminum wires have to be carefully even out with a help of a rounded plastic rod and a rubber hammer. Note that the intersection points need to have the same thickness as the rest of the wire. When wave formation along the wires is visible, the wire tension has to be carefully adjusted such that the intersection points do not shift.
- H. Place the chamber lid carefully on top of the chamber with the help of a crane. The lid should be gently let down and should be seated on the chamber flange evenly.

- I. Insert the fixing bolts at the chamber lid and tighten them in the usual way, all around the flange.

9.4.3 Lubrication schedule

DANGER

Moving parts: spindle lubrication



At the PGM there are several motion axis powered by electrical motors. Make sure that your way of working while maintenance procedure does not include any safety risk. Do not leave long hair untied, do not wear loose clothing or jewelry as this may involve the danger of becoming caught in the moving mechanisms.

All guidance parts and spindles of the in-air linear stages must be lubricated every 6 months or 1000 km whichever is the earlier. Recommended lubricants are ESSO Febis 68 or Alvania Grease S Nr.2 von Showa Shell Sekiyu. Lubricants may not be mixed. After lubrication the full length of the travel must be driven for equal distribution. In case of short repeated scans over the same region a weekly "lubrication run" over the full length of travel is recommended.

9.5 Appendix

9.5.1 Overview of technical specifications

Table 20: Overview of technical specifications (M6)

M6 mirror rotation (M6P)	
Range relative to horizontal	- 3.51° to 1.26°
Offset	544192 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.09 µrad Roll: 0.09 µrad
M6 mirror pitch stability (CLC)	0.016 µrad
Linear drive	PE2.4.0100GRT 12x1
Motor	Phytron ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	240
Sine-bar connecting rod	115 mm
Angle offset	35°
Step resolution (full step)	0.42 µrad /step
Encoder	Renishaw Tonic Encoder scale: RGSZ20-S Readhead (2x): T2641-30M Interpolator (2x): TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 µrad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-M6P	- 5759381 counts
FWD LSW-M6P	2806550 counts
M6 mirror translation (M6TZ)	
Range	± 10 mm
Repeatability of mirror rotation upon rotation (CLC) - FAT	Pitch: 0.08 µrad Roll: 0.16 µrad
M6 mirror pitch stability (CLC) - FAT	0.022 µrad
Motor	PI PiezoMike N-470.42U
Travel range	26 mm
Typical step size	20 nm (max. 30 nm)

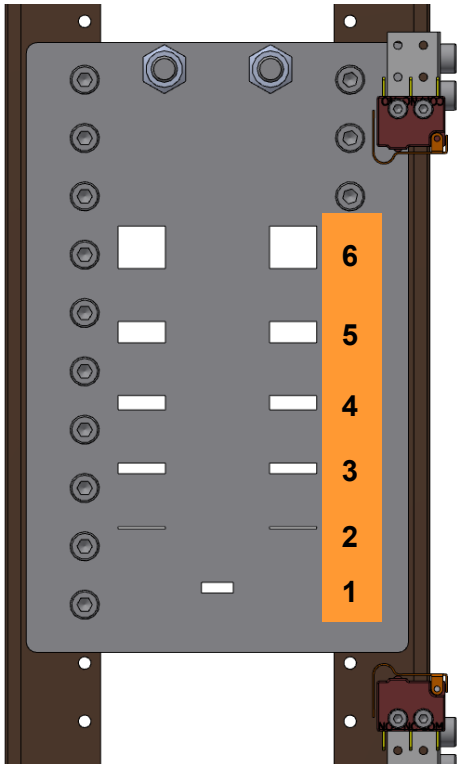
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B30V
Encoder resolution	0.005 μm /count
Limit switches	2 x Huber LS01
REV LSW-M6TZ	19600441 counts
FWD LSW-M6TZ	23642188 counts
M6 masks (M6M)	
Total translation range	0 – 90 mm
Motor	Phytron VSS32.200.1,2-Pt100-BESTEC
Spindle pitch	1.25 mm
Step resolution (full step)	6.25 μm /step
Microstepping 1/64	0.098 μm /step
Limit switches	2 x Huber LS01
REV LSW-M6M	0
FWD LSW-M6M	917423 counts
	
Offset (1) = 178000 steps Offset (2) = 296000 steps Offset (3) = 412000 steps Offset (4) = 540000 steps Offset (5) = 676000 steps Offset (6) = 844000 steps (middle positions of the apertures)	

Table 21: Overview of technical specifications (vacuum vessel, M7, gratings)

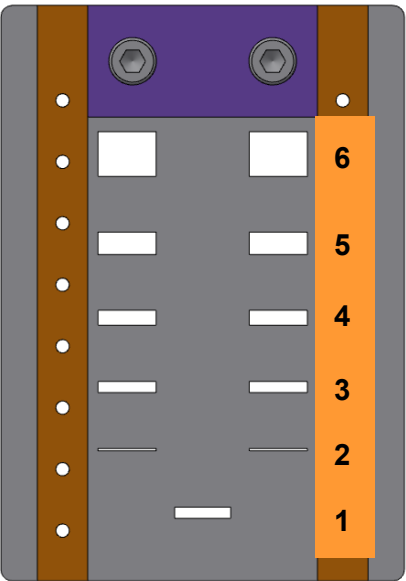
General	
Included components	M6 mirror mechanics M7 mirror mechanics Gratings mechanics M7 masks
Total weight of the system	~2000 kg (without lower granite block)
Vacuum vessel material	1.4404 (C < 0.02 %)
Chamber type	Stadium shaped
Main flange (chamber lid)	At the top (wire sealing)
Number of deflecting mirrors	1
Deflecting direction	Downward in vertical plane
Substrate dimensions (L x W x H) Active optical surface (L x W)	540 mm x 60 mm x 70 mm 525 mm x 40 mm
Number of gratings	3
Substrate dimensions (L x W x H) Active optical surface (L x W)	220 mm x 55 mm x 50 mm 200 mm x 40 mm
Base pressure of empty chamber after bakeout	3.5×10^{-10} mbar
M7 mirror and grating axis position (Reimer-Torge parameters)	
Beam offset	20 mm
Horizontal position of the plane mirror axis with respect to the grating axis	0.53 mm (closer to the source)
Vertical position of the plane mirror axis with respect to the grating axis	10.072 mm (below grating rotation axis)
Radius of rotation axis	20.0856 mm
Distance between mirror and mirror axis	61 mm
M7 mirror rotation (M7P)	
Range relative to horizontal	- 12.03° to 1.31°
Offset	475859 counts
Repeatability of mirror rotation upon rotation (CLC)	Pitch: 0.16 μ rad Roll: 0.09 μ rad
M7 mirror pitch stability (CLC)	0.029 μ rad
Linear drive	ERO MOLS 100.100 (with threaded spindle 12 x 1)

Motor	ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	495 mm
Sine-bar connecting rod	115 mm
Angle	15°
Step resolution (full step)	0.20 μ rad /step
Encoder	Renishaw Tonic Renishaw ring: REST20USA206 Readhead (2x): T2601-15M Interpolator (2x): Tonic TI20KD-E10A DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-M7P	- 21162794 counts
FWD LSW-M7P	2831215 counts
Gratings rotation (GP)	
Range relative to horizontal	- 14.19° to 1.74°
Offset	- 1081502 counts
Repeatability of grating rotation upon rotation (CLS)	Pitch: 0.10 μ rad Roll: 0.09 μ rad
Repeatability of grating rotation upon translation (CLS)	Pitch: 0.17 μ rad Roll: 0.18 μ rad
Grating pitch stability	0.032 μ rad
Linear drive	PE2.4.0100GRT 12x1
Motor	ZSS52.200.2,5-HD14/50
Spindle pitch	1 mm
Rotation lever length	340 mm
Sine-bar connecting rod	115 mm
Angle offset	12°
Step resolution	0.3 μ rad /step
Encoder	Renishaw Tonic Renishaw ring: REST20USA206 Readhead (2x): T2601-15M Interpolator (2x): Tonic TI20KD-E10A

	DSI-QUM10
Encoder resolution	0.0097087 μ rad /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
Collision switches	2 x Huber LS01
REV LSW-GP	- 26588173 counts
FWD LSW-GP	2044496 counts
Gratings translation (GT)	
Range	0 mm to 175.85 mm
Linear drive	Bestec linear drive
Motor	Phytron ZSS57.200.2,5
Spindle pitch	2 mm
Step resolution (full step) Belt gear 2:1	10 μ m /step 5 μ m /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BVT005B30V
Encoder resolution	0.005 μ m /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-GT	259900631 counts
FWD LSW-GT	295071198 counts
Optics base plate translation (BTZ)	
Range	- 11.5 mm to 11.5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μ m /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μ m /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-BTZ	15092500 counts

FWD LSW-BTZ	14630600 counts
2-theta rotation (3AAR2T)	
Total range	112°
Offset (at this position the incident beam on M6 is perpendicular to the incident beam on the sample)	44605742 counts
Motor	ZSH87/3.200.5.PLE80-80:1
Step resolution (full step)	4.45686×10^{-4} °/step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B10A
Encoder resolution	4.98311×10^{-6} °/count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Free switch	2 x Schaltbau S880
Collision switch	4 x Schaltbau S880
REV LSW-3AAR2T	33278342 counts
FWD LSW-3AAR2T	55927802 counts
3AA 2Theta circle Offset from origin point	x = 1.9 µm z = 3.5 µm
Circle of minimum confusion of the 3AA 2Theta radial bearing	Max. radius = 27 µm RMS radius = 7.9 µm
Height deviation along 3AA 2Theta circle of the 3AA 2Theta radial bearing	6.4 µm
Circle of the minimum confusion of the 3AA 2Theta - coupled motion of 3AA and DAA	Max. radius = 135 µm RMS radius = 66 µm (inner circle) Max. radius = 155 µm RMS radius = 57 µm (outer circle)
Height deviation of the 3AA 2Theta - coupled motion of 3AA and DAA	14 µm (inner circle) 14.3 µm (outer circle)
Wedge leveler outboard (OVY)	
Range	- 5.19 mm to 5.12 mm
Offset	19736085 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 µm /step

Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-OVY	19632356 counts
FWD LSW-OVY	19838414 counts
Wedge leveler inboard (IVY)	
Range	- 4.79 mm to 5.41 mm
Offset	17718638 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	2 mm
Step resolution (full step)	0.2 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S A-9763 Readhead: RL32BAT050B30A
Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Metrol STS080PB
REV LSW-IVY	17622865 counts
FWD LSW-IVY	17826826 counts
M7 masks (M7M)	
Total translation range	0 – 96 mm
Motor	Phytron ZSS57.200.2,5
Linear drive	VAb LD40-100
Spindle pitch	3 mm
Step resolution (full step)	15 μm /step
Microstepping 1/64	0.234 μm /step
Limit switches	2 x Schaltbau 880
REV LSW-M7M	- 408989 steps
FWD LSW-M7M	0

	Offset (1) = -72000 steps Offset (2) = -119280 steps Offset (3) = -168285 steps Offset (4) = -220400 steps Offset (5) = -277170 steps Offset (6) = -346000 steps (middle positions of the apertures)
---	---

9.5.2 Sine-bar mechanism

The rotation of the optical elements (M6, M7 and gratings) is driven in air by motorized high-precision linear feedthroughs and transferred into UHV through edge-welded bellows. The linear movement is translated into rotation using a sine-bar mechanism (separate for M6, M7 and gratings rotation). Sketches of the sine-bar mechanisms for this PGM are shown in Figure 89, Figure 90 and Figure 91.

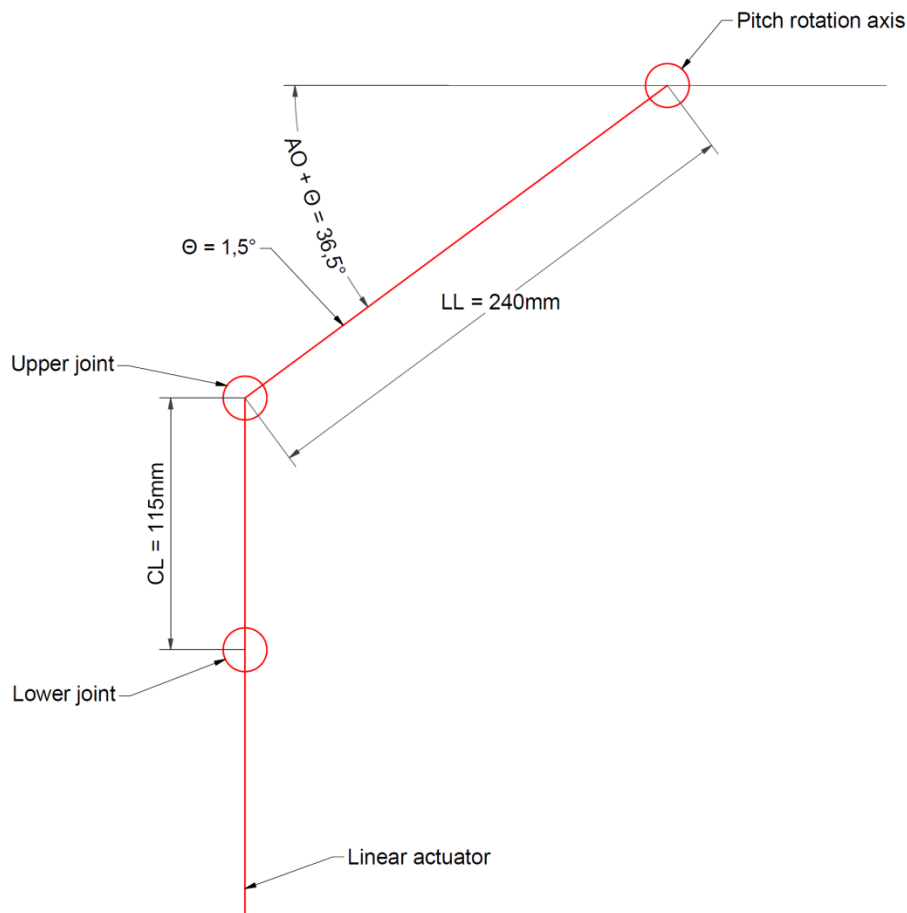


Figure 89: Sketch of the sine-bar mechanism driving M6 mirror pitch

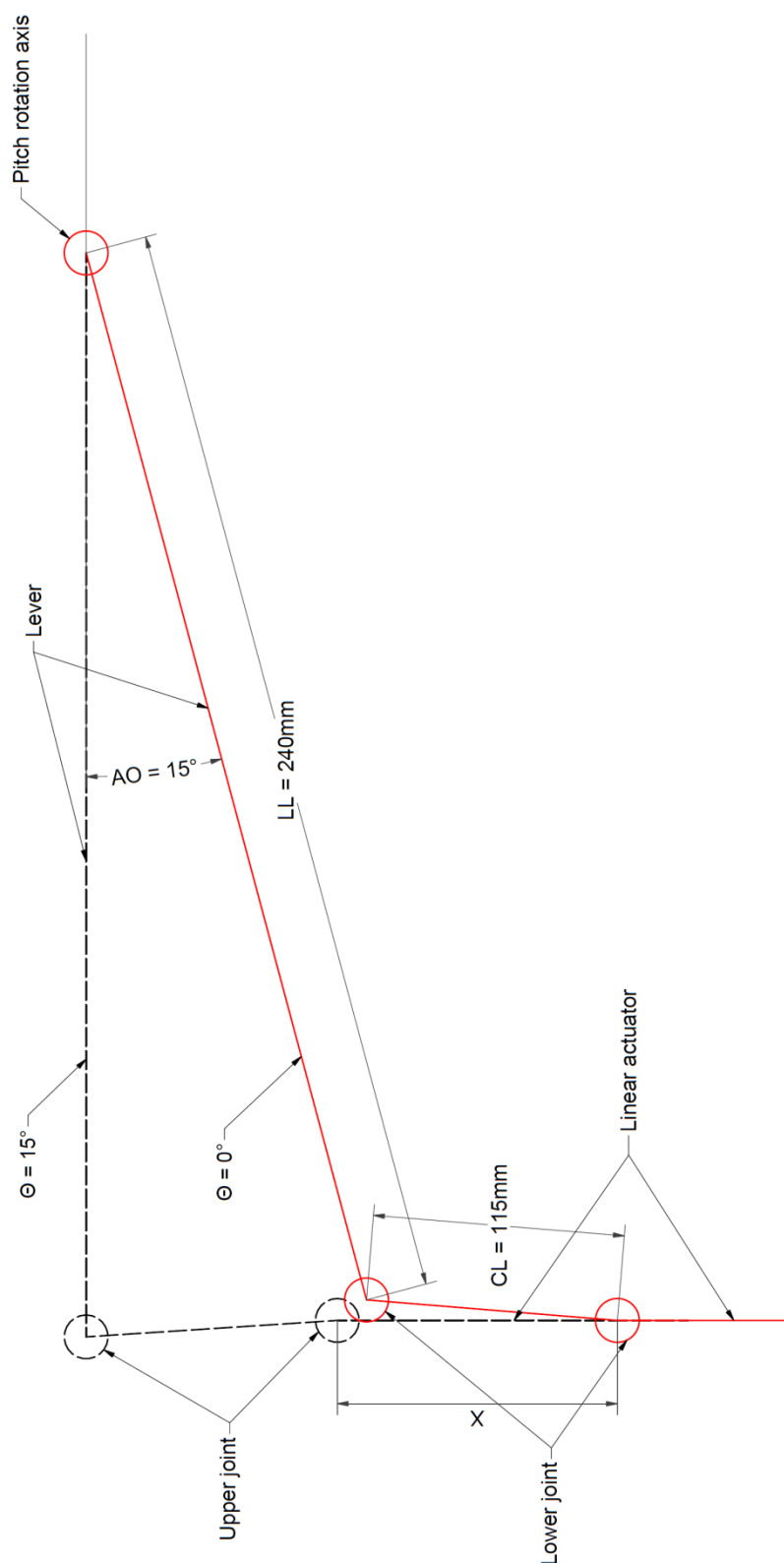


Figure 90: Sketch of the sine-bar mechanism driving M7 mirror pitch

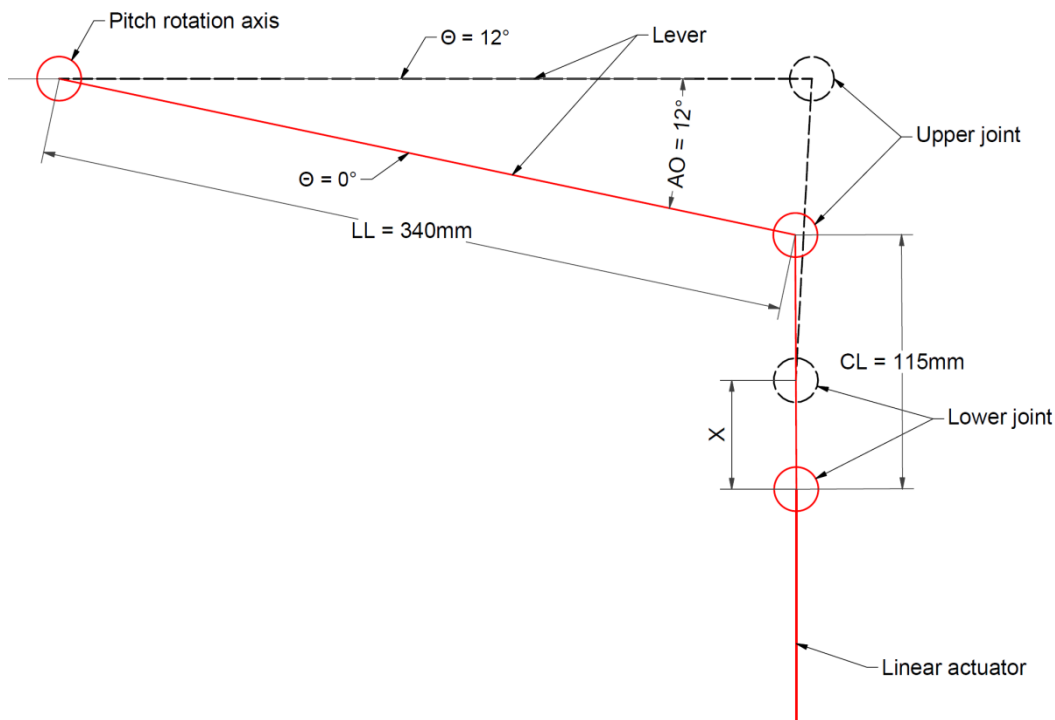


Figure 91: Sketch of the sine-bar mechanism driving gratings pitch

The relation between Rotation Angle θ and Actuator Position x is given in equation [1].

$$x = 2LL \sin\left(\frac{\theta}{2}\right) \cos\left(AO + \frac{\theta}{2}\right) - CL + \sqrt{CL^2 - 4LL^2 \sin^2\left(\frac{\theta}{2}\right) \sin^2\left(AO + \frac{\theta}{2}\right)} \quad [1]$$

Lever Length LL	Nominal distance between rotation axis and upper joint of the connecting rod.
Connecting rod Length CL	Nominal distance between upper and lower joint of the connecting rod.
Angle Offset AO	Nominal angle between lever arm and grating/mirror surface.
Actuator Position x	Linear travel of the actuator rod with respect to the position, where the lever arm orientation is AO with respect to the horizon and the mirror/grating surface is nominally horizontal.
Rotation Angle θ	Angle of the mirror/grating surface.

9.5.3 Location of in-vacuum components and sensors

Table 22: Location of in-vacuum components and sensors

In air (feedthrough)	In vacuum	Wiring scheme
TC7	At M6 mirror holder base plate	444 - 7M - 346a
TC8	At M7 mirror cradle	444 - 7M - 346a
TC9	At the gratings translation slide	444 - 7M - 346a
TC10	At M7 / gratings support granite block	444 - 7M - 346b
TC11	At M6 support granite block	444 - 7M - 346b
TC12	At the lower granite block	444 - 7M - 346b

9.5.4 List of supplied wiring schemes

Table 9 contains a list of the provided wiring schemes. For detailed schemes, please refer to Customer Documentation folder.

Table 23: List of wiring schemes for the 3AA system

Scheme number	Description
444 - 7M - 500a	Motion Axes - Overview
444 - 7M - 500b	Motion Axes - Overview
444 - 7M - 500c	Motion Axes - Overview
444 - 7M - 121	Layout Collision Detection Box
444 - 7M - 562a	M6P - Kill switches
444 - 7M - 571a	M6P - Encoder
444 - 7M - 560b	M7P - Motor + Limit switches
444 - 7M - 562b	M7P - Kill switches
444 - 7M - 571b	M7P - Encoder
444 - 7M - 560c	GT - Motor + Limit switches
444 - 7M - 562c	GT - Kill switches
444 - 7M - 566c	GT - Free switches
444 - 7M - 571c	GT - Encoder
444 - 7M - 560d	GP - Motor + Limit switches
444 - 7M - 562d	GP - Kill switches
444 - 7M - 348d	GP - CSW

444 - 7M - 959	GP - Collision detection box
444 - 7M - 571d	GP- Encoder
444 - 7M - 561e	M6TZ - Linear Actuator + LSW
444 - 7M - 583b	M6TZ - Limit switches
444 - 7M - 571e	M6TZ - Encoder
444 - 7M - 562f	M6M - Motor + Limit Switches
444 - 7M - 583a	M6M - Motor + Limit Switches
444 - 7M - 560g	M7M - Motor + Limit Switches
444 - 7M - 560f	BTZ - Motor + Limit Switches
444 - 7M - 570f	BTZ - Kill switches + Encoder
444 - 7M -	OVY - Motor + Limit switches
444 - 7M - 570g	OVY - Kill switches + Encoder
444 - 7M - 560h	IVY - Motor + Limit switches
444 - 7M - 570h	OVY - Kill switches + Encoder
444 - 7M - 560k	3AAR2T - Motor + Limit switches
444 - 7M - 570k	3AAR2T - Kill switches + Encoder
444 - 7M - 566d	3AAR2T - Free Switches
444 - 7M - 348b	3AAR2T - Collision Switches
444 - 7M - 587	Cables - Tonic DSI Adapter 587
444 - 7M - 356	TMTR1 - Tiltmeter
444 - 7M - 346a	Thermocouples
444 - 7M - 346b	Thermocouples Optics Support
444 - 7M - 345a	Air Bearing 3AAR2T
444 - 7M - 345b	Air Bearing 3AAR2T
444 - 7M - 345d	Air Bearing 3AAR2T
444 - 7M - 401a	Bakeout 3AA
444 - 7M - 401b	Bakeout 3AA

444 - 7M - 410c	Heaters Ion Getter Pumps
444 - 7M - 300	PLC Box 2 Layout
444 - 7M - 347a	Air Bearing
444 - 7M - 347b	Air Bearing
444 - 7M - 347c	PT100 Motor M6 Masks
444 - 7M - 348	Gate Valves

9.5.5 List of supplied assembly drawings

Table 10 contains a list of the provided assembly drawings. For detailed drawings, please refer to Customer Documentation folder.

Table 24: List of assembly drawings for the 3AA system

Drawing number	Description
N4440C0000	3AA optics tank assembly
N4441C0000	3AA vacuum chamber
N4441C0006	Gratings mask assembly
B04C0232A1	Linear feedthrough
N4441C0005	Gratings mask
01-0429-00	Aperture
N4441C0001	Chamber lid assembly
N4441C0002	3AA vacuum chamber
N4444C0013	3AA support structure
N4444C0005	Granite plate M6
N4444C0004	Granite plate M7_gratings
N4444C0020	Granite plates connector
N4442C0000	Wedge leveler inboard
N4442C0009	Wedge leveler outboard
N4442C0020	3AA carrier
N4444C0011	Drive unit 3AA 2Theta
N4444C0001	3AA 2Theta support
N4443C0012	M6 mirror mechanics
N4443C0032	M6 assembly base plate
N4443C0013	M6 bearing block left
N4443C0015	M6 bearing block right
N4443C0034	M6 mirror cradle
N4443C0023	M6 mask
N4443C0014	M6 mirror cradle
N4443C0033	M6 mirror translation slide

B04C0228A1	Linear actuator
N4443C0003	3AA internal mechanics (M7_gratings)
N4443C0036	M7_gratings base plate
N4443C0017	Bearing block
N4443C0018	Bearing block
N4443C0019	M7 mirror cradle
N4443C0039	M7 mirror cradle
N4443C0037	M7 mirror holder
N4443C0005	Gratings cradle
N4443C0035	Gratings translation slide
N4443C0020	Grating holder
N4443C0041	Grating holder
B04C0231A1	Linear actuator
B04C0229A1	Linear feedthrough
N4442C0010	Grating translation rod

9.5.6 List of recommended spare parts

If replacement of system components is necessary, please contact Bestec GmbH. Please note that self-repair of system components without Bestec GmbH consultation can lead to loss of warranty claim.

Table 25: List of recommended spare parts

Spare part	Manufacturer	Part number
Encoder readhead	Renishaw	T2641-30M
		T2601-15M
		RL32BVT005B30V
		RL32BAT050B30A
		RL32BAT050B10A
Interpolator	Renishaw	Ti20KD-E10A
Limit switch	Schaltbau	S880
Collision switch	Huber	LS01
Kill switch	Metrol	STS080PB

9.5.7 List of supplied subcontractor documents

Table 13 contains a list of provided subcontractor documents. For detailed manuals please refer to the provided Customer Documentation folder.

Table 26: List of provided subcontractor documents

Part	Manufacturer	File name
Heating cable	BriskHeat	Heating cable.pdf
Stepper motor	Phytron	ZSS.pdf VSS.pdf
Piezo motor	PI	PiezoMike linear actuator.pdf
Linear drive for M6 rotation and gratings rotation	LTK	Positioniereinheiten.pdf
Linear drive for M7 rotation	ERO	Power driven slides.pdf
Linear feedthrough	VAb	Linear feedthroughs.pdf
Limit / collision switch	Schaltbau	Snap action switches.pdf
	Huber	Limit switches.pdf
Kill switch	Metrol	Stopper bolt with switch.pdf
Encoder (absolute)	Renishaw	Resolute L-9517-9530-01-K.pdf Resolute L-9517-9448-04-G.pdf Resolute L-9517-9418-07-A.pdf Resolute M-9553-9433-06-C.pdf
Encoder (incremental)	Renishaw	Tonic L-9517-9426-02-B.pdf Tonic L-9517-9154-05-B.pdf Tonic L-9517-9348-01-B.pdf Tonic L-9517-9466-01-A.pdf Tonic M-9653-9161-03-D.pdf

10 DAA - detector arm and stand assembly

10.1 Safety information

10.1.1 Used symbols

The following symbols found in the present document are used to identify hazards. An additional purpose of them is to draw the user's attention to safety instructions or other useful information.

	Information on preventing any kind of physical injury.
	Information on preventing extensive equipment and environmental damage.
	Information on correct handling or use. Disregard can lead to malfunctions or equipment damage.
	General warning
	Pinch point hazard warning
	High voltage hazard warning
	Hot surface hazard warning
	Watch your head hazard warning
	Pinch point/entanglement hazard warning
	Gloves required
	Eye protection required
	Hard hat required

10.1.2 Safety instructions

General

In normal operation, there is no danger when working with the M5 mirror system. The operator of the mirror unit should however be aware of potential dangers that can occur by improper handling of the system and that can cause personnel injury or damage to the equipment.

When the system is operated without fulfilling the recommendations described in this document and the legal health and safety regulation, safety of personnel and equipment can be affected. Potential dangers and measures to avoid them are described in this section.

Electrical safety

In normal operation no hazards can occur associated with electrical voltage and current used by the equipment. Following accidents may happen if the electrical installation of the mirror unit is not properly operated: electric shock and burns, fire and electric arcing.

⚠ DANGER

Electrical voltage



1. Voltage and high voltage that may cause injury or death are used in the setup. Service or maintaining work on the electrical installation must only be performed by qualified trained personnel for this type of work.
2. The statutory regulations regarding electrical installations must be respected.
3. The entire electrical equipment supplied as part of the system is provided with a protective ground. Grounding cables must not be removed or manipulated. It is very important that the vacuum system is properly grounded at all times.
4. Fuses may not be bridged, electrical connections must not be separated or otherwise manipulated, protective covers and other electrical insulating material may not be removed or modified.
5. Do not attempt to administer first aid to someone who has suffered electric shock until the source of the shock has been isolated.

Safe handling of gases

In normal operation no dangers through use of gases may appear as long as the recommended precautions are followed. In case of gas handling contrary to the instructions, unconsciousness and associated injuries may occur.

⚠ DANGER

Gases



1. If pure nitrogen is used for venting the vacuum system one has to be aware that its inhalation can induce an oxygen deficiency after only few breaths, which may lead to unconsciousness.
2. Areas in which the oxygen concentration is less than 18% must be abandoned immediately.
3. For venting the vacuum system it is recommended to use dry air with normal oxygen concentration instead of pure nitrogen.

Safe handling of mechanical components

In normal operation only few mechanical works on the M5 mirror unit is required. For service and maintenance extensive mechanical work may be necessary. Only qualified personnel should attempt to dismantle, modify or repair equipment. The following hazards may arise when dealing with mechanical components: injury when transporting heavy loads (mirror unit, chamber lid), danger of crushing by moving parts (motors at the motion axes), injury by protruding parts at the mirror unit, severe damage of internal mechanics and optical elements (when changing motors at hexapod motion axes).

⚠ DANGER

Heavy loads



1. Only qualified personnel should lift and carry heavy loads.
2. Use a hard hat when working with heavy loads.
3. Lifting eyes are available at the individual components of the system and they must be used for transportation.
4. Parts of the equipment that have to be lifted may only be subject to gravity. Additional forces generated by tightening or vacuum must be removed before lifting.

⚠ DANGER

Moving parts



1. At the mirror unit there are a number of moving parts, which are powered by electrical motors. Make sure that your way of working does not include any safety risk. Do not leave long hair untied, do not wear loose clothing or jewelry as this may involve the danger of becoming caught in the moving mechanisms.
2. Do not remove or modify covers of the moving parts.

⚠ CAUTION

Components in UHV



1. In-UHV components can only be touched with powder/lint-free gloves.
2. Tools use for work on UHV components must be clean and free of grease.
3. The parts which are to be mounted in the vacuum chamber have to be material-specific cleaned according to the UHV-vacuum requirements.

⚠ CAUTION

Optical elements



1. Use protective gloves and clothing when working with optical elements.
2. Be careful not to drop anything on/touch the optical surface.
3. A suitable cover for the optical surface is recommended.

Safe handling of hot surfaces

During bakeout procedure the system or parts of it become hot ($>100^{\circ}\text{C}$) and can cause burns when touched.

⚠ DANGER Hot surfaces

1. The bakeout of the system has to be indicated by warnings. Contact data of the responsible operator have to be placed clearly visible and close to the system.
2. Wear protective gloves when touching the system while bakeout and cooling down of the system.

Safe handling of vacuum systems

By appropriate use of and compliance with the safety information during normal operation, no dangers occur from the vacuum system of the mirror unit. The following hazards may arise when dealing with vacuum systems: injury from “flying” pieces caused by exploding or imploding parts of the vacuum system; if the vacuum chamber is over pressurized, viewports may shatter and cause person injury. Use viewport covers and wear eye protection glasses.

⚠ DANGER Vacuum system

1. The walls of the vacuum system can withstand a pressure difference from outside to inside of at least 1 bar, from inside to outside only a pressure difference of about 0.1 bar. An increase in internal pressure of 1.1 bar can lead to the damage of the view ports and must therefore be avoided. Never exceed an overpressure of 0.1 bar inside the vacuum system.
2. The viewports destruction of a chamber under vacuum leads to implosion. There is a risk of injury from flying glass shards. Use viewport and eye protection.
3. Areas in which the oxygen concentration is less than 18% must be immediately abandoned.

10.1.3 Modifications and service

Safe operation, reliability and performance of the equipment may be impaired if assembly operations, extensions, re-adjustments, modifications or repairs are not carried out in accordance with the instructions provided in this document and with any other instructions issued by the manufacturer. If you wish to modify the equipment please contact Bestec GmbH for future advice.

10.2 Design overview

Figure 92 illustrates the complete DAA system: the radial guidance of the detector arm, the detector arm and the detector chamber and its 2-theta radial guidance.

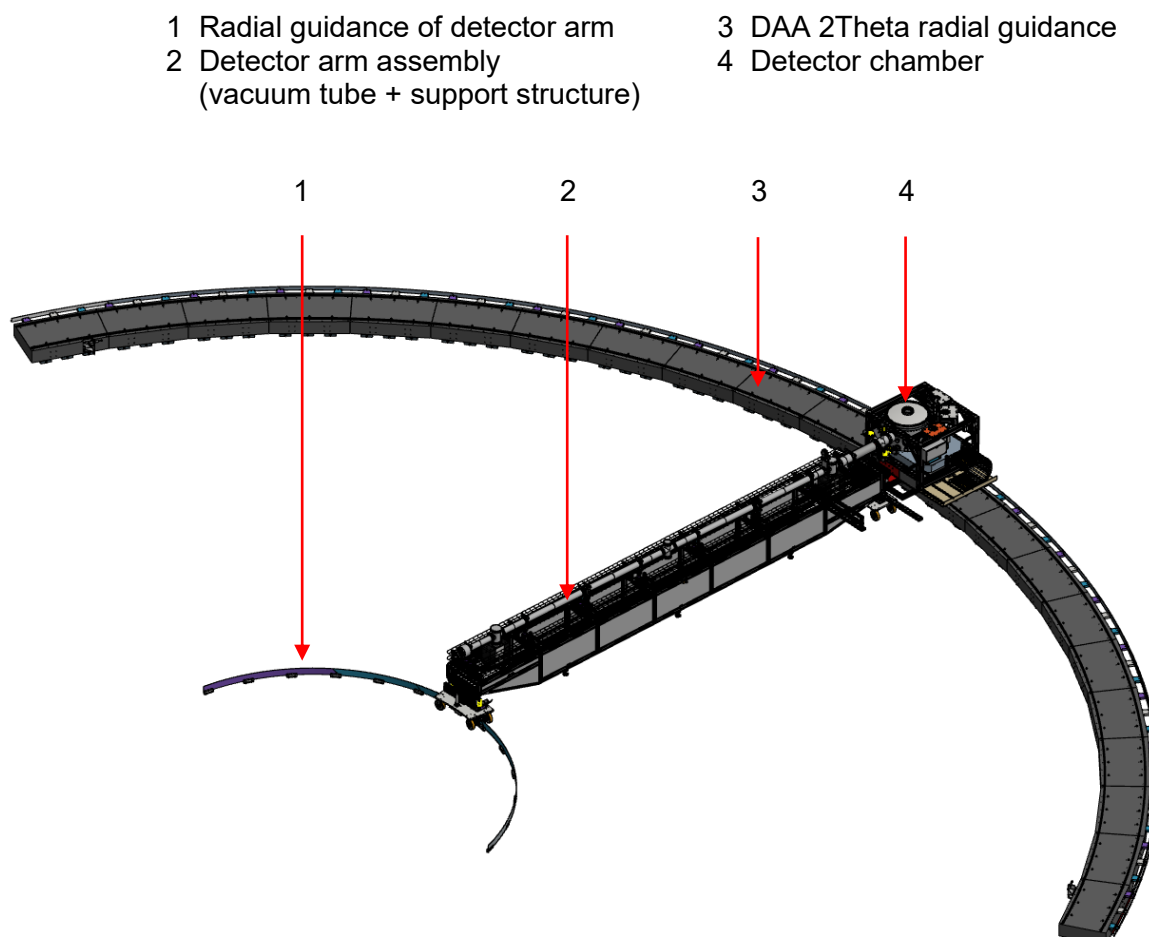


Figure 92: Design overview of the DAA system

Figure 93 shows the design overview of the detector chamber.

- | | |
|-------------------------------|--|
| 1 Aperture unit | 5 Terminal box for bakeout heater |
| 2 Detector chamber | 6 Support frame for camera stabilization |
| 3 Gate valve (3x) | 7 PLC Box |
| 4 Holders for the He cryostat | 8 Drag chain |

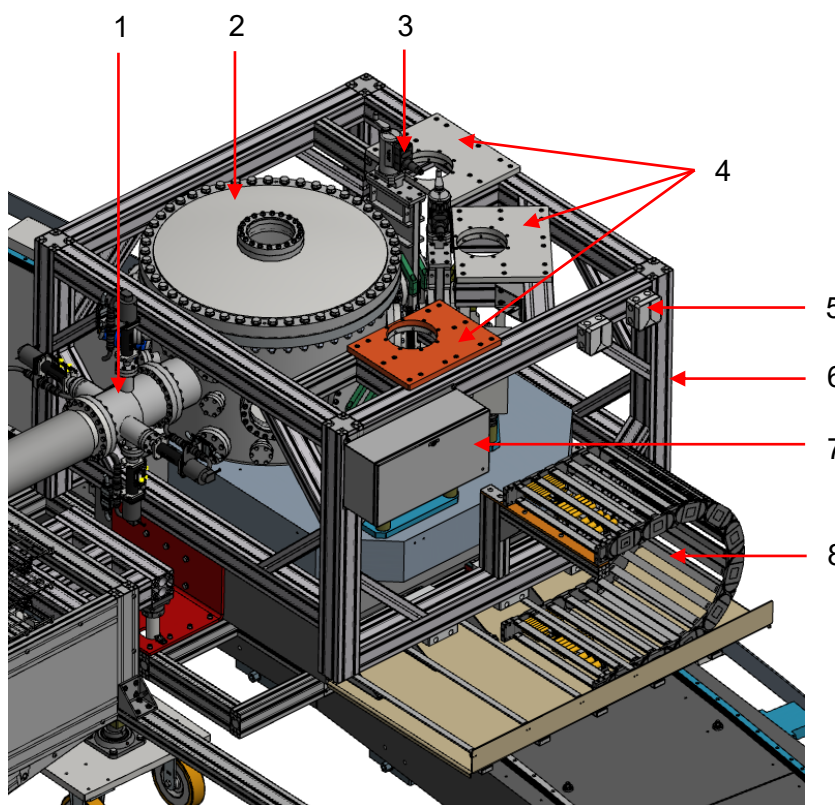


Figure 93: Design overview of the detector chamber

The Helium cryostat is connected to the corresponding detection camera by an edge welded bellow which is meant to decouple the vibrations of the cryostat from the detection camera. This is realized by connecting the cryostat holder (#4 in Figure 93) with a provided external support frame (#6 in Figure 93) which is decoupled from the detector chamber.

A set of four independent and motorized OFHC blades are available at the entrance flange of the detector chamber.

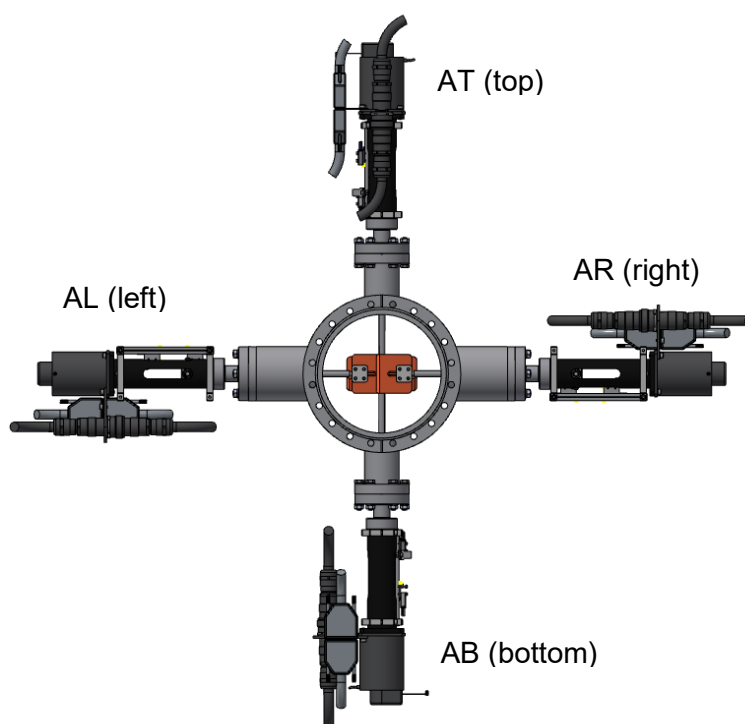


Figure 94: Aperture unit at the entrance of the detector chamber, downstream view

The aperture blades are staggered in order to allow the requested overlap between them.

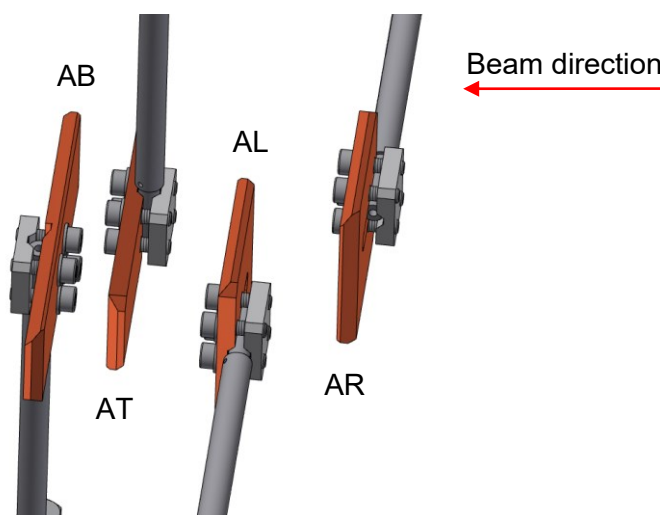


Figure 95: Arrangement of the aperture blades at the entrance of the detector chamber

Figure 96 illustrates the linear actuator assembly of the right aperture blade.

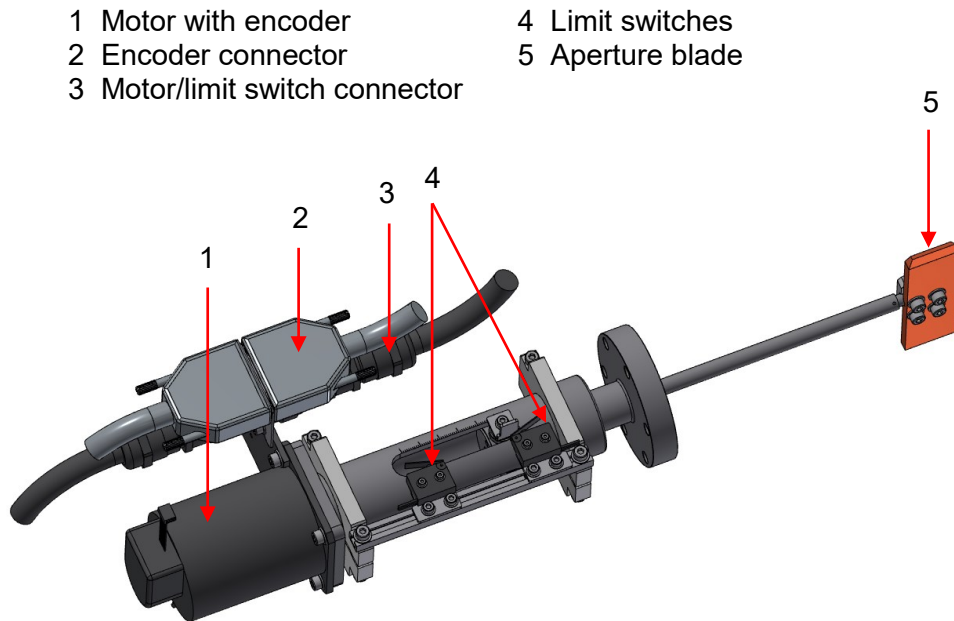


Figure 96: Linear actuator assembly

The detector arm connects the optics tank with the detector chamber. It consists of a vacuum pipe and its support structure. The support structure rests at both ends on two carriages equipped with motors which allow for height correction of the arm during 2-theta motion. Note that the detector arm is tilted by 1° in downstream direction.

- | | |
|------------------------------|------------------------|
| 1 Upstream carriage | 8 Ion getter pump |
| 2 Ion getter pump | 9 Aperture |
| 3 Pneumatic foot (6x) | 10 Cable tray |
| 4 Aperture | 11 Gate valve |
| 5 Support structure | 12 Ion getter pump |
| 6 Vacuum pipe | 13 Downstream carriage |
| 7 Emergency stop push button | |

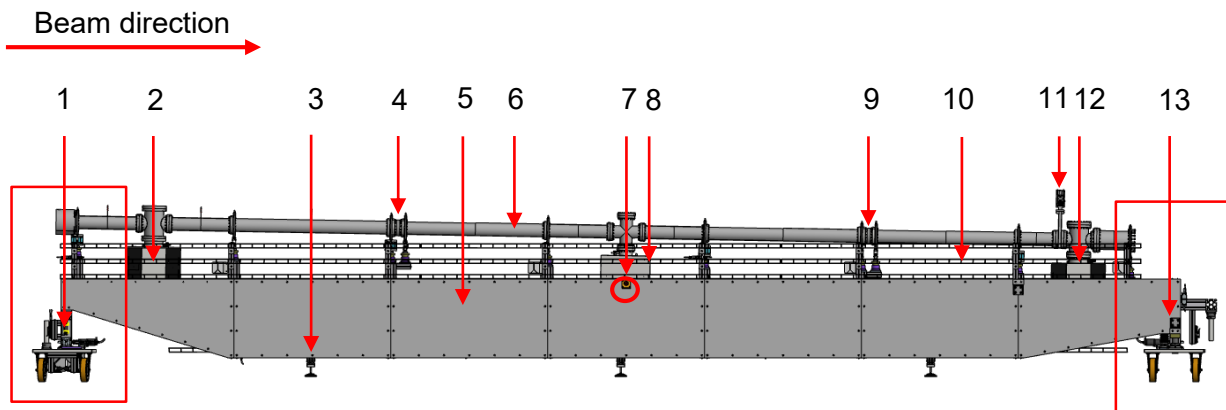


Figure 97: Detector arm assembly, side view

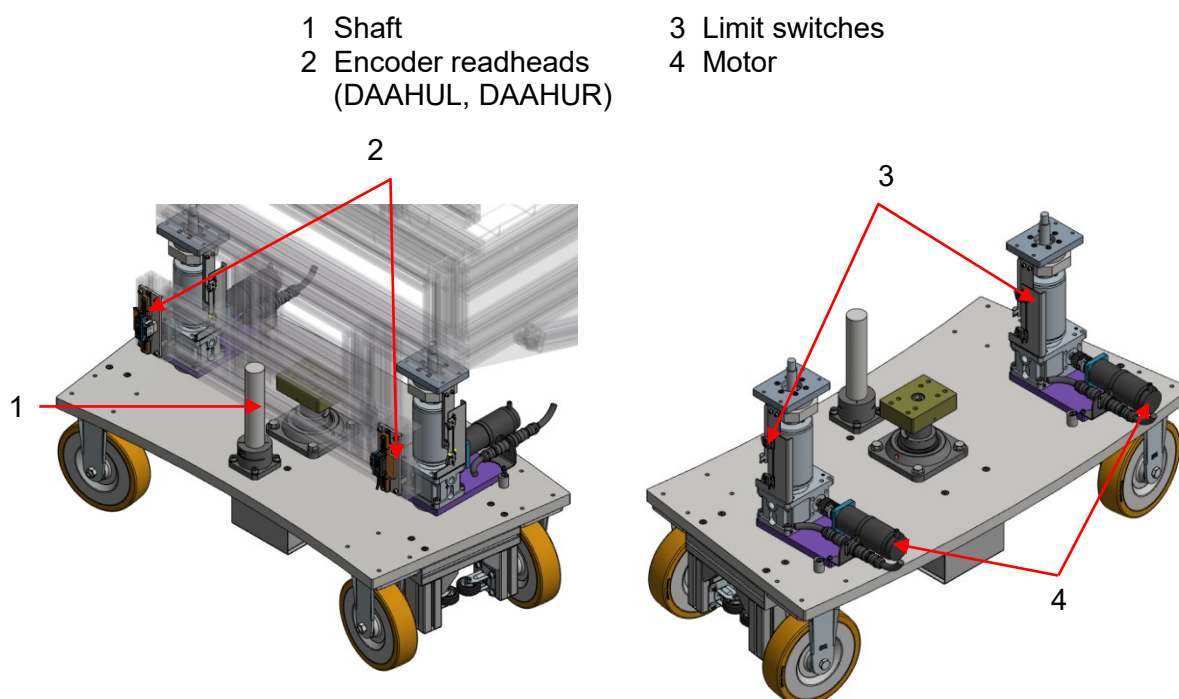


Figure 98: Upstream carriage, upstream view (left) and downstream view (right)

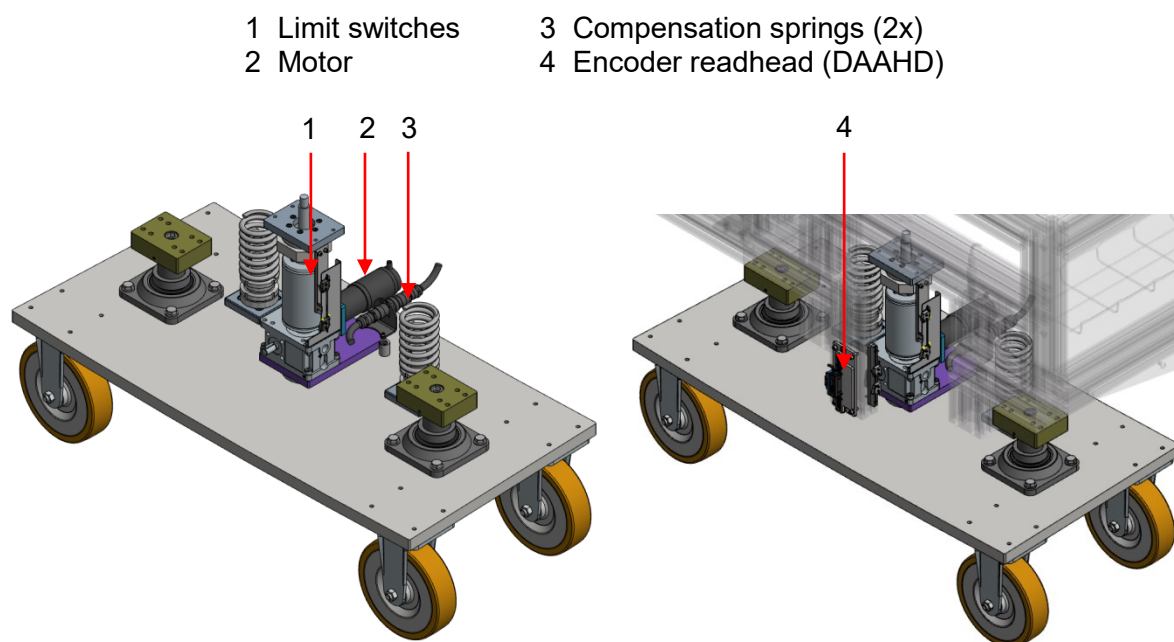


Figure 99: Downstream carriage, upstream view

In Figure 100 the six collision switches available at the DAA system are illustrated.

1, 2 Collision switches at the spectrometer arm

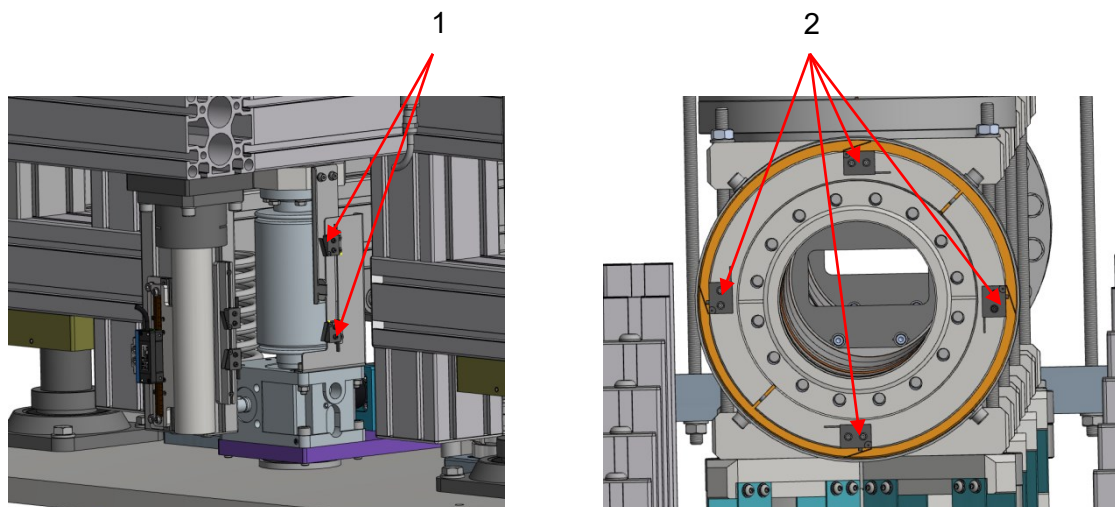


Figure 100: Collision switches (2x) at the downstream carriage (left), collision switches (4x) at the detector arm (upstream)

After performing the 2-theta motion the detector arm can be mechanically decoupled from the 3AA and DDA units. For this purpose two pairs of free switches are available at both ends of the spectrometer arm, as shown in Figure 101.

1, 2 Free switches at the spectrometer arm

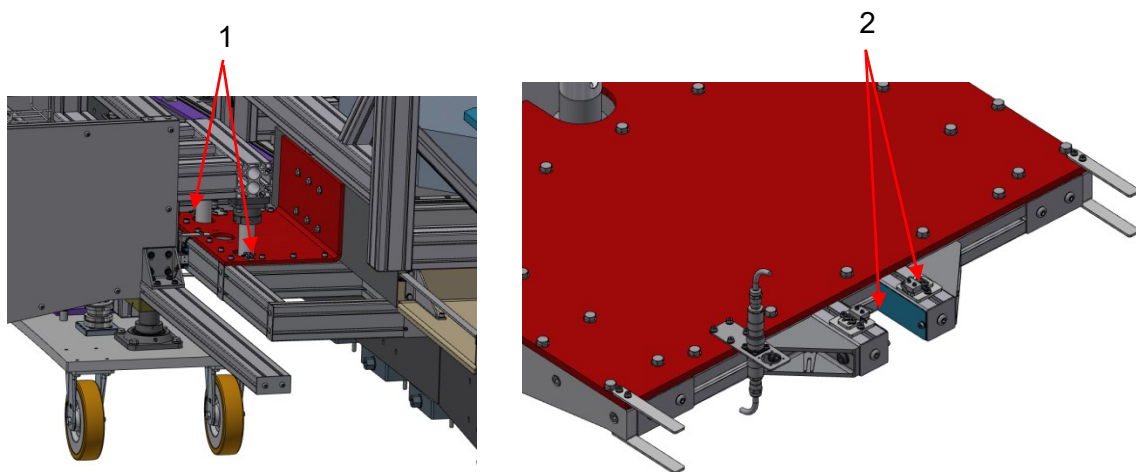


Figure 101: Free switches at the spectrometer arm

The detector chamber can be translated along the detector arm (beam direction) and synchronously rotated with 3AA and the detector arm along 2-theta. The corresponding motion axes are the detector translation (DT) and detector 2-theta rotation (DAAR2T).

The stacking order of the detector chamber support structure is illustrated in Figure 102.

- | | |
|--|--|
| 1 Rail for external frame translation | 5 Motor |
| 2 Rail for detector chamber translation | 6 Granite block (@ 2-theta stage) |
| 3 Support structure for detection camera | 7 Granite base fixed to floor |
| 4 Granite block (@ translation stage) | (2-theta radial guidance of the chamber) |
| | 8 Wedge leveler with WASI adjuster |

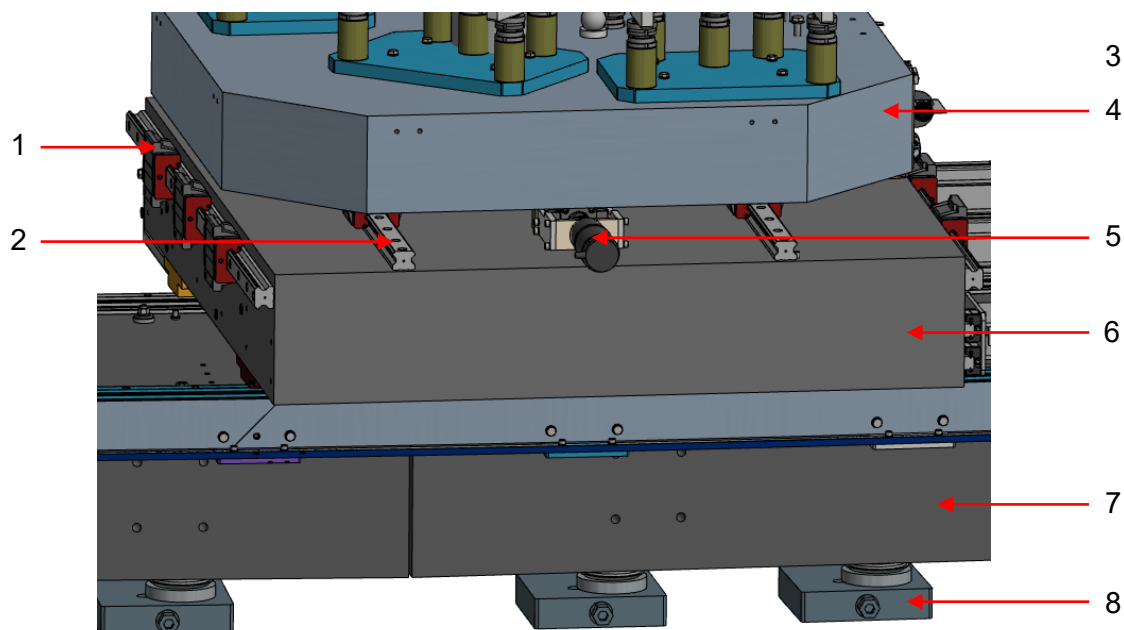


Figure 102: Stacking order of the detector chamber support structure

For the translation motion, the detector chamber and its underlying granite block (#4 in Figure 102) are moved on a pair of linear guideways which are fixed on the granite block of the 2-theta motion (#6 in Figure 102). The limit switches define the two extreme positions of the translation motion.

- | | |
|----------------------------------|---------------------------------------|
| 1 Motor/limit switches connector | 5 Encoder connector |
| 2 Kill switches connector | 6 Encoder tape |
| 3 Limit/kill switches | 7 Granite block (@ translation stage) |
| 4 Encoder readhead | 8 Linear guide for support frame |
- 3 4 5 6

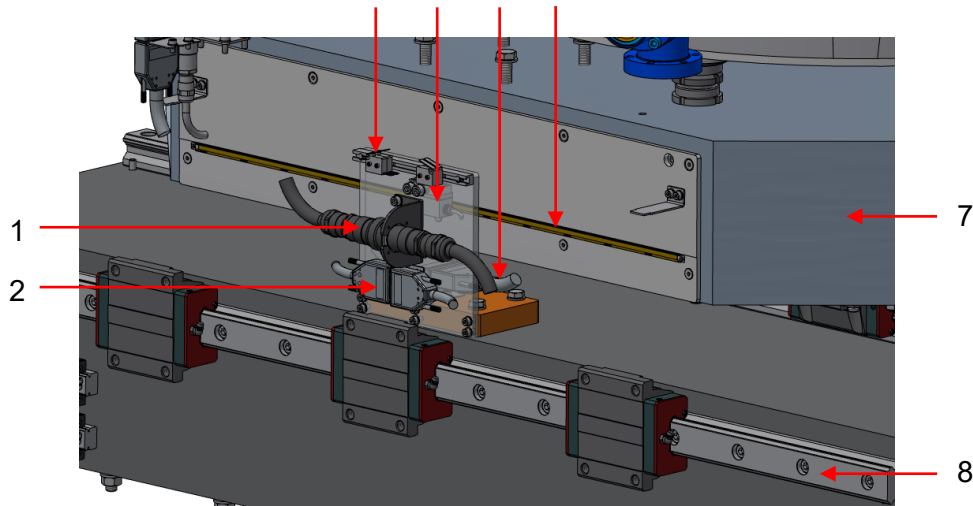


Figure 103: Encoder and switches at the DT motion axis

The support frame follows the translation motion of the detector chamber. A shaft, fixed at the support frame (#6 in Figure 104) is pushed during translation by a metal plate (#4 in Figure 104) directly connected to the granite block located underneath the detector chamber (# 7 in Figure 104). At the end of the translation motion the shaft can be decoupled from the metal plate with the help of two provided free switches (#5 in Figure 104). In this way it is assured that the support frame will be decoupled from the detector chamber. A tiltmeter is provided for measuring the tilt of the granite/chamber during translation motion.

- | | |
|--------------------|--------------------------------------|
| 1 Tiltmeter | 4 Fixed mounted shaft |
| 2 Metal plate | 5 Granite block at translation stage |
| 3 Free switch (2x) | 6 Support frame |

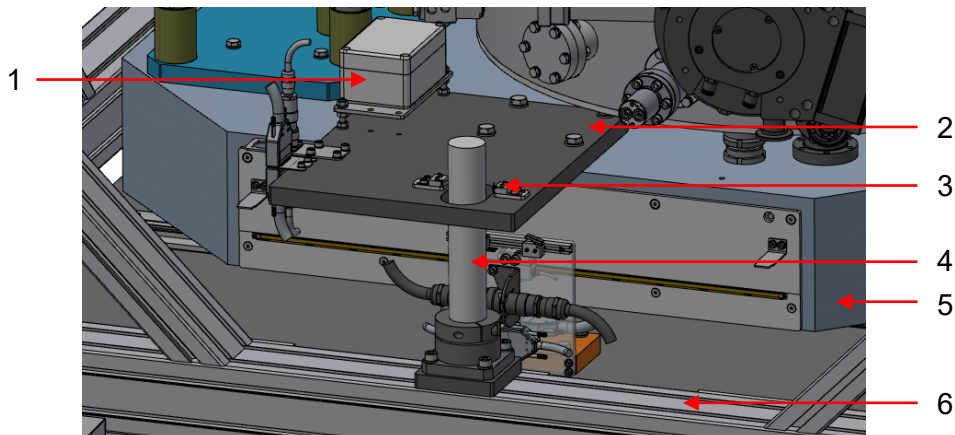


Figure 104: Free switches at the translation axis of the detector

The 2-theta rotation stage of the detector chamber is located below the chamber translation stage. The granite block of the 2-theta motion moves on two bended linear guides (#7 in Figure 105), which are fixed onto the granite base. The friction wheel drive is shown in Figure 105.

- | | |
|--|-------------------------------|
| 1 Encoder readhead with protection plate | 5 Motor with gear |
| 2 Encoder tape | 6 Counter wheel |
| 3 Granite base fixed to laboratory floor | 7 Linear guide |
| 4 Motorized guide wheel | 8 Metal sheet linear guidance |

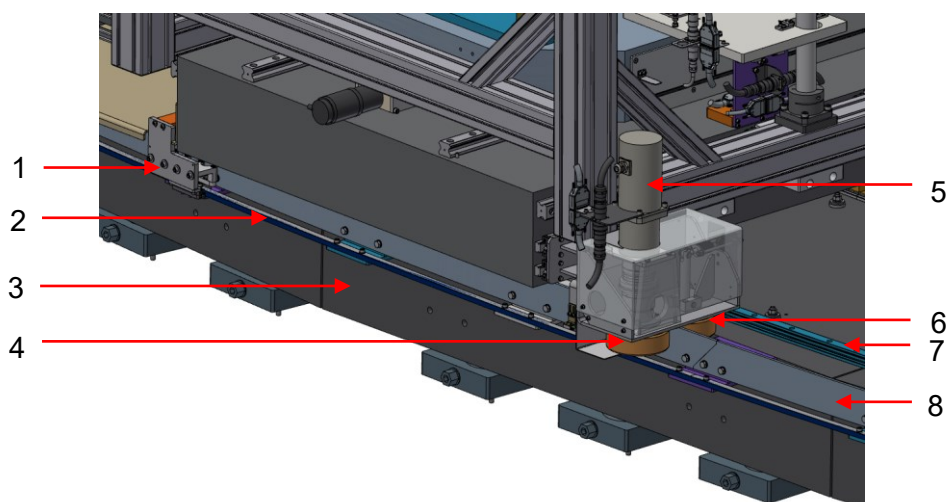


Figure 105: 2-theta rotation stage of detector chamber

For a complete assignment of all motors, limit switches and collision switches available at the system please refer to the provided assembly drawings 01-0744-00 and N4453C0000.

10.3 Appendix

10.3.1 Overview of technical specifications

Table 27: Overview of technical specifications (DAA)

General	
Vacuum vessel material	1.4404 (C < 0.02 %)
Chamber type	cylindrical, vertical axis
Main flange (chamber lid)	COF-type, diameter 700 mm
DAA height upstream left (DAAHUL)	
Total range	51.5 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~0.244 μm /count
Limit switches	2 x Schaltbau S880
DAA height upstream right (DAAHUR)	
Total range	51.6 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~0.244 μm /count
Limit switches	2 x Schaltbau S880
DAA height downstream (DAAHD)	
Total range	35.15 mm
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	4 mm
Step resolution (full step)	0.4 μm /step
Encoder	Renishaw/RLS

	Encoder scale: AS10A00120C00 Readhead: LMA10DC13B10A00
Encoder resolution	~0.244 μm /count
Limit switches	2 x Schaltbau S880
2-theta rotation (DAAR2T)	
Total range	113.4°
Offset (at this position the incident beam on DAA is perpendicular to the incident beam on the sample)	15747638 counts
Motor	Phytron ZSH87/3.200.5-PLE80/20
Step resolution (full step)	3.8957×10^{-4} °/step
Encoder	Balluff Encoder scale: BML-M02-A55-A0-M3300 Readhead: BML-S1G0-B7ED-M5EZ-90-S284
Encoder resolution	3.9566798×10^{-6} ° /count
Limit switches	2 x Schaltbau S880
Free switch	2 x Schaltbau S880
Collision switch	2 x Schaltbau S880
REW LSW-DAAR2T	1453122 counts
FWD LSW-DAAR2T	30111482 counts
DAA 2Theta circle Offset from origin point	x = -614 μm z = -24 μm
Circle of minimum confusion	Max. radius = 324 μm RMS radius = 114 μm
Height deviation along DAA 2Theta circle	35.5 μm
Detector translation (DT)	
Range	- 253 mm to 253 mm
Offset (middle position)	6025499 counts
Motor	Phytron ZSS57.200.2,5-GPL52/50
Spindle pitch	5 mm
Linear drive	HIWIN HGW35CC3R1220ZAPSS
Step resolution (full step)	0.5 μm /step
Encoder	Renishaw Resolute Encoder scale: RTLA-S Readhead: RL32BAT050B10A

Encoder resolution	0.05 μm /count
Limit switches	2 x Schaltbau S880
Kill switches	2 x Schaltbau S880
Free switch	2 x Schaltbau S880
REW LSW-DT	959216 counts
FWD LSW-DT	11091782 counts
Aperture unit (AT, AB, AL, AR)	
Blade material	OFHC
Translation range AT	-1.9 mm to 31.9 mm
Translation range AB	-35.23 mm to 4.14 mm
Translation range AL	-1.25 mm to 46.66 mm
Translation range AR	-45.35 mm to 2.48 mm
Offset AT (at beam height 1161.585 mm)	- 30896 counts (47 mm - scale at actuator)
Offset AB (at beam height 1161.585 mm)	33203 counts (43.8 mm - scale at actuator)
Offset AL (beam center)	- 44400 counts (47.2 mm - scale at actuator)
Offset AR (beam center)	43402 counts (46.2 mm - scale at actuator)
Linear feedthrough	VAb MLS40-50
Stroke	50 mm
Motor	Phytron ZSS57.200.2,5-I with encoder
Spindle pitch	1 mm
Step resolution (full step)	10 μm /step at linear actuator
Encoder resolution	1 μm /count
Limit switches	2 x Schaltbau S880
Initialisation AT	Forward limit switch, negative search direction for reference mark
REV LSW-AT	- 32787 counts (49 mm - scale at actuator)
FWD LSW-AT	1022 counts (14.8 mm - scale at actuator)
Initialisation AB	Reverse limit switch, positive search direction for reference mark
REV LSW-AB	- 2027 counts (8.2 mm - scale at actuator)
FWD LSW-AB	37339 counts (47.8 mm - scale at actuator)
Initialisation AL	Forward limit switch, negative search direction for reference mark
REV LSW-AL	- 45745 counts (48.6 mm - scale at actuator)
FWD LSW-AL	2255 counts (0.5 mm - scale at actuator)

Initialisation AR	Reverse limit switch, positive search direction for reference mark
REV LSW-AR	-1943 counts (0.8 mm - scale at actuator)
FWD LSW-AR	45877 counts (47.8 mm - scale at actuator)

10.3.2 List of supplied wiring schemes

Table 9 contains a list of the provided wiring schemes. For detailed schemes, please refer to Customer Documentation folder.

Table 28: List of wiring schemes for the complete spectrometer arm system

Scheme number	Description
445 - 7M - 500a	Motion Axes - Overview
445 - 7M - 113	Emergency Stop
445 - 7M - 560a	DAAHUL - Motor + Limit switches
445 - 7M - 570a	DAAHUL - Encoder
445 - 7M - 560b	DAAHUR - Motor + Limit switches
445 - 7M - 570b	DAAHUR - Encoder
445 - 7M - 560c	DAAHD - Motor + Limit switches
445 - 7M - 570c	DAAHD - Encoder
445 - 7M - 560d	DAAR2T - Motor + LSW
445 - 7M - 5801	DAAR2T - Encoder
445 - 7M - 566a	DAAR2T - FSW
445 - 7M - 566d	DAAR2T - CSW
445 - 7M - 560e	DT - Motor + LSW
445 - 7M - 570e	DT - KSW + Encoder
445 - 7M - 566b	DT - FSW
445 - 7M - 356	TMTR2 - Tiltmeter
445 - 7M - 345d	Support Feet DAA
445 - 7M - 401a	Bakeout DC
445 - 7M - 401b	Bakeout DAA
445 - 7M - 401b	Bakeout DAA
445 - 7M - 401b	Bakeout DAA
445 - 7M - 410a	Heaters Ion Getter Pumps

445 - 7M - 410b	Heating Cables
445 - 7M - 300	PLC Box 3 Layout
445 - 7M - 347a	Gate Valves
445 - 7M - 347b	Gate Valves
445 - 7M - 347c	Support Feet DAA

10.3.3 List of supplied assembly drawings

Table 10 contains a list of the provided wiring schemes. For detailed drawings, please refer to Customer Documentation folder.

Table 29: List of assembly drawings for the complete spectrometer arm system

Drawing number	Description
N4450C0001	DAA detector arm assembly
N4454C0000	Floor unit
N4453C0000	Detector unit
N4454C0009	Driving unit
01-1430-00	Camera holder
01-1798-00	Linear unit
01-2158-00	Frame
N4555C0000	Chamber with extensions
N4555C0001	Aperture unit
01-0725-00	Linear unit
01-1406-00	Aperture recipient
01-1041-00	Detector chamber
01-1079-00	Chamber lid
01-1777-00	Support structure
01-2160-00	External frame
01-0744-00	Beamline assembly
01-1416-00	Flange_tube support
01-1417-00	Flange_tube support
01-1417-01	Flange_tube support
01-1542-00	Collision detection unit

N4451C0000	Beamline with extensions
01-1778-00	Beamline support with extensions
01-2692-00	Bridge frame
01-0743-01	Carriage with lifting unit
01-0690-00	Lifting unit
01-0743-00	Carriage with lifting units
N4454C0011	Guide unit
N4420C0002	Beam with mirrors

10.3.4 List of recommended spare parts

If replacement of system components is necessary, please contact Bestec GmbH. Please note that self-repair of system components without Bestec GmbH consultation can lead to loss of warranty claim.

Table 30: List of recommended spare parts

Spare part	Manufacturer	Part number
Encoder readhead	Renishaw/RLS	LMA10DC13B10A00
Encoder readhead	Renishaw	RL32BAT050B10A
Encoder readhead	Balluff	BML-S1G0-B7ED-M5EZ-90-S284
Limit / kill / collision / free switch	Schaltbau	S880

10.3.5 List of supplied subcontractor documents

Table 13 contains a list of provided subcontractor documents. For detailed manuals please refer to the provided Customer Documentation folder.

Table 31: List of provided subcontractor documents

Part	Manufacturer	File name
Heating cable	BriskHeat	Heating cable.pdf
Linear guideway	HIWIN	Linear guideway.pdf
Linear feedthrough	VAb	Linear feedthroughs.pdf
Stepper motor	Phytron	ZSS.pdf ZSH.pdf
Limit / kill / collision / free switch	Schaltbau	Snap action switches.pdf

Revised Instruction Manual

P442 Spectrometer Arm System • SIX Beamline • NSLS-II



Encoder	Renishaw/RLS	LMA10D01_04.pdf
Encoder	Renishaw	L-9517-9448-04-G.pdf L-9517-9418-07-A.pdf
Encoder	Balluff	PV124422.pdf



11 Instructions and hints for synchronized 2-theta motion of the system

- Drive up all six pneumatic feet underneath the detector arm.
- Drive the detector translation axis (DT) to its middle position.
- Make sure that the height correction motors of the detector arm (DAAHUL, DAAHUR, DAAHD) are activated during the 2-theta motion.
- Make sure that the 2-theta motion of the optics tank (3AA chamber) synchronously follows the 2-theta motion of the detector chamber. The six air bearings located below the 3AA unit have to be activated during the 3AA 2-theta motion.
- As long as the triple rotating flange is not available at the sample chamber, the 3AA chamber has to be disconnected from the sample chamber before performing the synchronized 2-theta motion.
- Make sure that all collision switches activate and are set to a safe position.

12 Ambient condition and media supply

Temperature	Between 20 K and 25 °C, with ± 0.5 K accuracy
Humidity	40 - 60%
Air pressure	850 - 1150 mbar
Area load	3 tons / m ²
Water temperature	$>RT \pm 0.5^{\circ}\text{C}$
Max. inlet water pressure	10 bar
Pressured air	For operating gate valves
Helium	Suitable for leak tests

Nr. / No.	Abk. / Abbr.	Bezeichnung deutsch / Title German	Bezeichnung englisch / Title English	Artikel Nr. / Stock No.
1	D-11i	Vakuumbür zweite Stufe - Hochvakuumanschluss	Vacuum door 2nd stage - high vacuum port	
2	D-11o	Vakuumbür erste Stufe - Grobvakuumanschluss	Vacuum door 1st stage - roughing port	
3	G-11	Pirani Druckmessröhre	Pirani pressure gauge	0000-01188
4	G-12	Druckmessröhre, Kalkkathode	Pressure gauge, cold cathode	0000-01187
5	G-21	Pirani Druckmessröhre	Pirani pressure gauge	0000-01188
6	G-22	Druckmessröhre, Kalkkathode	Pressure gauge, cold cathode	0000-01187
7	G-41	Pirani Druckmessröhre	Pirani pressure gauge	0000-01188
8	G-42	Druckmessröhre, Kalkkathode	Pressure gauge, cold cathode	0000-01187
9	RS-11i	Drehdurchführung mit Schrittmotorvorbereitung	Rotary feedthrough with stepper motor adaption	0001-02081
10	RS-11o	Drehdurchführung mit Schrittmotorvorbereitung	Rotary feedthrough with stepper motor adaption	0001-02081
11	RS-12o1	Dreifach rotierbarer Flansch - 1 - erster Grobvakuumanschluss	Triple rotating flange - 1 - 1st roughing port	
12	RS-12o2	Dreifach rotierbarer Flansch - 1 - zweiter Grobvakuumanschluss	Triple rotating flange - 1 - 2nd roughing port	
13	RS-13o1	Dreifach rotierbarer Flansch - 2 - erster Grobvakuumanschluss	Triple rotating flange - 2 - 1st roughing port	
14	RS-13o2	Dreifach rotierbarer Flansch - 2 - zweiter Grobvakuumanschluss	Triple rotating flange - 2 - 2nd roughing port	
15	RS-14i	Dreifach rotierbarer Flansch - zweite Stufe - Hochvakuumanschluss	Triple rotating flange - 2nd stage - high vacuum port	
16	RS-14o	Dreifach rotierbarer Flansch - erste Stufe - Grobvakuumanschluss	Triple rotating flange - 1st stage - roughing port	
17	P-11	Membranpumpe MZZ-NT	Diaphragm Pump	0000-05676
18	P-12	Turbopumpe HiPace 30	Turbopump HiPace 30	0000-05675
19	P-13	Ionen Getterpumpe	Ion getter pump	0000-06403
20	P-21	Ionen Getterpumpe	Low profile ion pump	0000-04916
21	P-31	Ionen Getterpumpe	Ion getter pump	0000-06403
22	V-11	Manuelles Eckventil KF16	Manual corner valve KF16	0000-05857
23	V-12	Manuelles Eckventil KF16	Manual corner valve KF16	0000-05857
24	V-13	Manuelles Eckventil KF16	Manual corner valve KF16	0000-05857
25	V-14	Vakuum-Eckventil, Ganzmetall	Vacuum angle valve, all metal	0000-01898
26	V-15	Vakuum-Plattenventil / Schieber	Vacuum plate valve	0000-05698
27	V-21	Vakuum-Eckventil, Ganzmetall	Vacuum angle valve, all metal	0000-01898
28	V-22	Vakuum-Plattenventil / Schieber	Vacuum Gate Valve	0000-04024
29	V-31	Vakuum-Plattenventil / Schieber	Vacuum Gate Valve	0000-04024
30	V-41	Vakuum-Eckventil, Ganzmetall	Vacuum angle valve, all metal	0000-01898
31	V-42	Vakuum-Plattenventil / Schieber	Vacuum gate valve	0000-05915
32	V-43	Vakuum-Plattenventil / Schieber	Vacuum gate valve	0000-05915
33	V-44	Vakuum-Plattenventil / Schieber	Vacuum Gate Valve	0000-04024

BESTEC GmbH Am Stude 2b D-12489 Berlin Tel. 0049-30 - 677 4376		Vakuumschema Vacuum scheme Zeichnungsnummer / drawing no. NSLS-2 SIX	
Anmerkungen Datum Name	Bearb. 14.08.2017 B. H.	Zeichnungsnummer / drawing no. 2 / 3	Drawing Revision 2 / 3

Figure 107: Vacuum scheme of the spectrometer arm system

Abk. / Abbr.	Symbol / Symbol	Bezeichnung deutsch / Title German	Bezeichnung englisch / Title English
		Kupferdichtung	Copper seal
		Vitondichtung	Viton seal
3AA		3-m Arm und Optikbank	3-m arm and optics tank assembly
D		Vakuurtür	Vacuum door
DAA		Detektorarm	Detector arm assembly
DC		Detektorkammer	Detector chamber
G		Druckmessröhre	Gauge
P		Vakuumpumpe	Vacuum pump
P		Membranpumpe	Membran pump
P		Turbomolekularpumpe	Turbo molecular pump
P		Ionengetterpumpe	Ion getter pump
RS		Rotationsdichtung	Rotary seal
SC		Probenkammer	Sample chamber
V		Ventil	Valve

Figure 108: Vacuum scheme of the spectrometer arm system

14 Declaration of conformity

Declaration of conformity according to the EC Directive on machinery 2006/42/EC

We hereby declare that the machinery named below in its design and construction and in the state when placed on the market is in conformity with the essential safety and health requirements of EU Directive 2006/42/EC. Upon modification of the machine without our prior permission, this declaration becomes invalid.

Manufacturer	Product
Bestec GmbH Am Studio 2b 12489 Berlin	Spectrometer Arm System SIX Beamline Project P442 12/2016 NSLS-II USA

EU Directives:

- 2006/42/EC Directive on machinery

Standards:

- EN 12100-1
Safety of machinery - Basic concepts, general principles for design
(Part 1: Basic terminology, methodology)
- EN 12100-2
Safety of machinery - Basic concepts, general principles for design
(Part 2: Technical principles)
- EN 60204-1
Safety of machinery - Electrical equipment of machines
(Part 1: General requirements)
- EN 13849-1
Safety of machinery - Safety-related parts of control systems
(Part 1: General principles for design)

Berlin, 8 October 2020

A handwritten signature in blue ink, appearing to be "J. Schell", written over a horizontal dotted line.